WSQ · FOUR DAYS · 32 HOURS

Autodesk Certified Professional (ACP) for Revit Mechanical Design

Mechanical BIM from model to issued evidence



TGS-2023036661 · v7.0 · 14 September 2026
Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Digital attendance

REQUIRED

AM · PM · Assessment

Learning outcomes

REQUIRED

LO1: Execute mechanical calculations and build conceptual duct and piping designs with a Basis of Design.
LO2: Produce coordinated mechanical schedules, plans, sections, sheets and revisions.
LO3: Create and validate MEP families, connectors and parameters for accurate BIM models.
LO4: Review analytical spaces, zones, energy inputs and calculated values against project criteria.
LO5: Coordinate linked models, clashes, standards and handover evidence efficiently.

Course route

REQUIRED

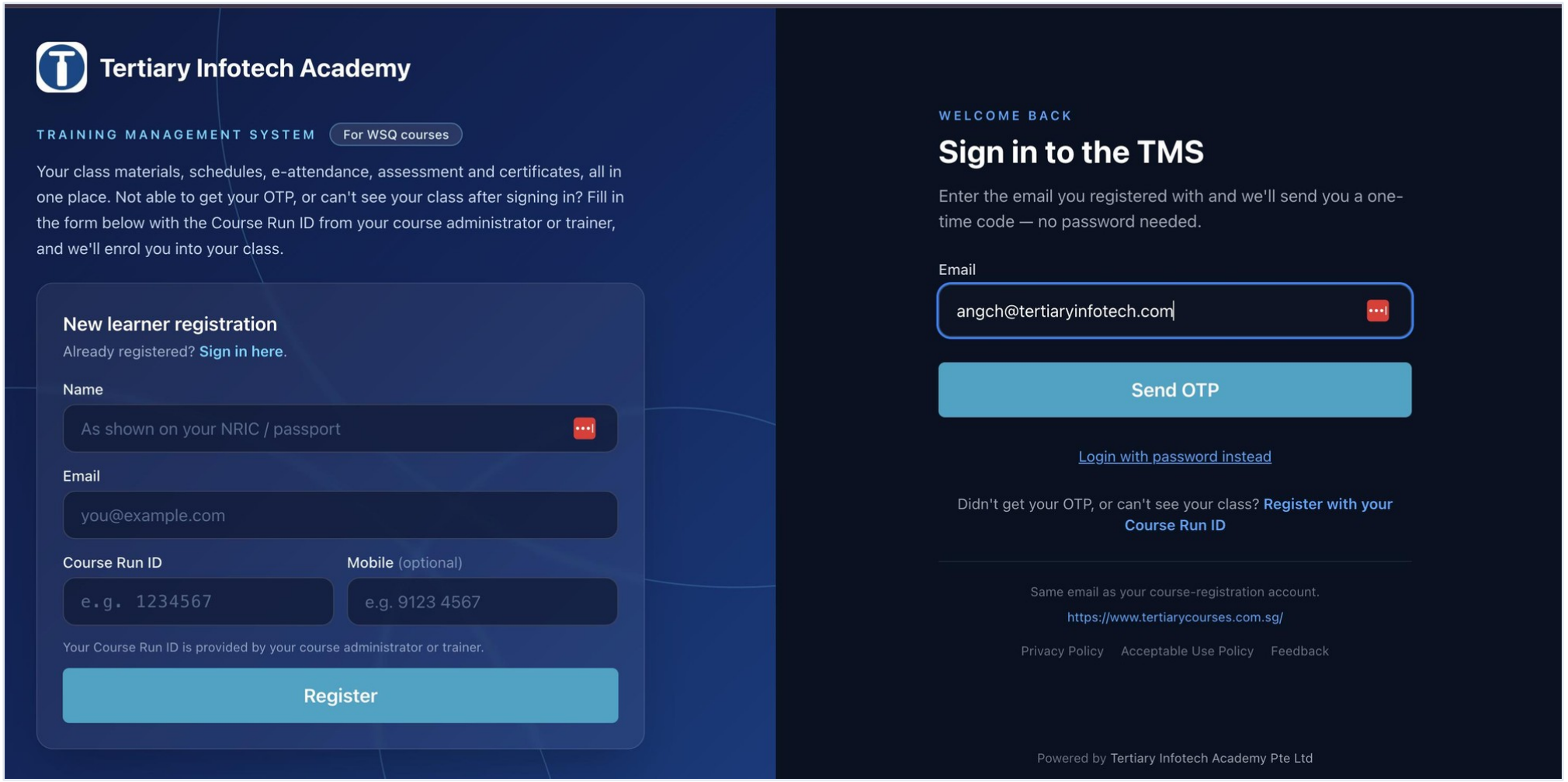
Model → Document → Family → Analyse → Coordinate

Assessment briefing

REQUIRED

Written SAQ 60 min · Practical Performance 120 min

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

Course material

REQUIRED

Open LMS-TMS → Courseware Link → current Learner Guide and Labs

Trainer profile template

TRAINER

Name · professional credentials · Revit/MEP project experience · contact channel

Dr Alfred Ang

TRAINER

Tertiary Infotech Academy · course delivery profile. Verify live biography and credentials before presenting.

Assessment flow

WA

K1–K8 · 60 min

PP

A1–A9 · 120 min

REVIEW

Evidence and C/NYC

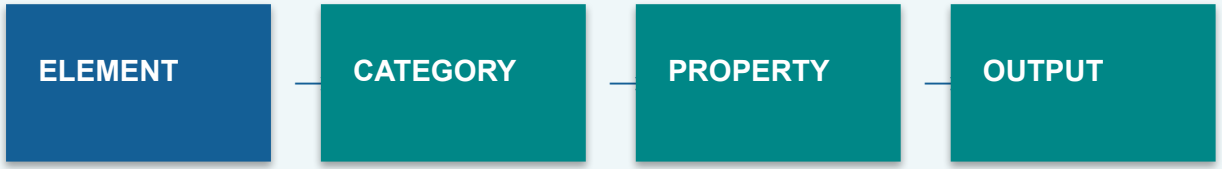
SIGN-OFF

Assessor and candidate

Mechanical equipment and system classification — Model anatomy

MECHANISM / ARTIFACT

Input: Mechanical equipment and system classification.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



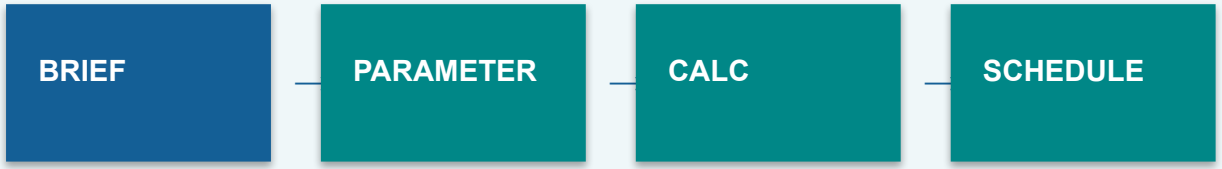
VERIFICATION

Check mechanical / equipment / and / system in its relevant Revit view, browser or schedule.

Mechanical equipment and system classification — Parameter chain

MECHANISM / ARTIFACT

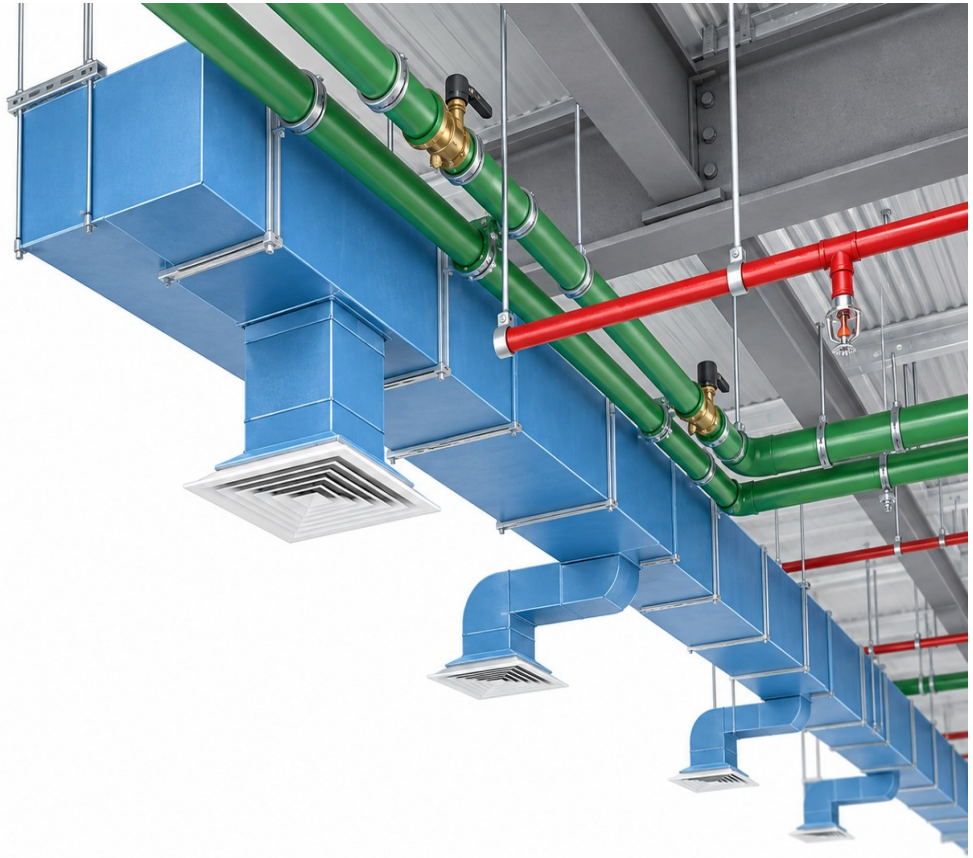
Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 145 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Mechanical equipment and system classification — Coordinated model state



COORDINATION CHECK

Architecture link provides levels and room bounds. Mechanical model owns Mechanical equipment and system classification.
A coordination view must expose both without duplicating source elements.
Check shared coordinates and linked-model visibility.

Mechanical equipment and system classification — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 145 L/s through two branches = 72 L/s each. At 0.08 m², face velocity = 1.81 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Mechanical equipment and system classification — Failure trace

MECHANISM / ARTIFACT

If Mechanical equipment and system classification is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Mechanical equipment and system classification — Decision rule

MECHANISM / ARTIFACT

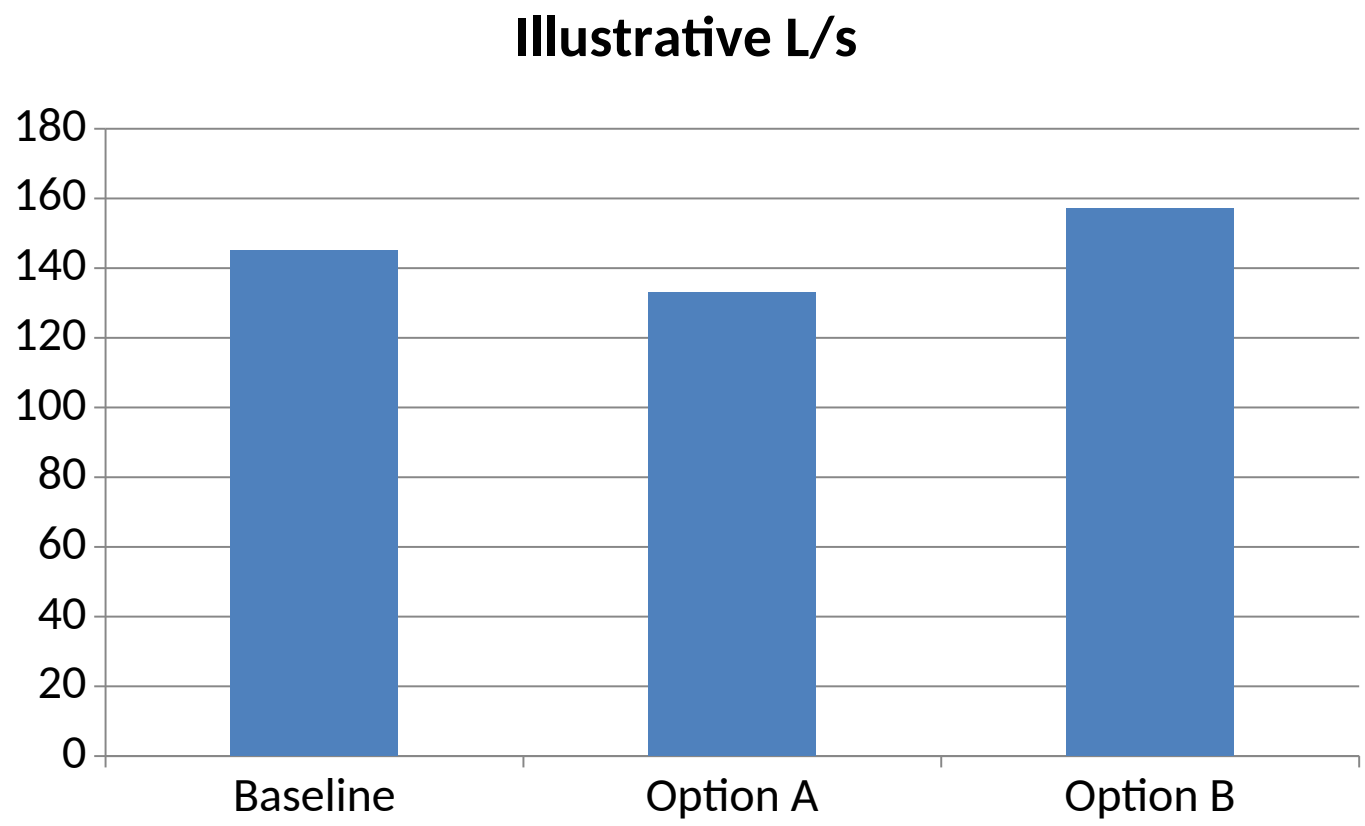
Accept Mechanical equipment and system classification only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Mechanical equipment and system classification — Quantitative review



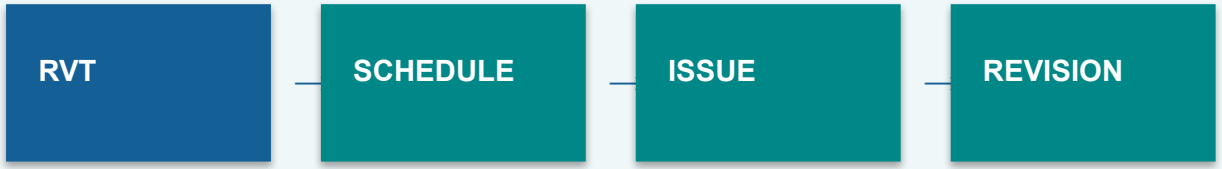
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Mechanical equipment and system classification — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Mechanical equipment and system classification.



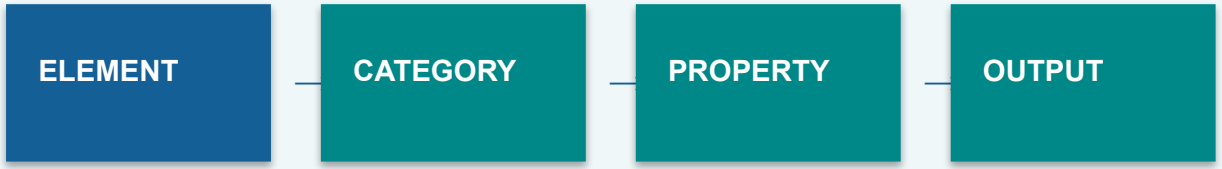
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 01.

Air terminals and design airflow — Model anatomy

MECHANISM / ARTIFACT

Input: Air terminals and design airflow.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



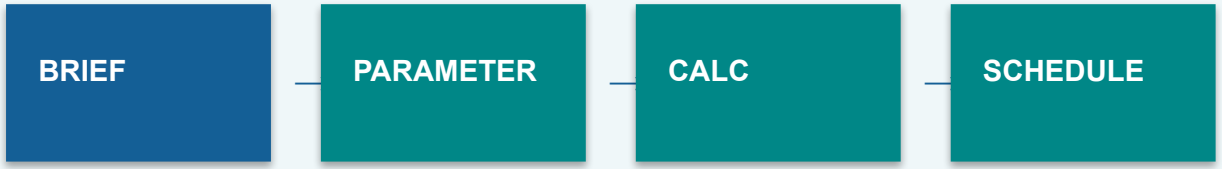
VERIFICATION

Check air / terminals / and / design in its relevant Revit view, browser or schedule.

Air terminals and design airflow — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 170 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Air terminals and design airflow — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Air terminals and design airflow. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Air terminals and design airflow — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 170 L/s through two branches = 85 L/s each. At 0.08 m², face velocity = 2.12 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Air terminals and design airflow — Failure trace

MECHANISM / ARTIFACT

If Air terminals and design airflow is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Air terminals and design airflow — Decision rule

MECHANISM / ARTIFACT

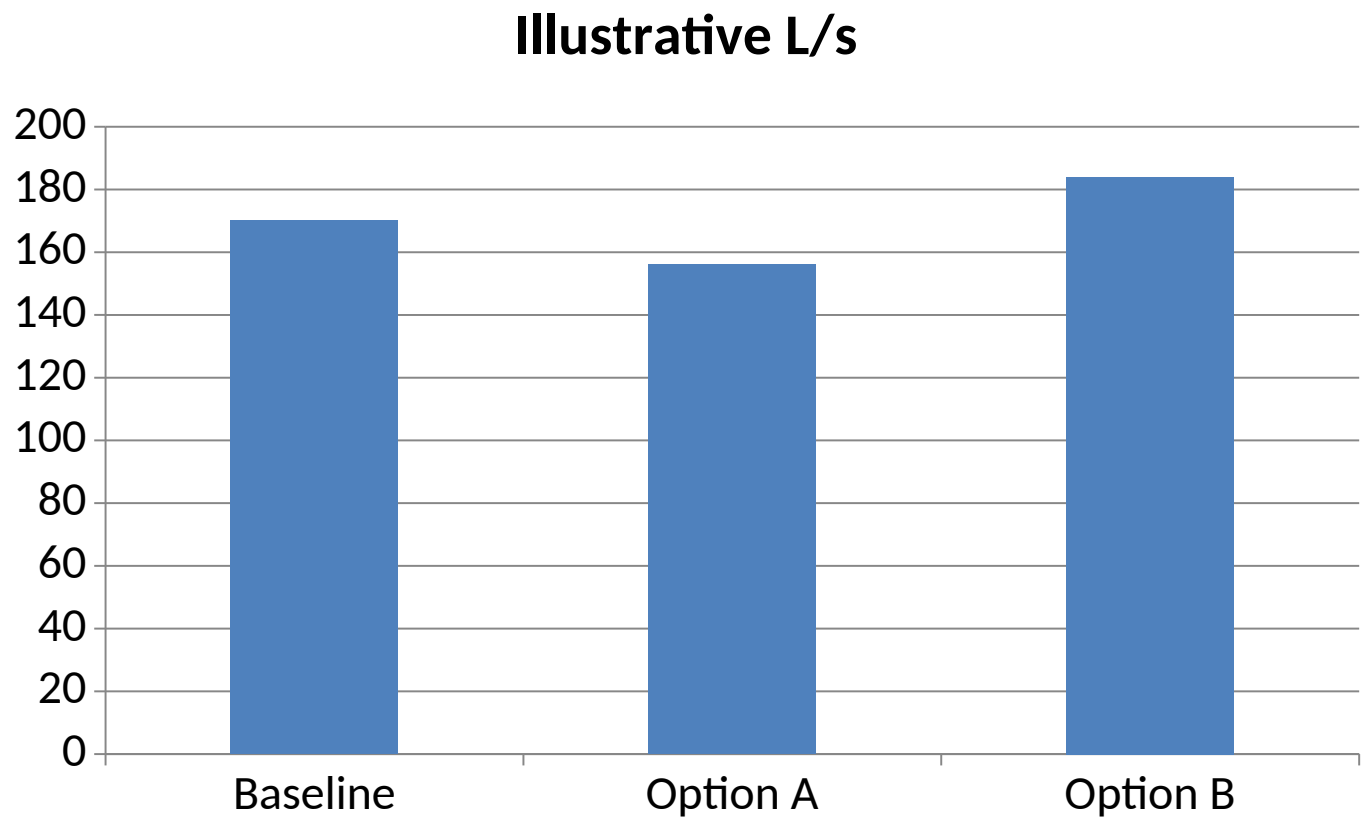
Accept Air terminals and design airflow only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Air terminals and design airflow — Quantitative review



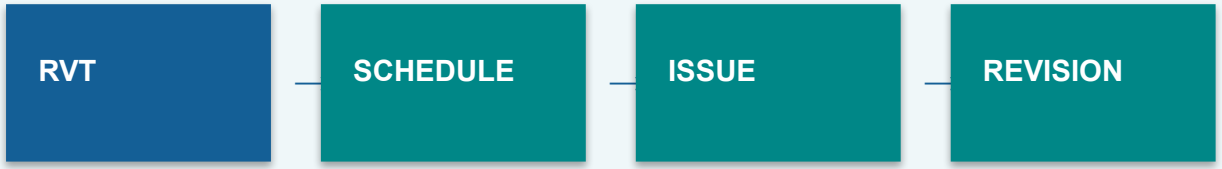
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Air terminals and design airflow — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Air terminals and design airflow.



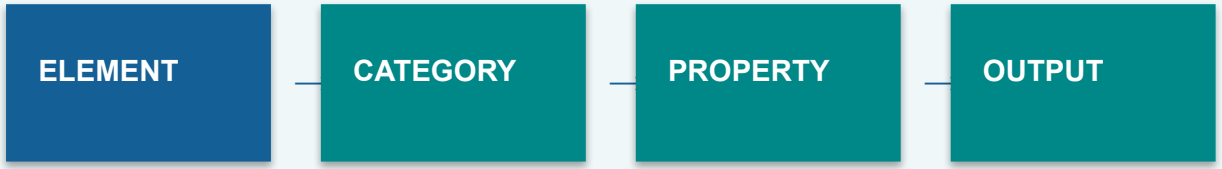
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 01.

Supply, return and exhaust duct systems — Model anatomy

MECHANISM / ARTIFACT

Input: Supply, return and exhaust duct systems.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



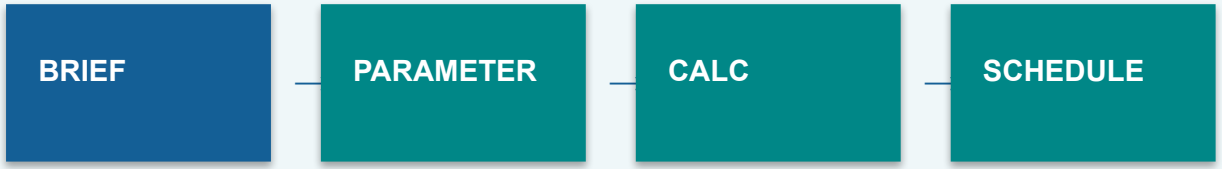
VERIFICATION

Check supply, / return / and / exhaust in its relevant Revit view, browser or schedule.

Supply, return and exhaust duct systems — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 195 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Supply, return and exhaust duct systems — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Supply, return and exhaust duct systems.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Supply, return and exhaust duct systems — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 195 L/s through two branches = 98 L/s each. At 0.08 m², face velocity = 2.44 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Supply, return and exhaust duct systems — Failure trace

MECHANISM / ARTIFACT

If Supply, return and exhaust duct systems is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Supply, return and exhaust duct systems — Decision rule

MECHANISM / ARTIFACT

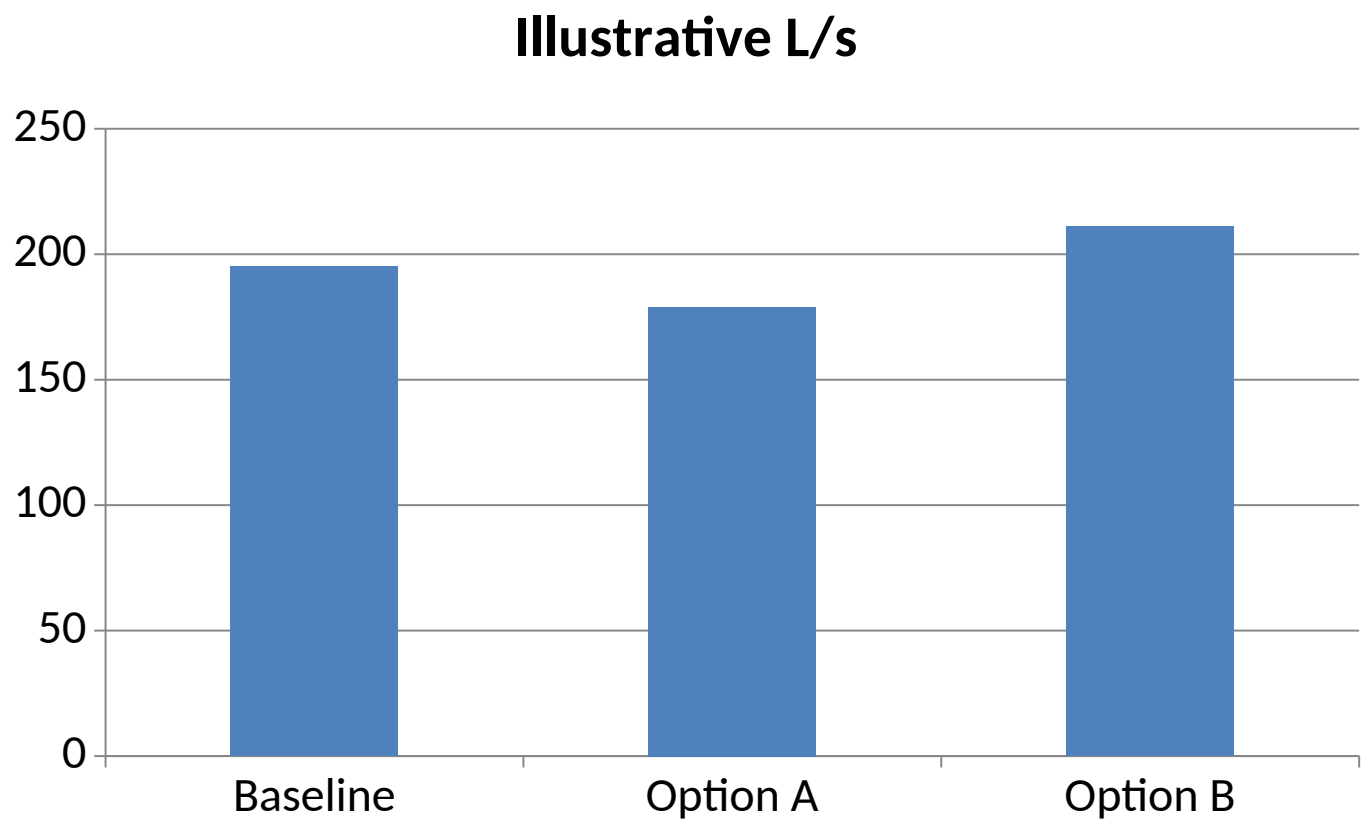
Accept Supply, return and exhaust duct systems only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Supply, return and exhaust duct systems — Quantitative review



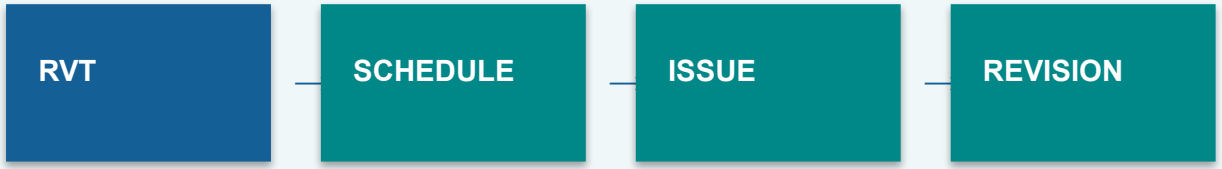
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Supply, return and exhaust duct systems — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Supply, return and exhaust duct systems.



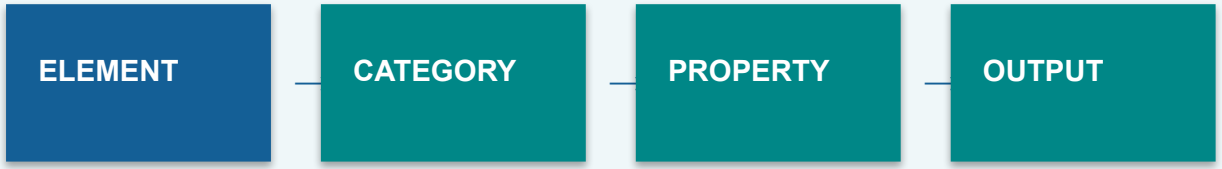
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 01.

Duct types, routing and fittings — Model anatomy

MECHANISM / ARTIFACT

Input: Duct types, routing and fittings.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



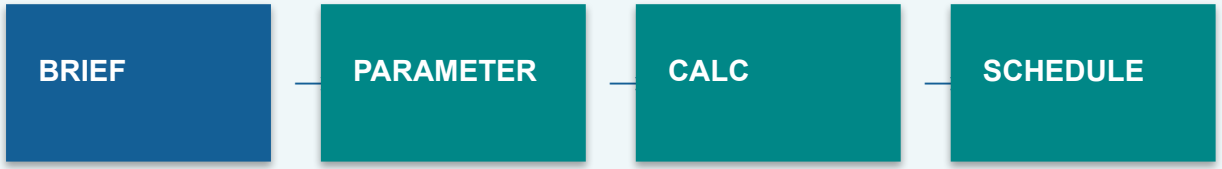
VERIFICATION

Check duct / types, / routing / and in its relevant Revit view, browser or schedule.

Duct types, routing and fittings — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 220 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Duct types, routing and fittings — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Duct types, routing and fittings.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Duct types, routing and fittings — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 220 L/s through two branches = 110 L/s each. At 0.08 m², face velocity = 2.75 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Duct types, routing and fittings — Failure trace

MECHANISM / ARTIFACT

If Duct types, routing and fittings is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Duct types, routing and fittings — Decision rule

MECHANISM / ARTIFACT

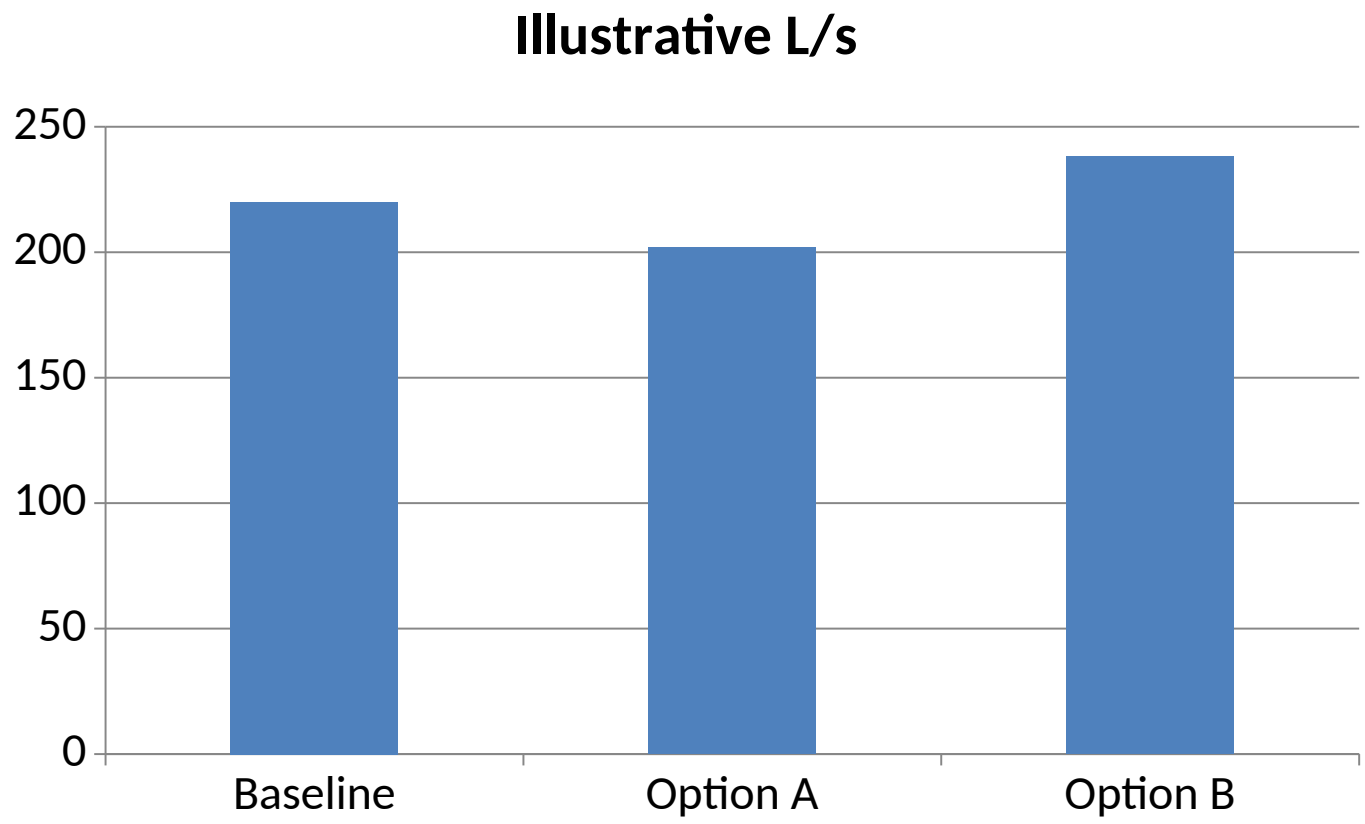
Accept Duct types, routing and fittings only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Duct types, routing and fittings — Quantitative review



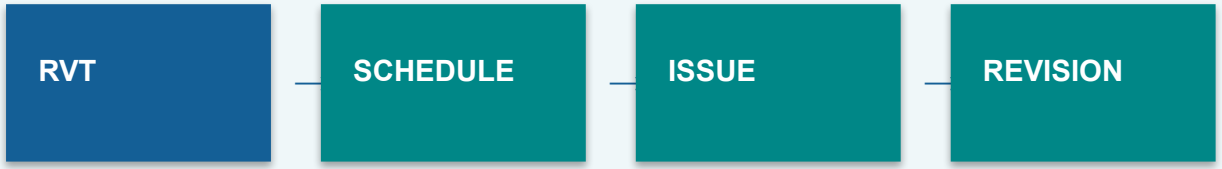
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Duct types, routing and fittings — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Duct types, routing and fittings.



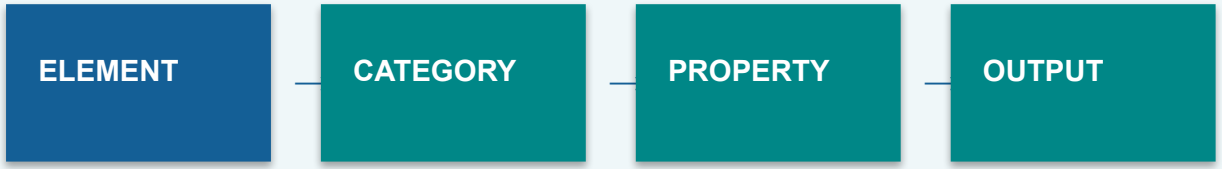
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 02.

Duct sizing and pressure loss — Model anatomy

MECHANISM / ARTIFACT

Input: Duct sizing and pressure loss.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



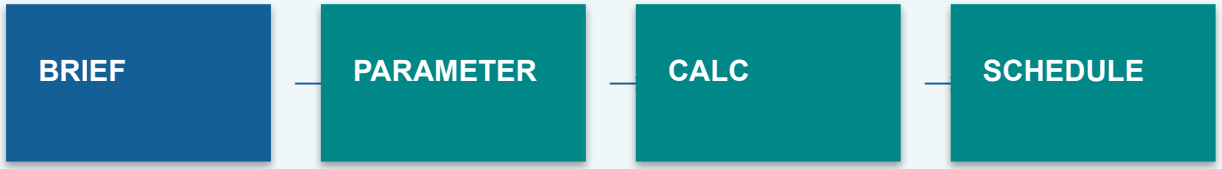
VERIFICATION

Check duct / sizing / and / pressure in its relevant Revit view, browser or schedule.

Duct sizing and pressure loss — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 245 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Duct sizing and pressure loss — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Duct sizing and pressure loss.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Duct sizing and pressure loss — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 245 L/s through two branches = 122 L/s each. At 0.08 m², face velocity = 3.06 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Duct sizing and pressure loss — Failure trace

MECHANISM / ARTIFACT

If Duct sizing and pressure loss is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Duct sizing and pressure loss — Decision rule

MECHANISM / ARTIFACT

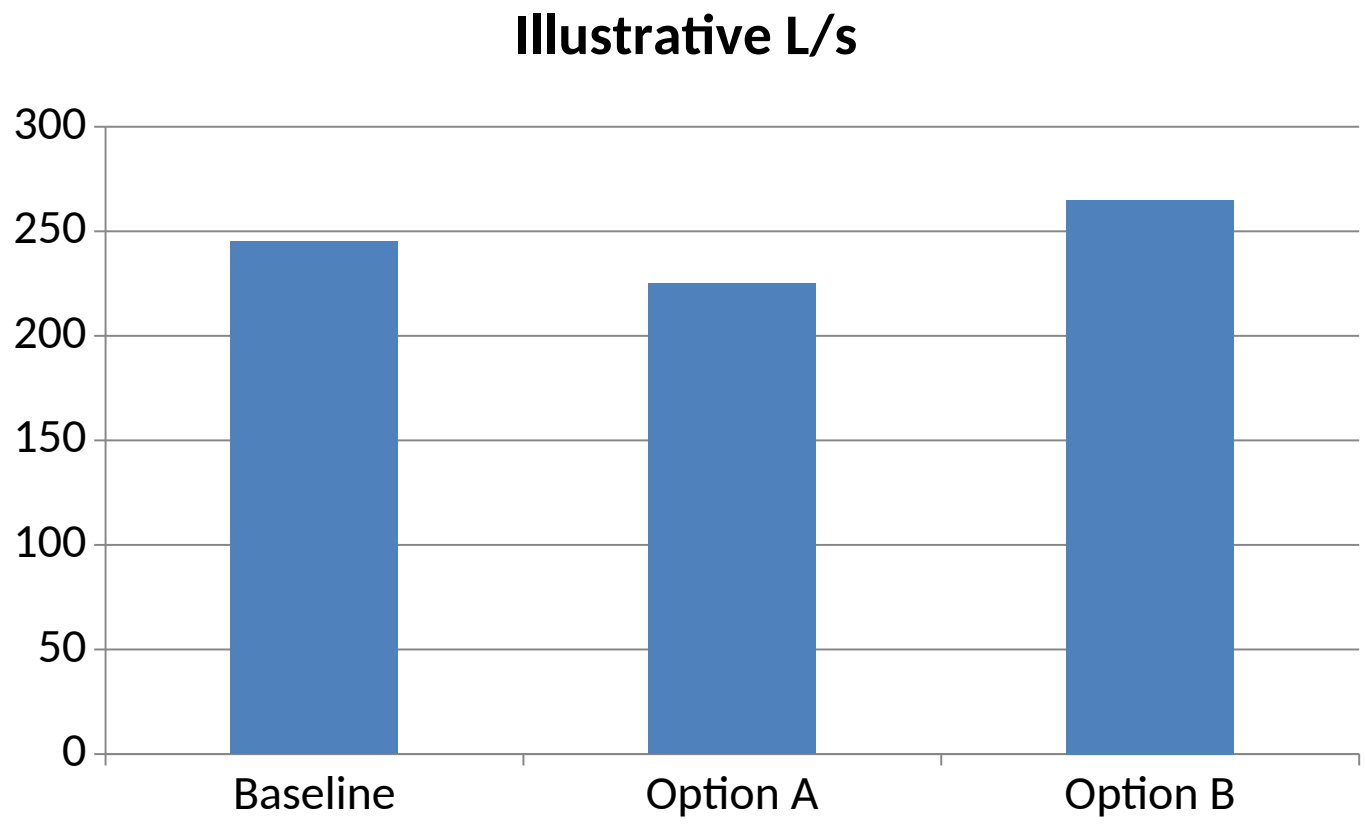
Accept Duct sizing and pressure loss only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Duct sizing and pressure loss — Quantitative review



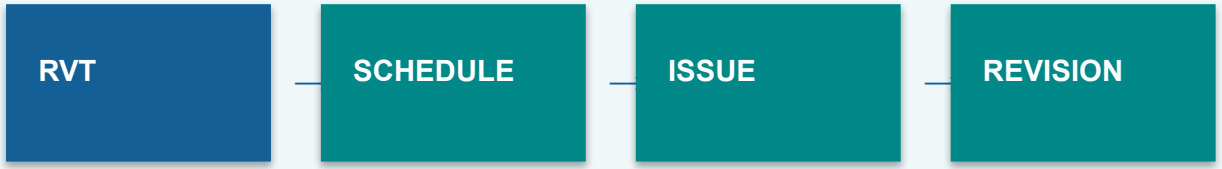
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Duct sizing and pressure loss — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Duct sizing and pressure loss.



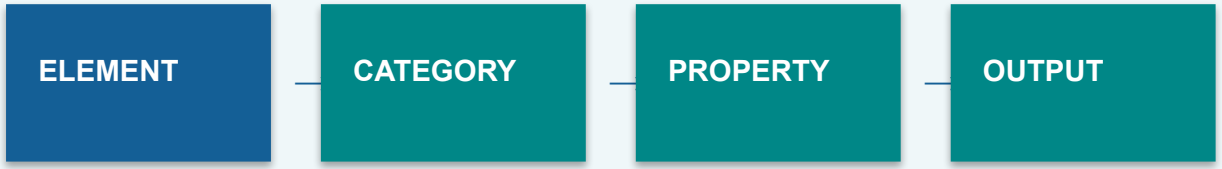
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 02.

Plumbing fixtures and pipe systems — Model anatomy

MECHANISM / ARTIFACT

Input: Plumbing fixtures and pipe systems.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



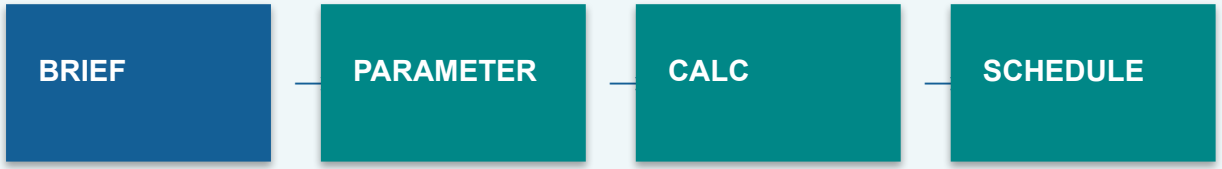
VERIFICATION

Check plumbing / fixtures / and / pipe in its relevant Revit view, browser or schedule.

Plumbing fixtures and pipe systems — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 270 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Plumbing fixtures and pipe systems — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Plumbing fixtures and pipe systems.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Plumbing fixtures and pipe systems — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 270 L/s through two branches = 135 L/s each. At 0.08 m², face velocity = 3.38 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Plumbing fixtures and pipe systems — Failure trace

MECHANISM / ARTIFACT

If Plumbing fixtures and pipe systems is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Plumbing fixtures and pipe systems — Decision rule

MECHANISM / ARTIFACT

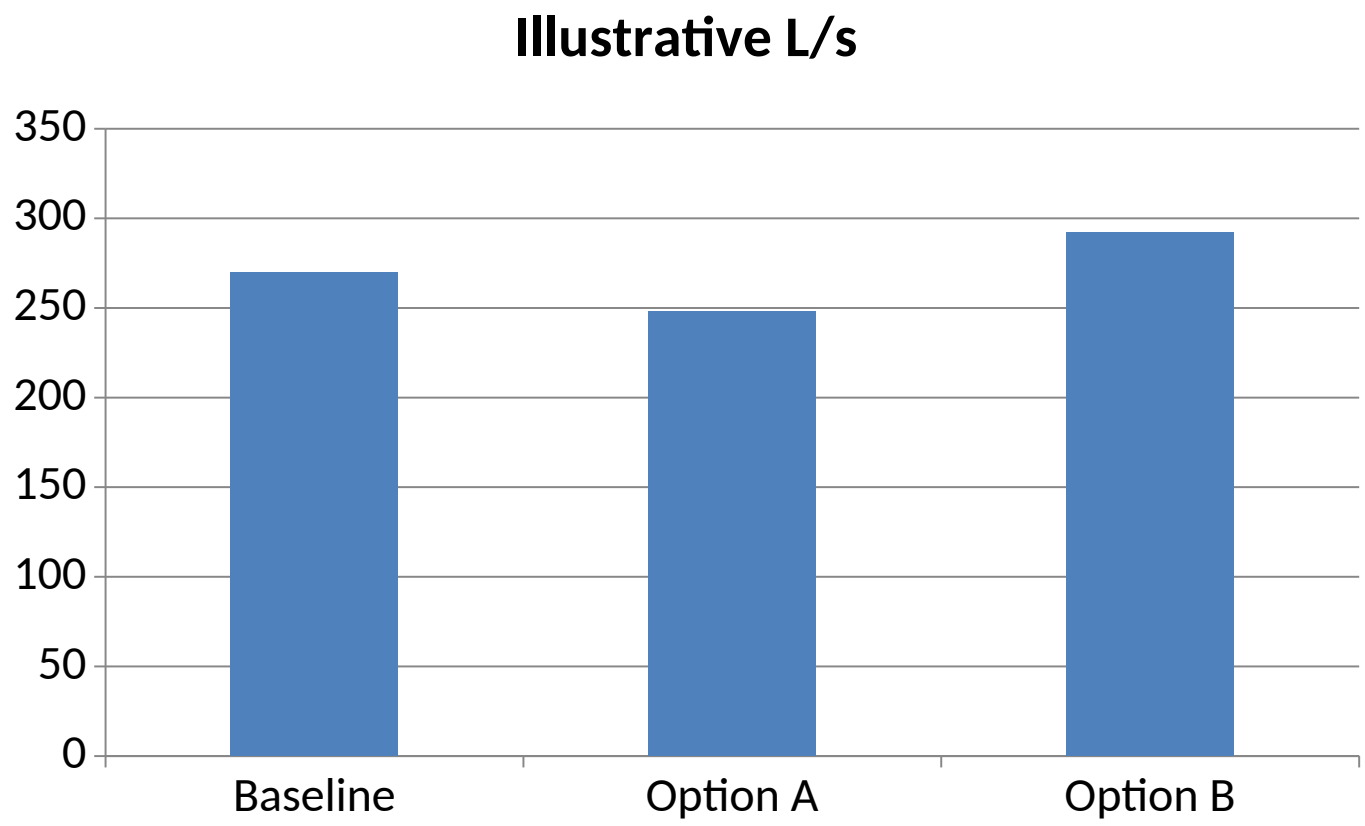
Accept Plumbing fixtures and pipe systems only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Plumbing fixtures and pipe systems — Quantitative review



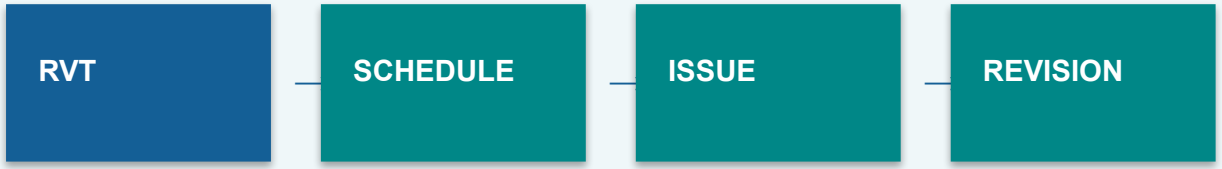
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Plumbing fixtures and pipe systems — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Plumbing fixtures and pipe systems.



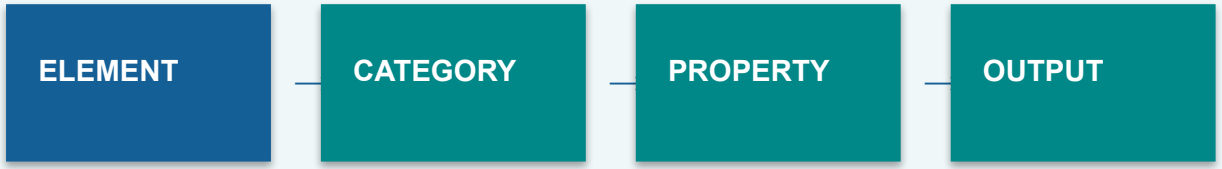
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 02.

Hydronic equipment and pipe sizing — Model anatomy

MECHANISM / ARTIFACT

Input: Hydronic equipment and pipe sizing.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



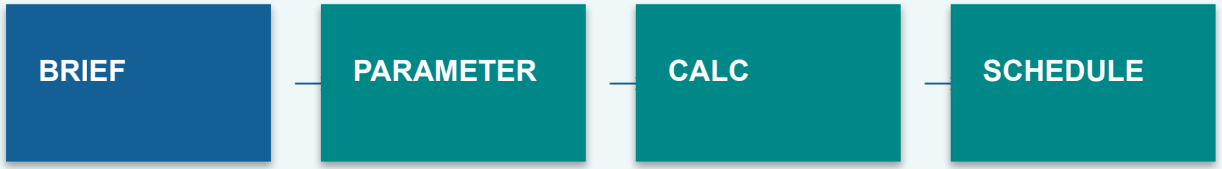
VERIFICATION

Check hydronic / equipment / and / pipe in its relevant Revit view, browser or schedule.

Hydronic equipment and pipe sizing — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 295 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Hydronic equipment and pipe sizing — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Hydronic equipment and pipe sizing.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Hydronic equipment and pipe sizing — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 295 L/s through two branches = 148 L/s each. At 0.08 m², face velocity = 3.69 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Hydronic equipment and pipe sizing — Failure trace

MECHANISM / ARTIFACT

If Hydronic equipment and pipe sizing is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Hydronic equipment and pipe sizing — Decision rule

MECHANISM / ARTIFACT

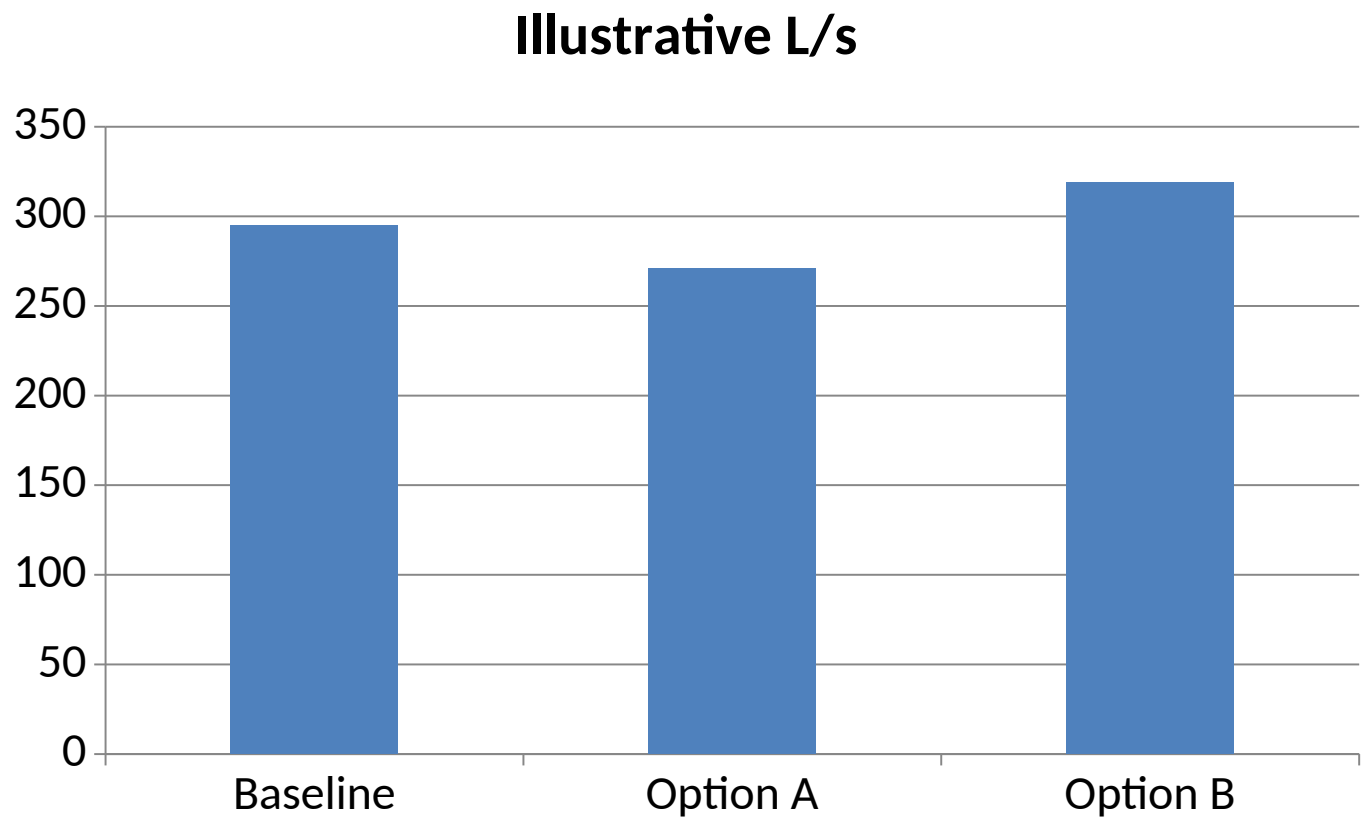
Accept Hydronic equipment and pipe sizing only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Hydronic equipment and pipe sizing — Quantitative review



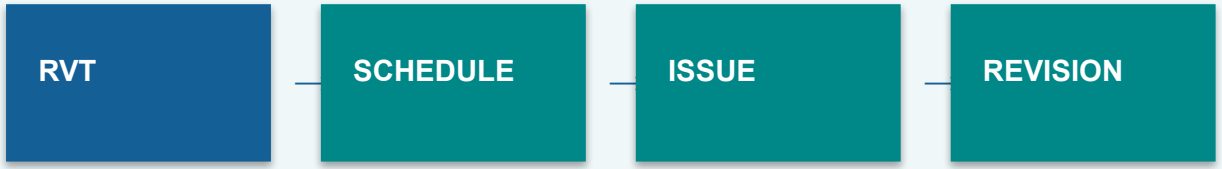
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Hydronic equipment and pipe sizing — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Hydronic equipment and pipe sizing.



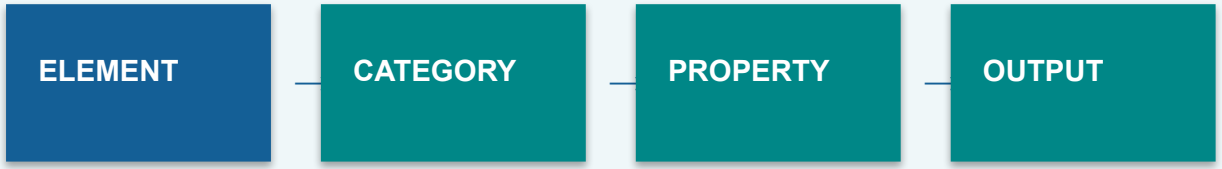
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 03.

Fire protection sprinklers and branch flow — Model anatomy

MECHANISM / ARTIFACT

Input: Fire protection sprinklers and branch flow.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



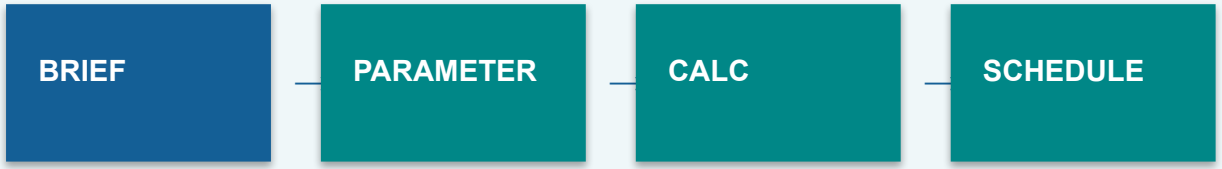
VERIFICATION

Check fire / protection / sprinklers / and in its relevant Revit view, browser or schedule.

Fire protection sprinklers and branch flow — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 320 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

Fire protection sprinklers and branch flow — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Fire protection sprinklers and branch flow.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Fire protection sprinklers and branch flow — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 320 L/s through two branches = 160 L/s each. At 0.08 m², face velocity = 4.00 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Fire protection sprinklers and branch flow — Failure trace

MECHANISM / ARTIFACT

If Fire protection sprinklers and branch flow is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

Fire protection sprinklers and branch flow — Decision rule

MECHANISM / ARTIFACT

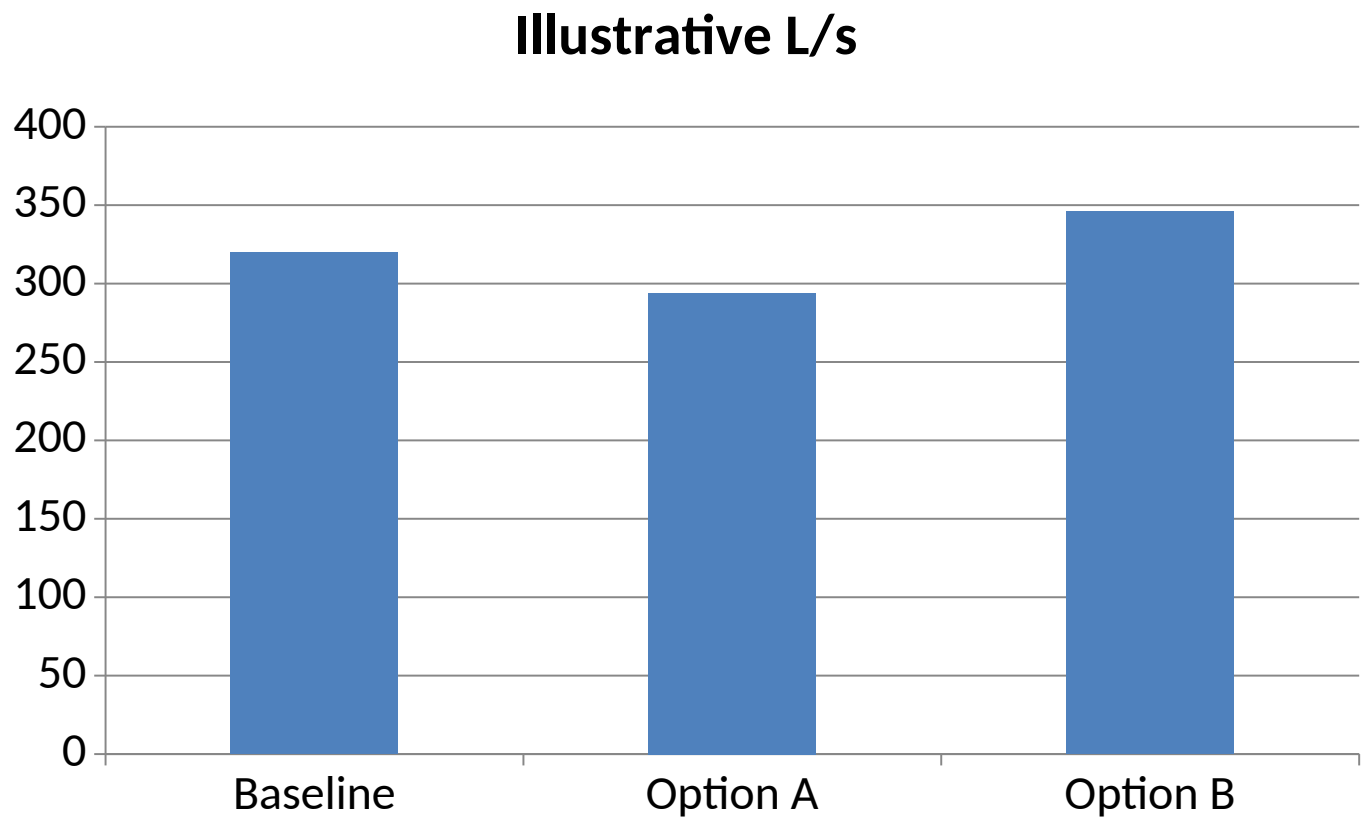
Accept Fire protection sprinklers and branch flow only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Fire protection sprinklers and branch flow — Quantitative review



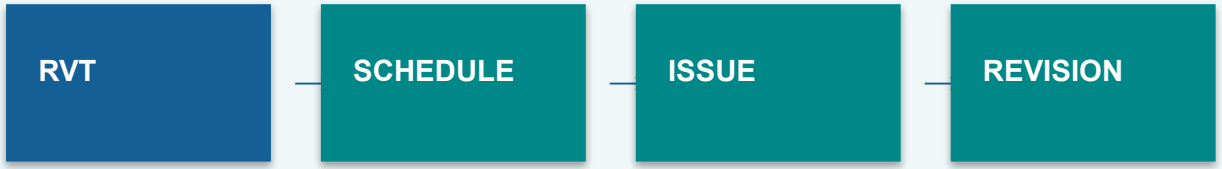
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Fire protection sprinklers and branch flow — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Fire protection sprinklers and branch flow.



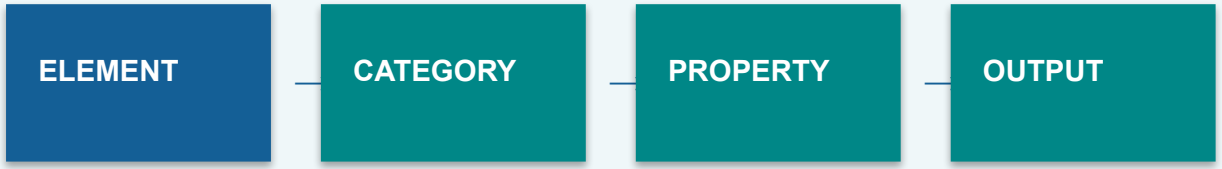
VERIFICATION

Criteria: K1 K2 A1 A2; Lab 03.

System Browser and unassigned connectors — Model anatomy

MECHANISM / ARTIFACT

Input: System Browser and unassigned connectors.
Control: category, system type, connector classification and flow.
Output: a named MEP element and logical system with a traceable element ID.



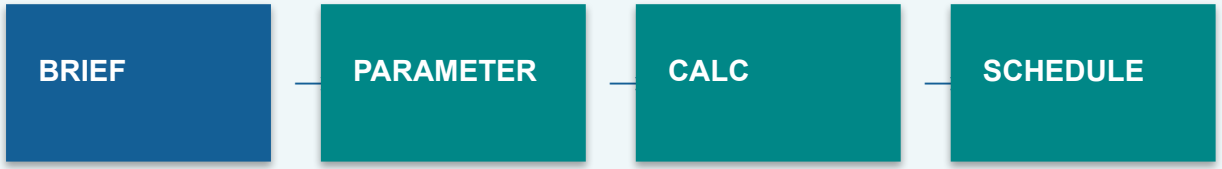
VERIFICATION

Check system / browser / and / unassigned in its relevant Revit view, browser or schedule.

System Browser and unassigned connectors — Parameter chain

MECHANISM / ARTIFACT

Project brief → category, system type, connector classification and flow → model state → drawing or report. Illustrative value: 345 L/s.



VERIFICATION

Check units, discipline and type versus instance scope.

System Browser and unassigned connectors — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns System Browser and unassigned connectors.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

System Browser and unassigned connectors — Worked design example

MECHANISM / ARTIFACT

Illustrative supply 345 L/s through two branches = 172 L/s each. At 0.08 m², face velocity = 4.31 m/s.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

System Browser and unassigned connectors — Failure trace

MECHANISM / ARTIFACT

If System Browser and unassigned connectors is missing or inconsistent, isolate connector classification, flow setting, sizing constraint and System Browser assignment.



VERIFICATION

Capture element ID and exact property before changing settings.

System Browser and unassigned connectors — Decision rule

MECHANISM / ARTIFACT

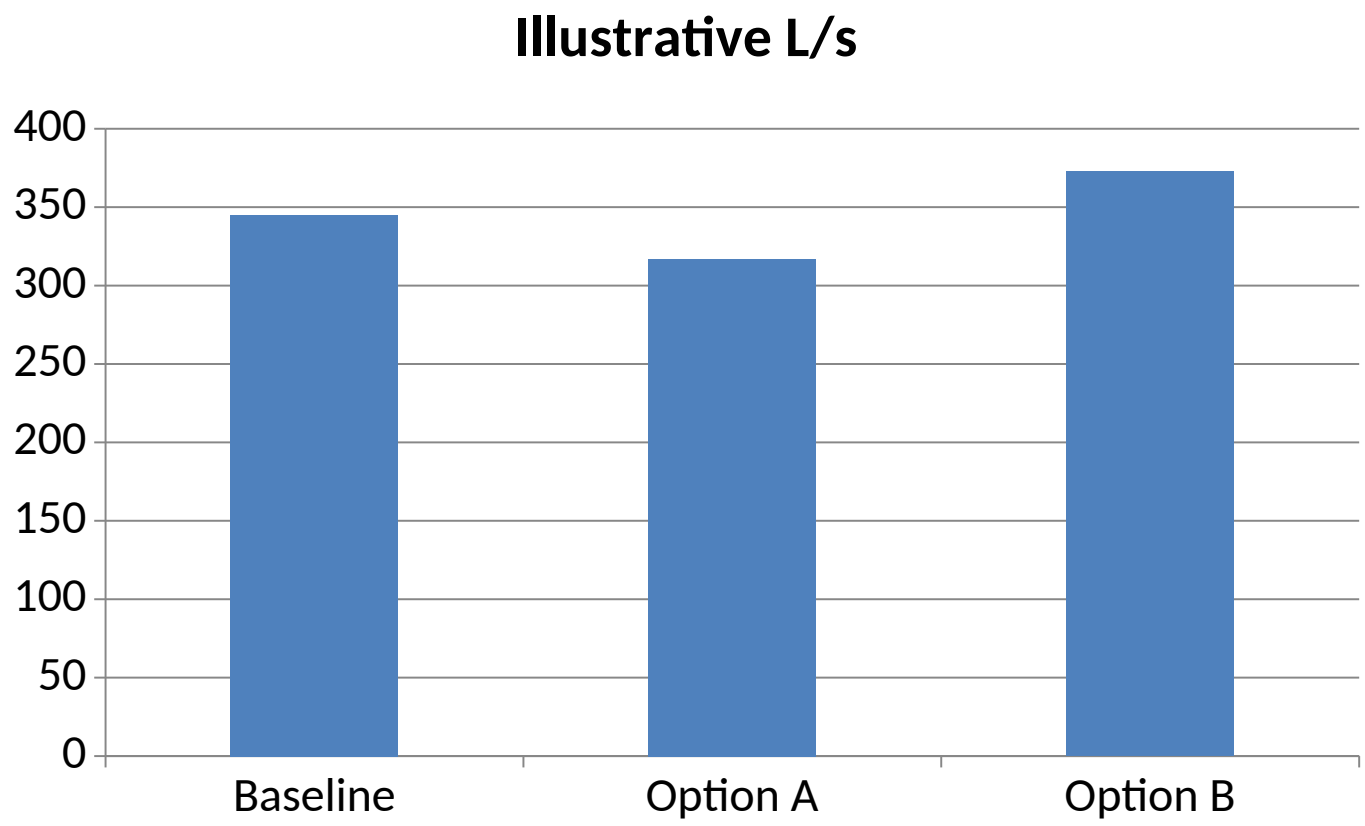
Accept System Browser and unassigned connectors only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

System Browser and unassigned connectors — Quantitative review



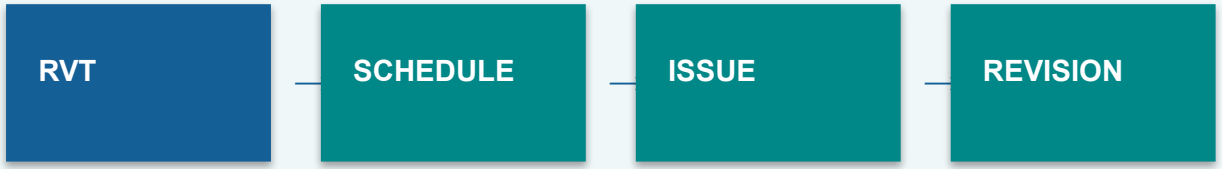
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

System Browser and unassigned connectors — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for System Browser and unassigned connectors.



VERIFICATION

Criteria: K1 K2 A1 A2; Lab 03.

Lab 01 — Establish MEP project and link architecture

MODEL OUTPUT

a clean mechanical project and link audit

EVIDENCE

linked architecture, levels, shared coordinates; working RVT, schedule and PDF

Lab 01 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 02 — Place equipment and air terminals

MODEL OUTPUT

equipment and terminal layout with a connector register

EVIDENCE

AHU, diffusers, flow assignment; working RVT, schedule and PDF

Lab 02 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 03 — Build and size a duct system

MODEL OUTPUT

connected supply system with calculated airflow

EVIDENCE

supply duct, branches, fittings and sizing; working RVT, schedule and PDF

Lab 03 — acceptance evidence

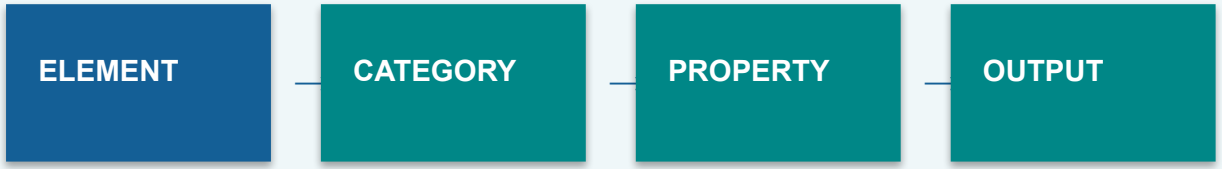
PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

View templates and discipline visibility — Model anatomy

MECHANISM / ARTIFACT

Input: View templates and discipline visibility.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



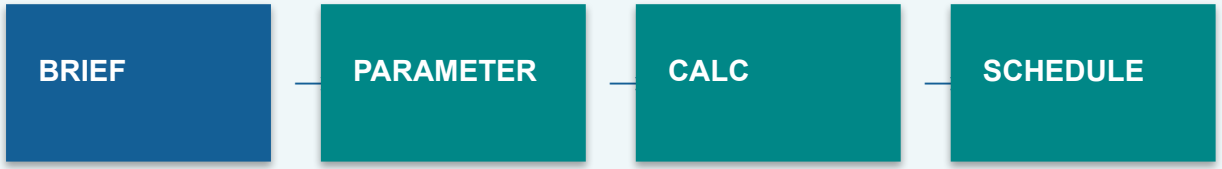
VERIFICATION

Check view / templates / and / discipline in its relevant Revit view, browser or schedule.

View templates and discipline visibility — Parameter chain

MECHANISM / ARTIFACT

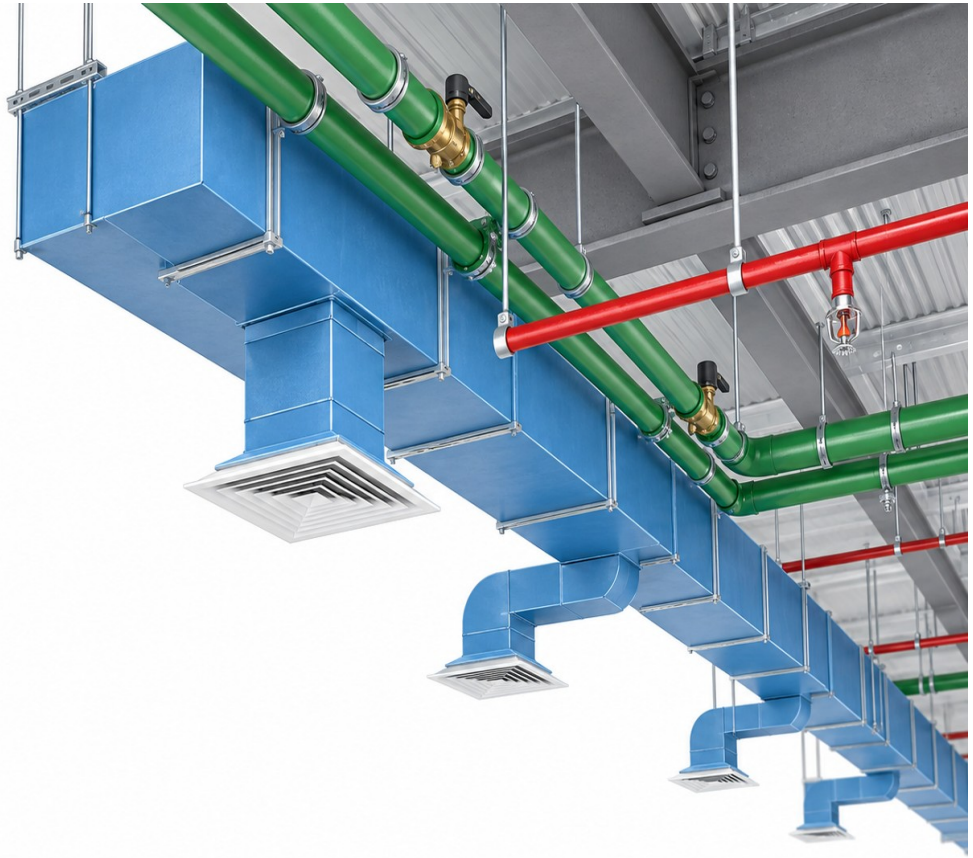
Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 6 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

View templates and discipline visibility — Coordinated model state



COORDINATION CHECK

Architecture link provides levels and room bounds. Mechanical model owns View templates and discipline visibility.
A coordination view must expose both without duplicating source elements.
Check shared coordinates and linked-model visibility.

View templates and discipline visibility — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 6 scheduled items, 5 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

View templates and discipline visibility — Failure trace

MECHANISM / ARTIFACT

If View templates and discipline visibility is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

View templates and discipline visibility — Decision rule

MECHANISM / ARTIFACT

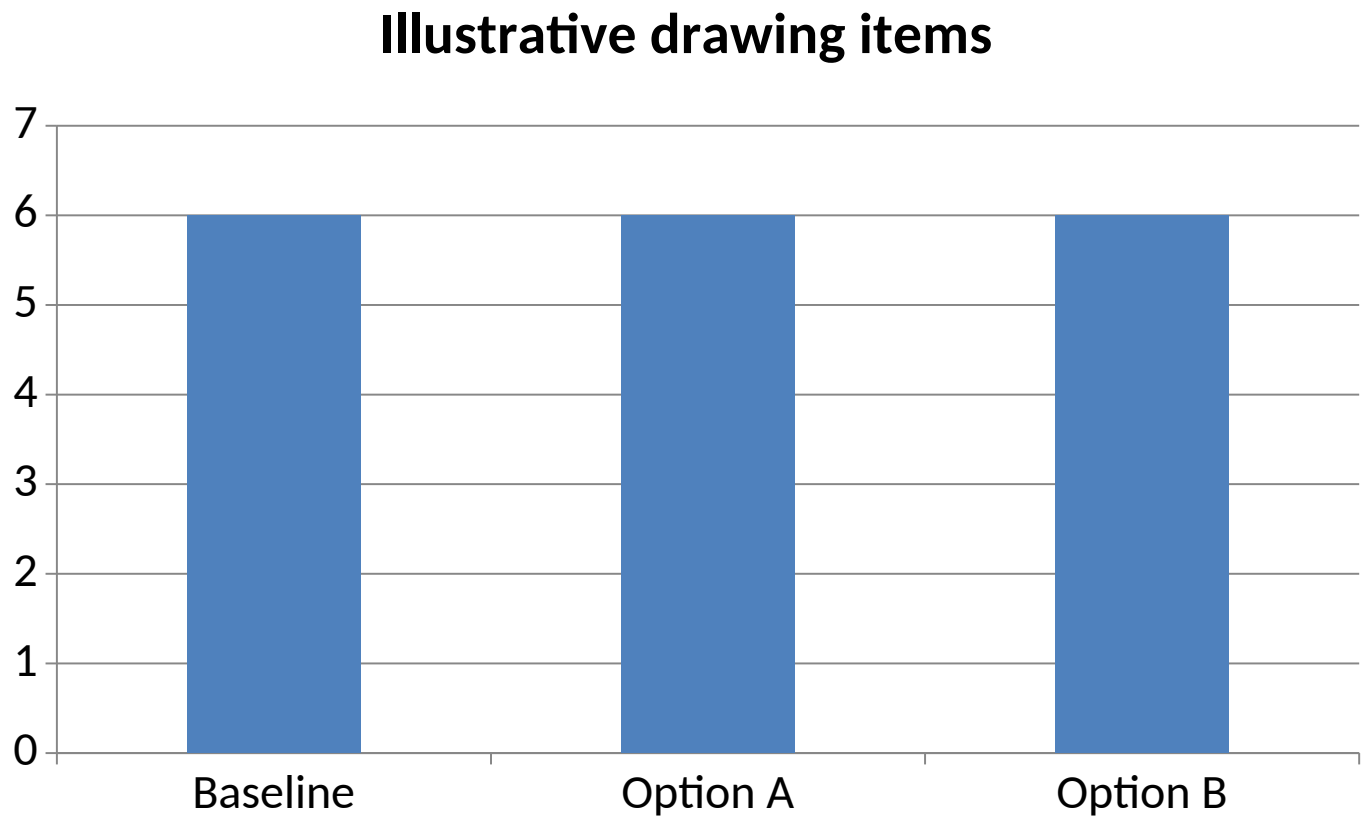
Accept View templates and discipline visibility only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

View templates and discipline visibility — Quantitative review



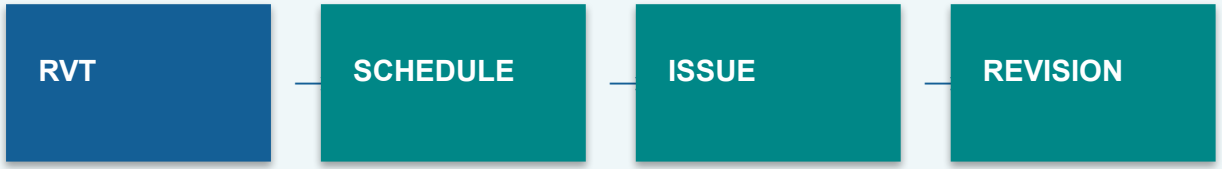
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

View templates and discipline visibility — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for View templates and discipline visibility.



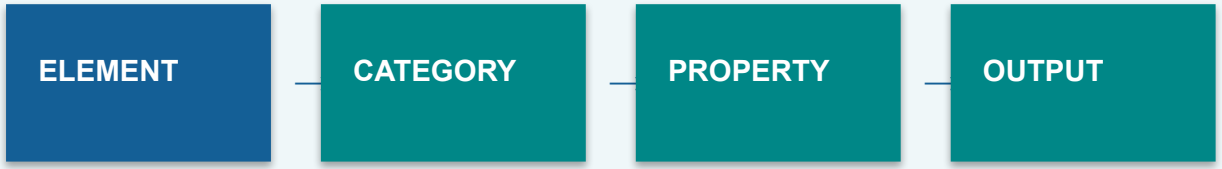
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 04.

Mechanical plans, sections and callouts — Model anatomy

MECHANISM / ARTIFACT

Input: Mechanical plans, sections and callouts.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



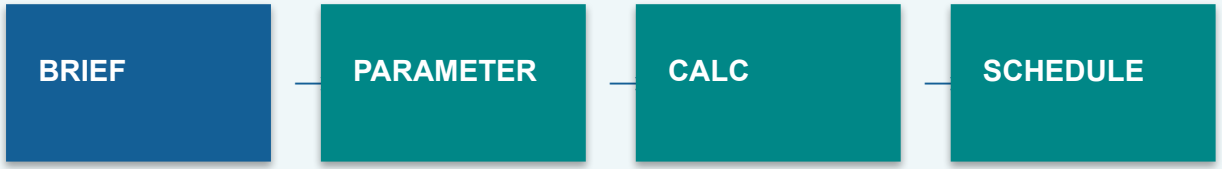
VERIFICATION

Check mechanical / plans, / sections / and in its relevant Revit view, browser or schedule.

Mechanical plans, sections and callouts — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 7 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Mechanical plans, sections and callouts — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Mechanical plans, sections and callouts.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Mechanical plans, sections and callouts — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 7 scheduled items, 6 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Mechanical plans, sections and callouts — Failure trace

MECHANISM / ARTIFACT

If Mechanical plans, sections and callouts is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Mechanical plans, sections and callouts — Decision rule

MECHANISM / ARTIFACT

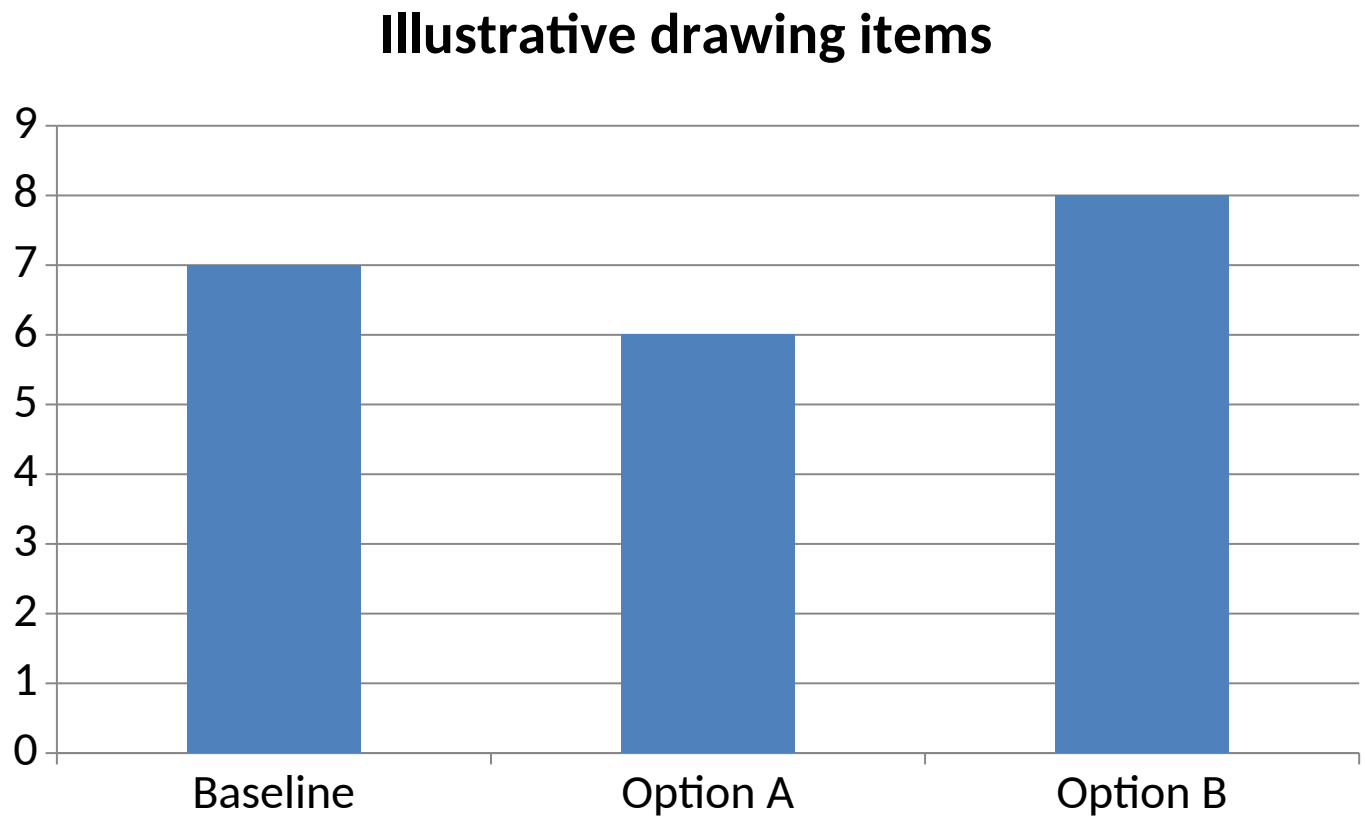
Accept Mechanical plans, sections and callouts only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Mechanical plans, sections and callouts — Quantitative review



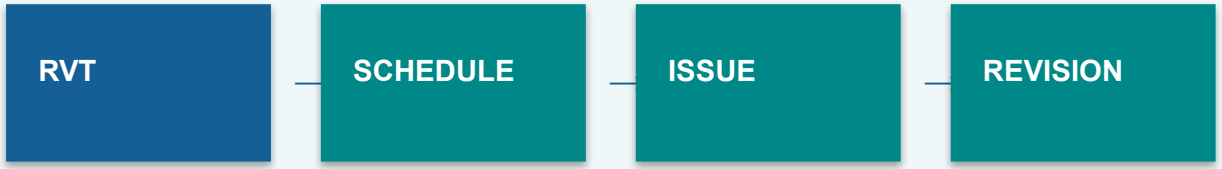
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Mechanical plans, sections and callouts — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Mechanical plans, sections and callouts.



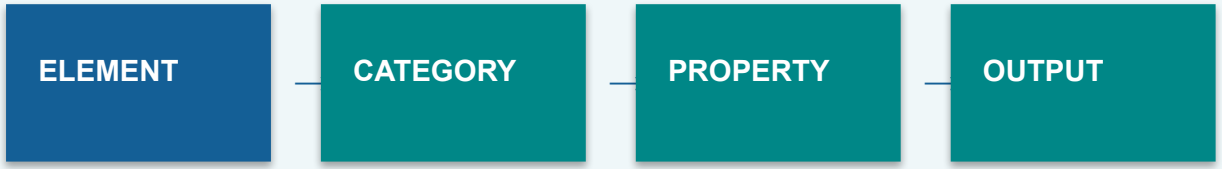
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 04.

Duct and pipe schedules — Model anatomy

MECHANISM / ARTIFACT

Input: Duct and pipe schedules.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



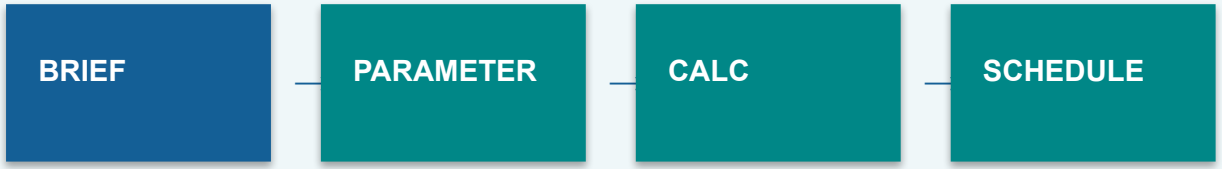
VERIFICATION

Check duct / and / pipe / schedules in its relevant Revit view, browser or schedule.

Duct and pipe schedules — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 8 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Duct and pipe schedules — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Duct and pipe schedules.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Duct and pipe schedules — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 8 scheduled items, 7 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Duct and pipe schedules — Failure trace

MECHANISM / ARTIFACT

If Duct and pipe schedules is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Duct and pipe schedules — Decision rule

MECHANISM / ARTIFACT

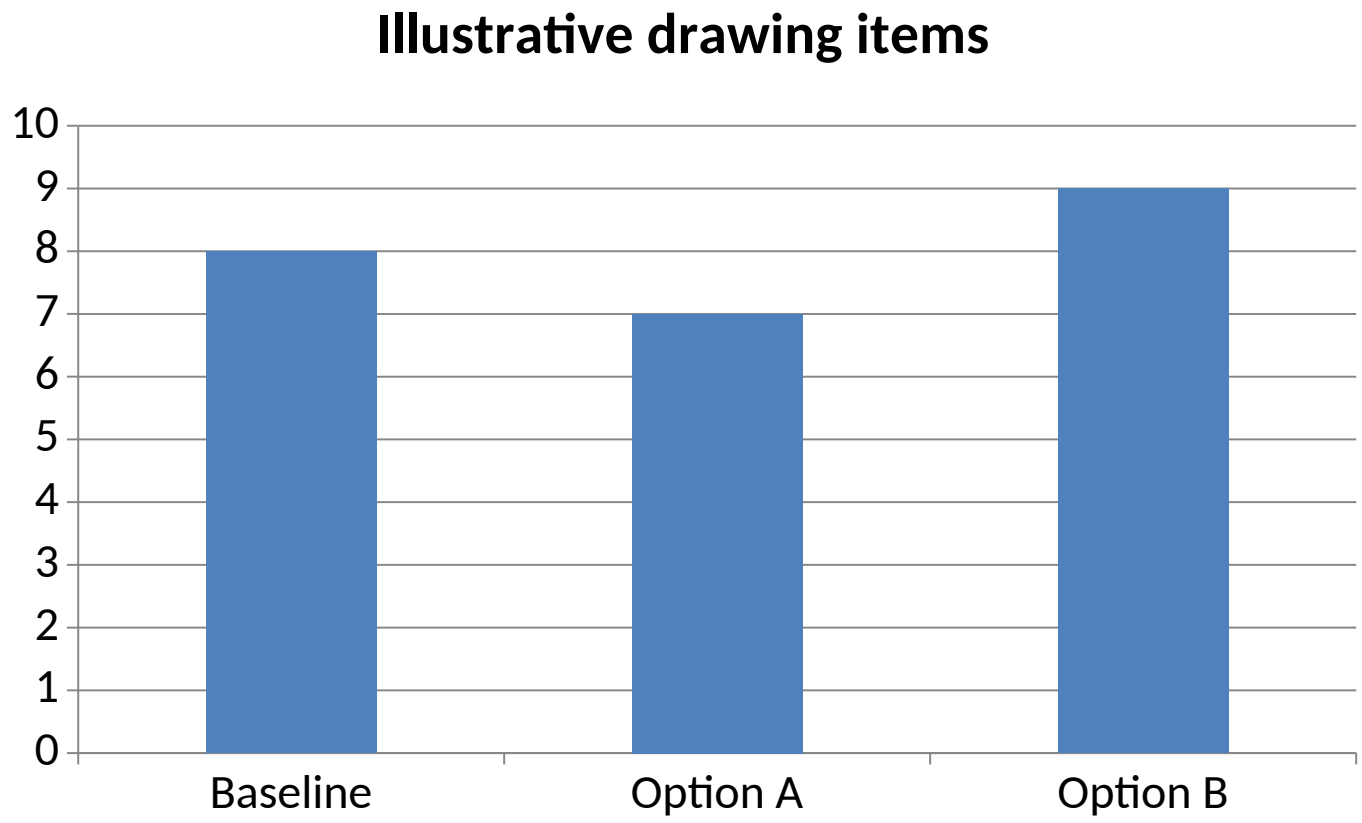
Accept Duct and pipe schedules only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Duct and pipe schedules — Quantitative review



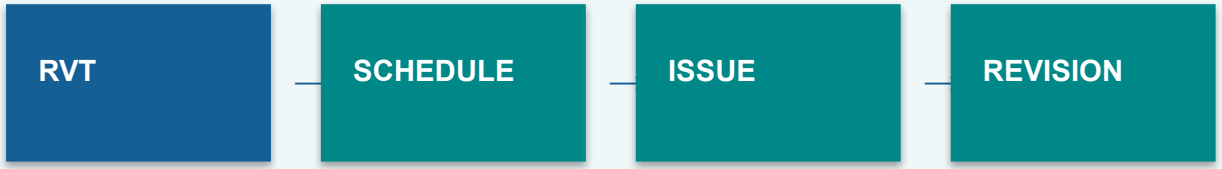
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Duct and pipe schedules — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Duct and pipe schedules.



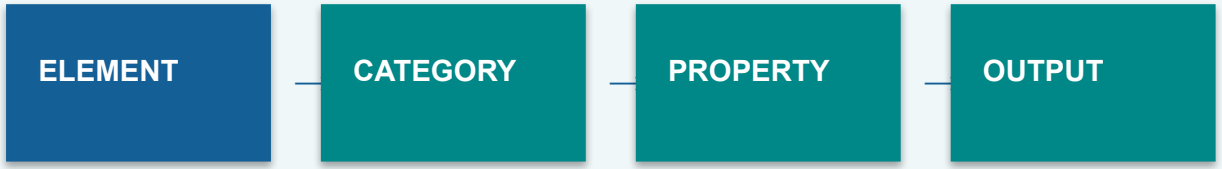
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 04.

Equipment schedules and calculated fields — Model anatomy

MECHANISM / ARTIFACT

Input: Equipment schedules and calculated fields.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



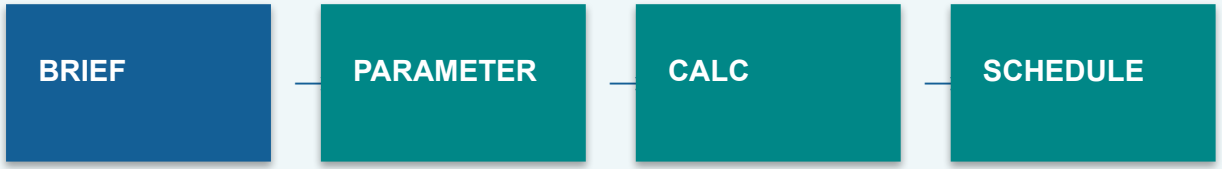
VERIFICATION

Check equipment / schedules / and / calculated in its relevant Revit view, browser or schedule.

Equipment schedules and calculated fields — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 9 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Equipment schedules and calculated fields — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Equipment schedules and calculated fields.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Equipment schedules and calculated fields — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 9 scheduled items, 8 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Equipment schedules and calculated fields — Failure trace

MECHANISM / ARTIFACT

If Equipment schedules and calculated fields is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Equipment schedules and calculated fields — Decision rule

MECHANISM / ARTIFACT

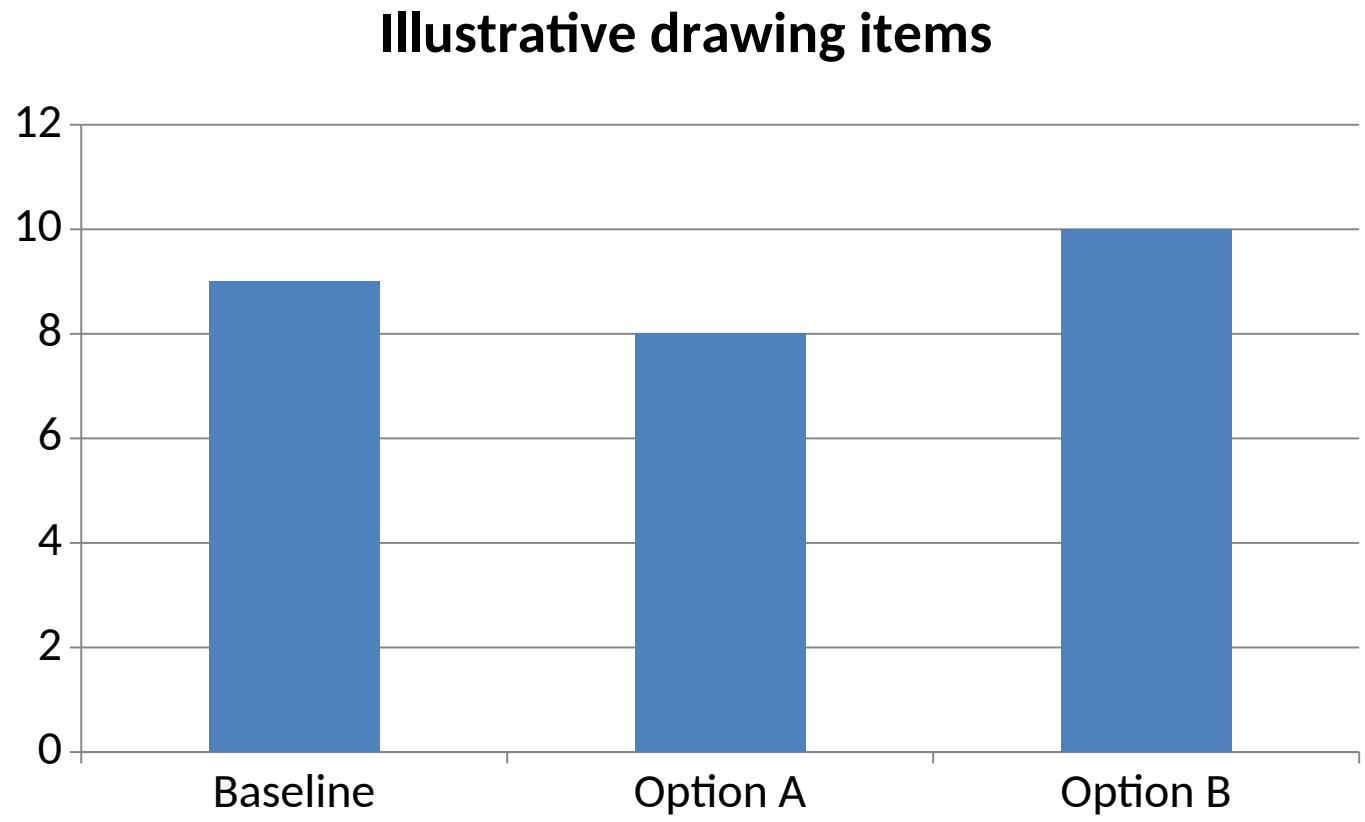
Accept Equipment schedules and calculated fields only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Equipment schedules and calculated fields — Quantitative review



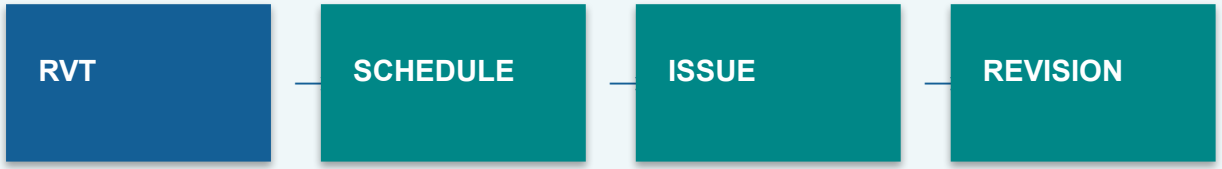
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Equipment schedules and calculated fields — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Equipment schedules and calculated fields.



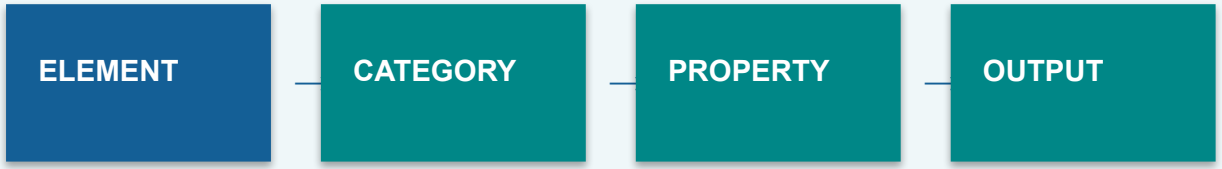
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 05.

Sheets, titleblocks and viewports — Model anatomy

MECHANISM / ARTIFACT

Input: Sheets, titleblocks and viewports.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



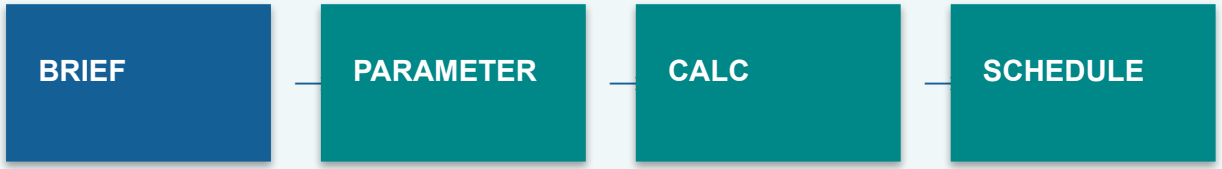
VERIFICATION

Check sheets, / titleblocks / and / viewports in its relevant Revit view, browser or schedule.

Sheets, titleblocks and viewports — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 10 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Sheets, titleblocks and viewports — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Sheets, titleblocks and viewports. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Sheets, titleblocks and viewports — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 10 scheduled items, 9 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Sheets, titleblocks and viewports — Failure trace

MECHANISM / ARTIFACT

If Sheets, titleblocks and viewports is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Sheets, titleblocks and viewports — Decision rule

MECHANISM / ARTIFACT

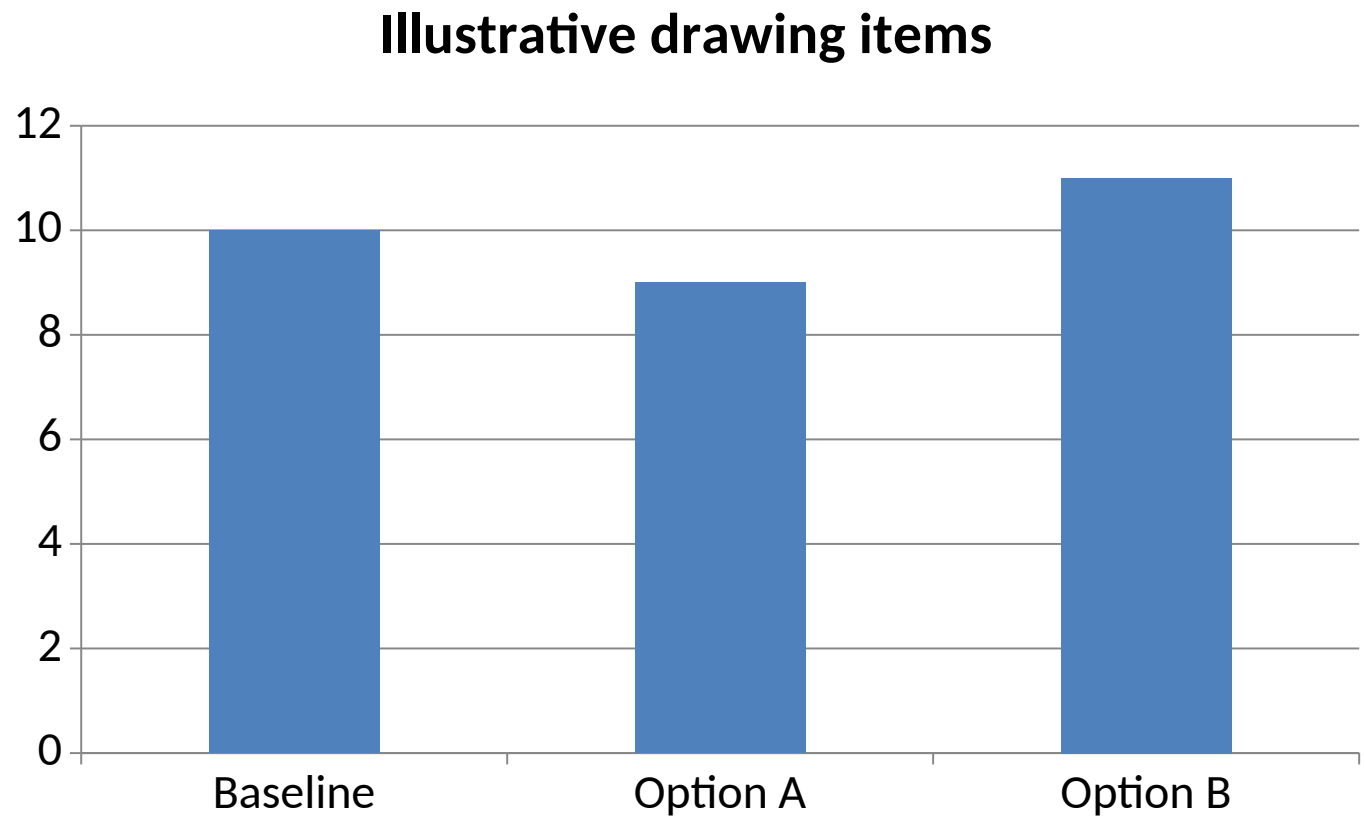
Accept Sheets, titleblocks and viewports only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Sheets, titleblocks and viewports — Quantitative review



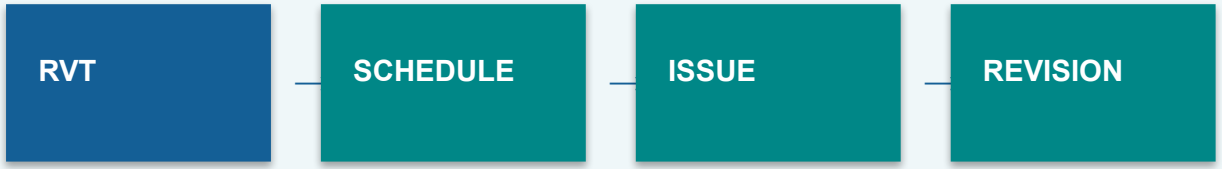
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Sheets, titleblocks and viewports — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Sheets, titleblocks and viewports.



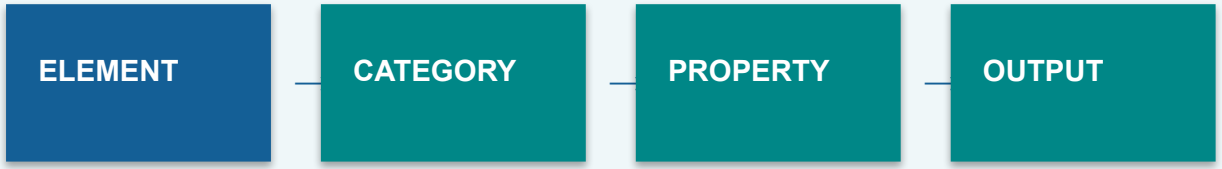
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 05.

Revision clouds and issue metadata — Model anatomy

MECHANISM / ARTIFACT

Input: Revision clouds and issue metadata.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



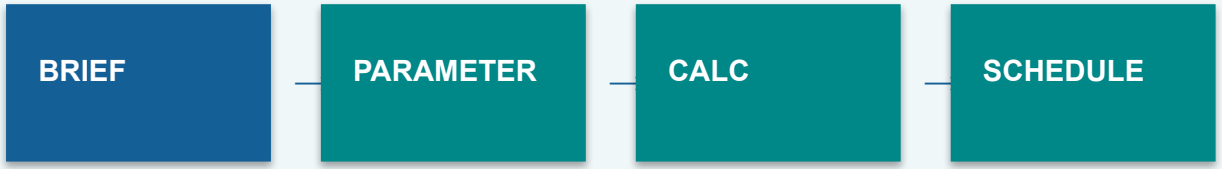
VERIFICATION

Check revision / clouds / and / issue in its relevant Revit view, browser or schedule.

Revision clouds and issue metadata — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 11 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Revision clouds and issue metadata — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Revision clouds and issue metadata.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Revision clouds and issue metadata — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 11 scheduled items, 10 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Revision clouds and issue metadata — Failure trace

MECHANISM / ARTIFACT

If Revision clouds and issue metadata is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Revision clouds and issue metadata — Decision rule

MECHANISM / ARTIFACT

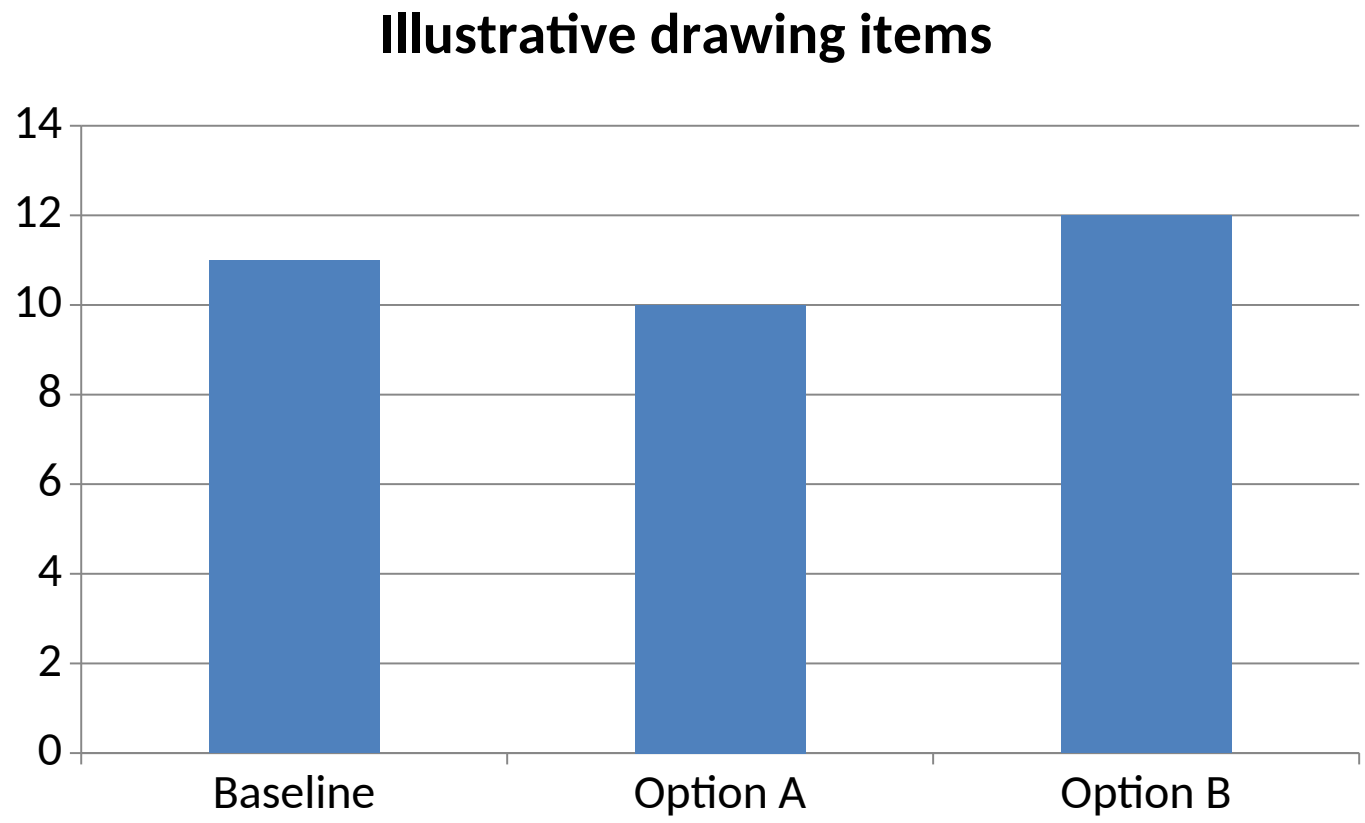
Accept Revision clouds and issue metadata only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Revision clouds and issue metadata — Quantitative review



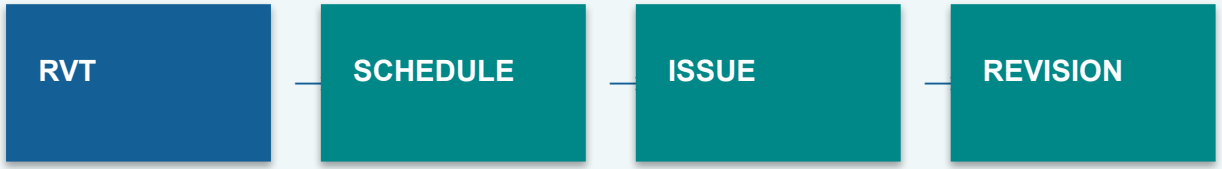
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Revision clouds and issue metadata — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Revision clouds and issue metadata.



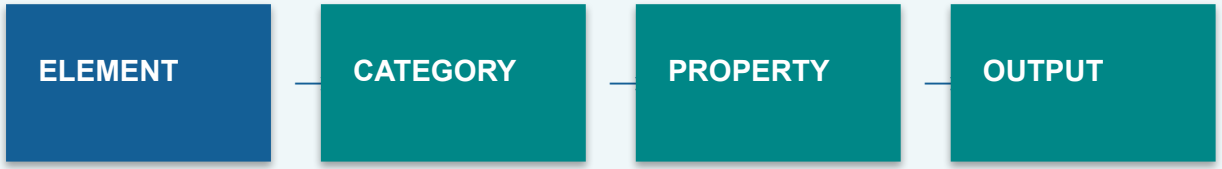
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 05.

Phasing for existing and new systems — Model anatomy

MECHANISM / ARTIFACT

Input: Phasing for existing and new systems.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



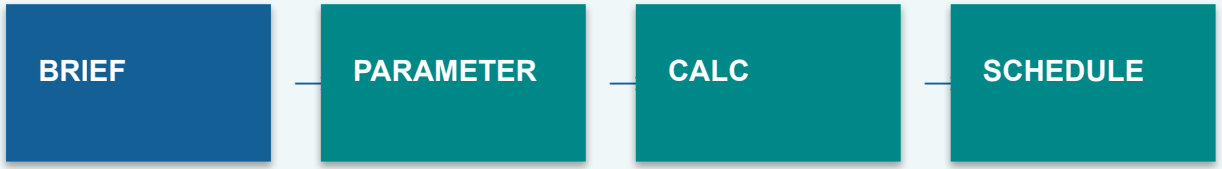
VERIFICATION

Check phasing / for / existing / and in its relevant Revit view, browser or schedule.

Phasing for existing and new systems — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 12 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Phasing for existing and new systems — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Phasing for existing and new systems.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Phasing for existing and new systems — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 12 scheduled items, 11 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Phasing for existing and new systems — Failure trace

MECHANISM / ARTIFACT

If Phasing for existing and new systems is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Phasing for existing and new systems — Decision rule

MECHANISM / ARTIFACT

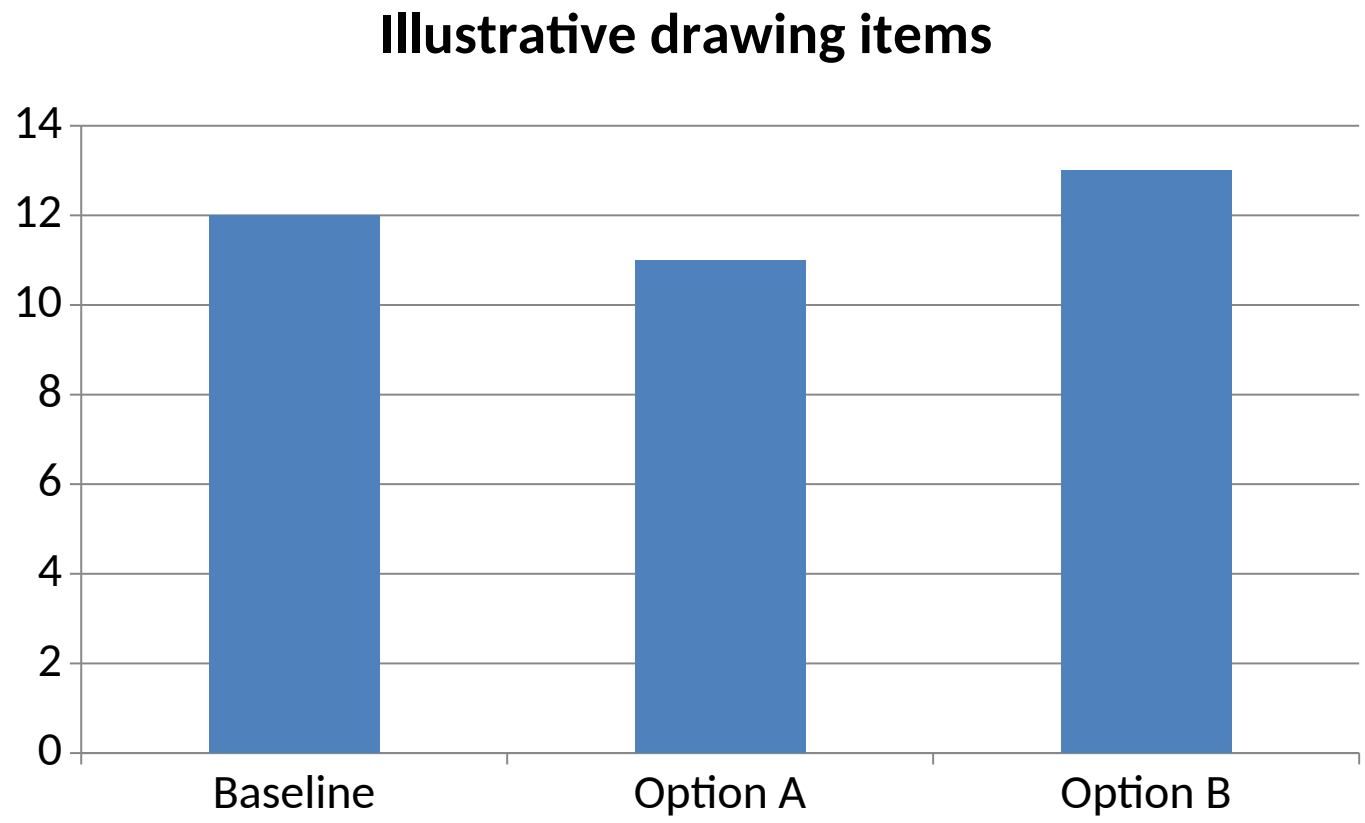
Accept Phasing for existing and new systems only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Phasing for existing and new systems — Quantitative review



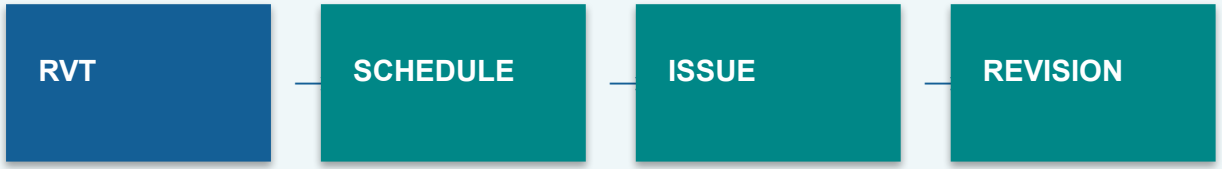
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Phasing for existing and new systems — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Phasing for existing and new systems.



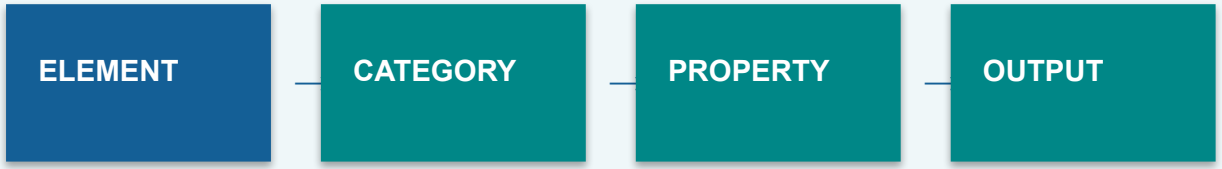
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 06.

Tags, keynotes and annotation families — Model anatomy

MECHANISM / ARTIFACT

Input: Tags, keynotes and annotation families.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



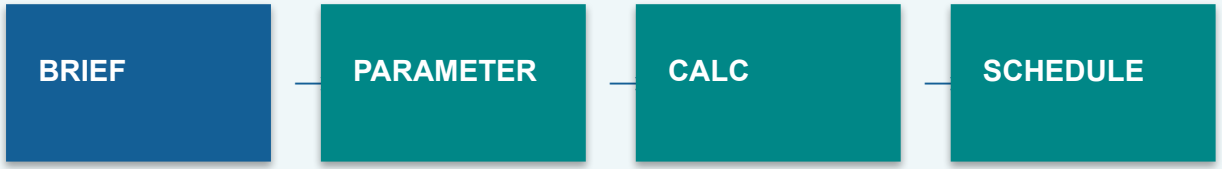
VERIFICATION

Check tags, / keynotes / and / annotation in its relevant Revit view, browser or schedule.

Tags, keynotes and annotation families — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 13 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Tags, keynotes and annotation families — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Tags, keynotes and annotation families.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Tags, keynotes and annotation families — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 13 scheduled items, 12 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Tags, keynotes and annotation families — Failure trace

MECHANISM / ARTIFACT

If Tags, keynotes and annotation families is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Tags, keynotes and annotation families — Decision rule

MECHANISM / ARTIFACT

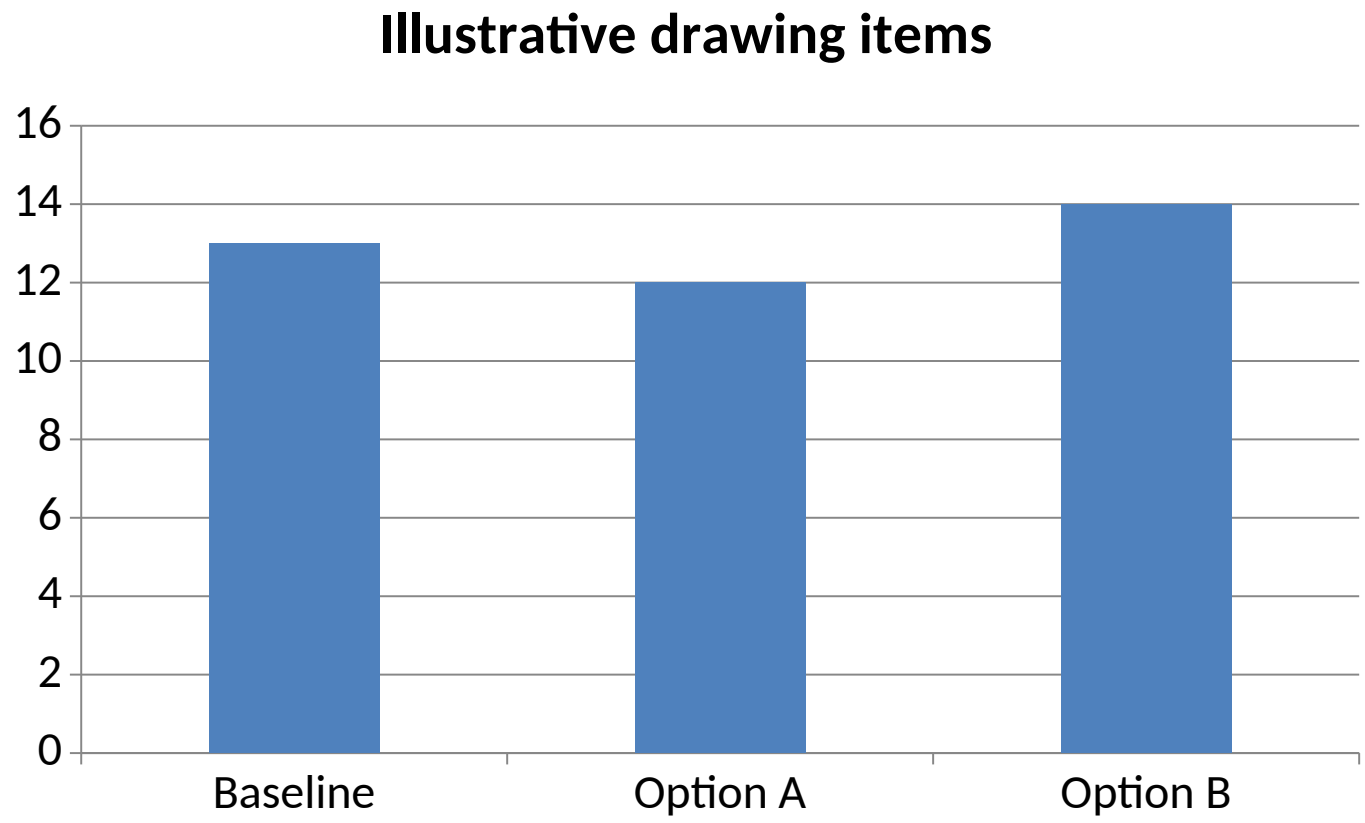
Accept Tags, keynotes and annotation families only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Tags, keynotes and annotation families — Quantitative review



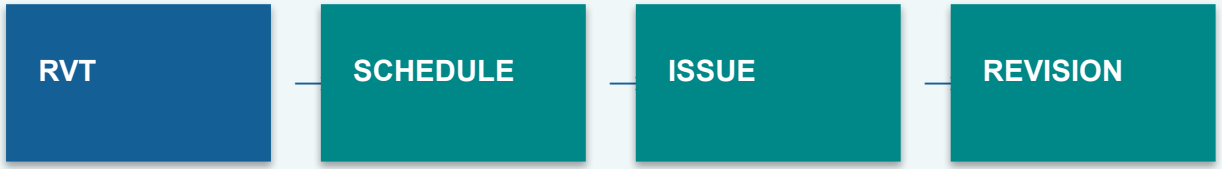
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Tags, keynotes and annotation families — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Tags, keynotes and annotation families.



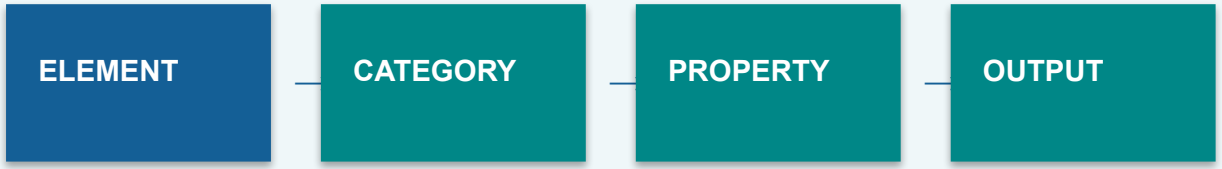
VERIFICATION

Criteria: K3 K4 A3 A4; Lab 06.

Drawing QA and coordination notes — Model anatomy

MECHANISM / ARTIFACT

Input: Drawing QA and coordination notes.
Control: view template, phase, discipline, filter and annotation.
Output: a named view, schedule or issued sheet with a traceable element ID.



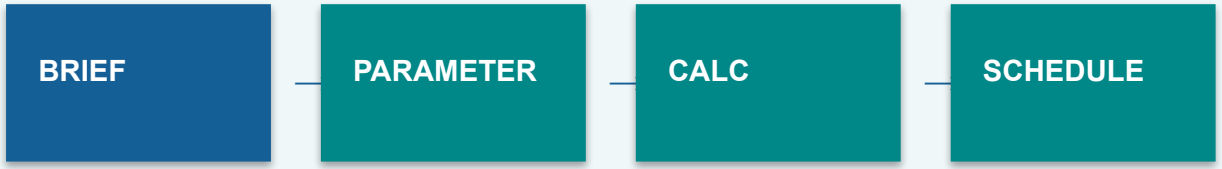
VERIFICATION

Check drawing / qa / and / coordination in its relevant Revit view, browser or schedule.

Drawing QA and coordination notes — Parameter chain

MECHANISM / ARTIFACT

Project brief → view template, phase, discipline, filter and annotation → model state → drawing or report.
Illustrative value: 14 drawing items.



VERIFICATION

Check units, discipline and type versus instance scope.

Drawing QA and coordination notes — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Drawing QA and coordination notes.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Drawing QA and coordination notes — Worked design example

MECHANISM / ARTIFACT

Illustrative issue check: 14 scheduled items, 13 tagged on sheet → 1 unresolved drawing discrepancy.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Drawing QA and coordination notes — Failure trace

MECHANISM / ARTIFACT

If Drawing QA and coordination notes is missing or inconsistent, isolate view template, phase filter, schedule filter and sheet revision.



VERIFICATION

Capture element ID and exact property before changing settings.

Drawing QA and coordination notes — Decision rule

MECHANISM / ARTIFACT

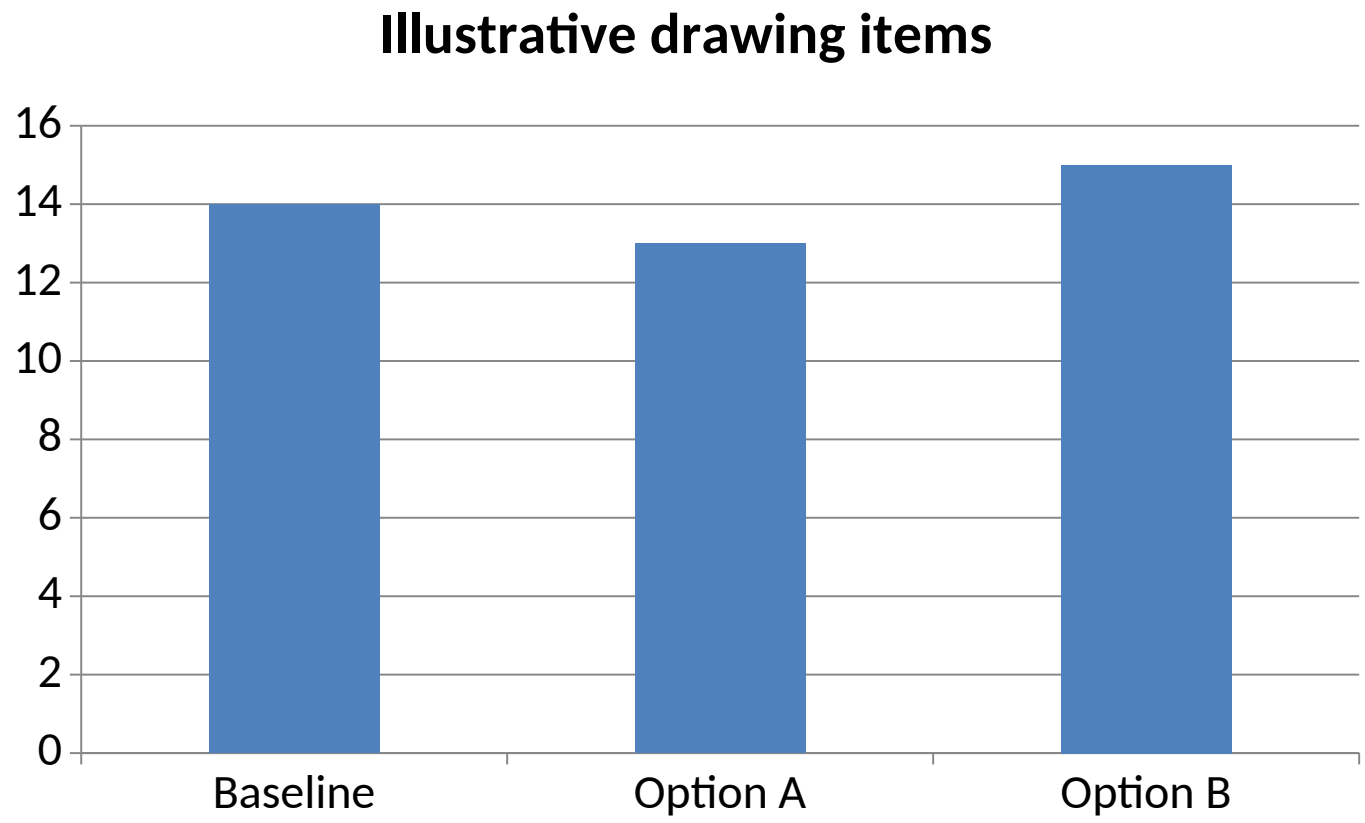
Accept Drawing QA and coordination notes only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Drawing QA and coordination notes — Quantitative review



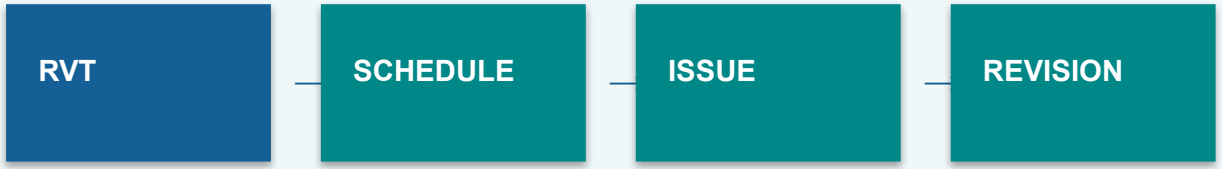
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Drawing QA and coordination notes — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Drawing QA and coordination notes.



VERIFICATION

Criteria: K3 K4 A3 A4; Lab 06.

Lab 04 — Create mechanical views and templates

MODEL OUTPUT

mechanical view set with consistent visibility

EVIDENCE

floor plan, section, view template; working RVT, schedule and PDF

Lab 04 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 05 — Schedule equipment ducts and pipes

MODEL OUTPUT

three auditable schedules with units and filters

EVIDENCE

equipment, duct and pipe schedule fields; working RVT, schedule and PDF

Lab 05 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 06 — Issue sheets tags and revisions

MODEL OUTPUT

issued drawing set with revision register

EVIDENCE

sheet, viewport, tags, revision cloud; working RVT, schedule and PDF

Lab 06 — acceptance evidence

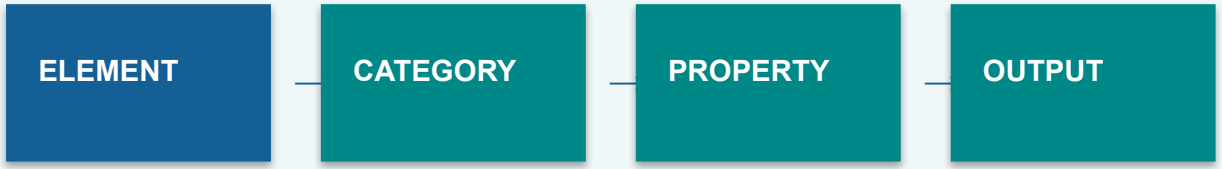
PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Family category and part type — Model anatomy

MECHANISM / ARTIFACT

Input: Family category and part type.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



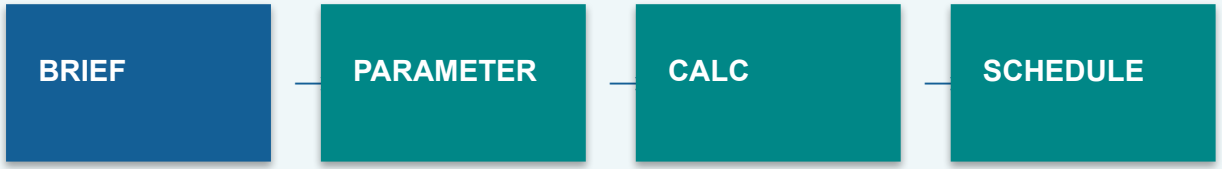
VERIFICATION

Check family / category / and / part in its relevant Revit view, browser or schedule.

Family category and part type — Parameter chain

MECHANISM / ARTIFACT

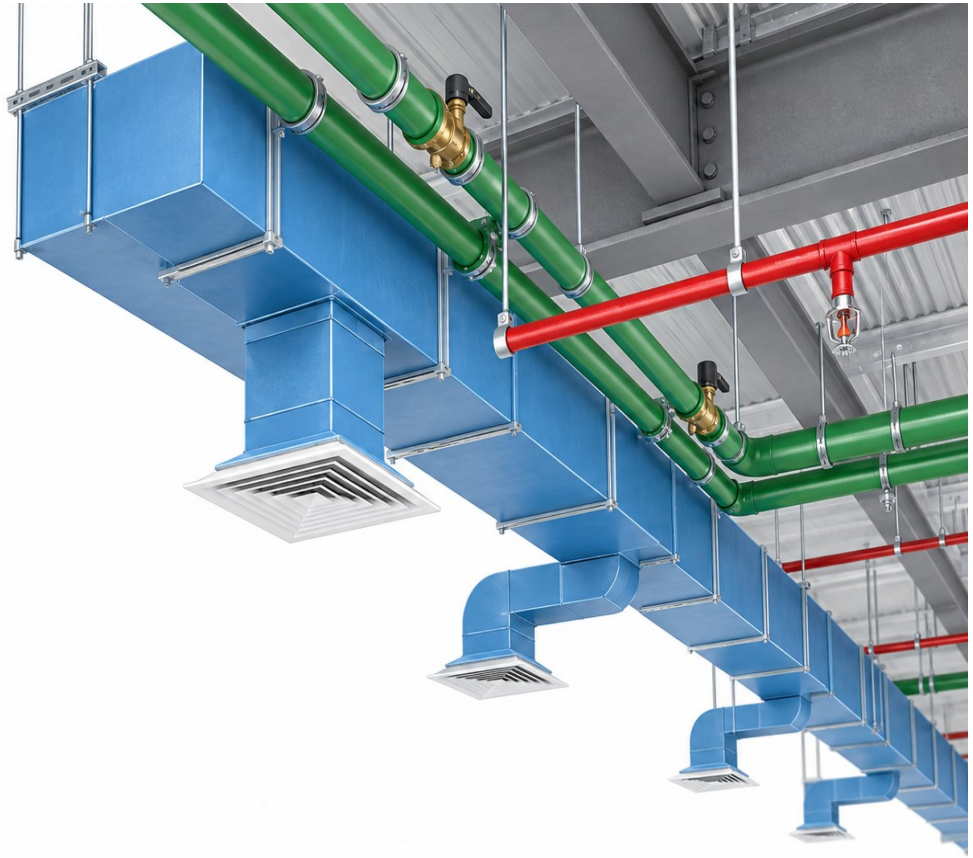
Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 110 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Family category and part type — Coordinated model state



COORDINATION CHECK

Architecture link provides levels and room bounds. Mechanical model owns Family category and part type.
A coordination view must expose both without duplicating source elements.
Check shared coordinates and linked-model visibility.

Family category and part type — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø110 mm: area = π×(0.11 m)²/4 = 0.010 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Family category and part type — Failure trace

MECHANISM / ARTIFACT

If Family category and part type is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Family category and part type — Decision rule

MECHANISM / ARTIFACT

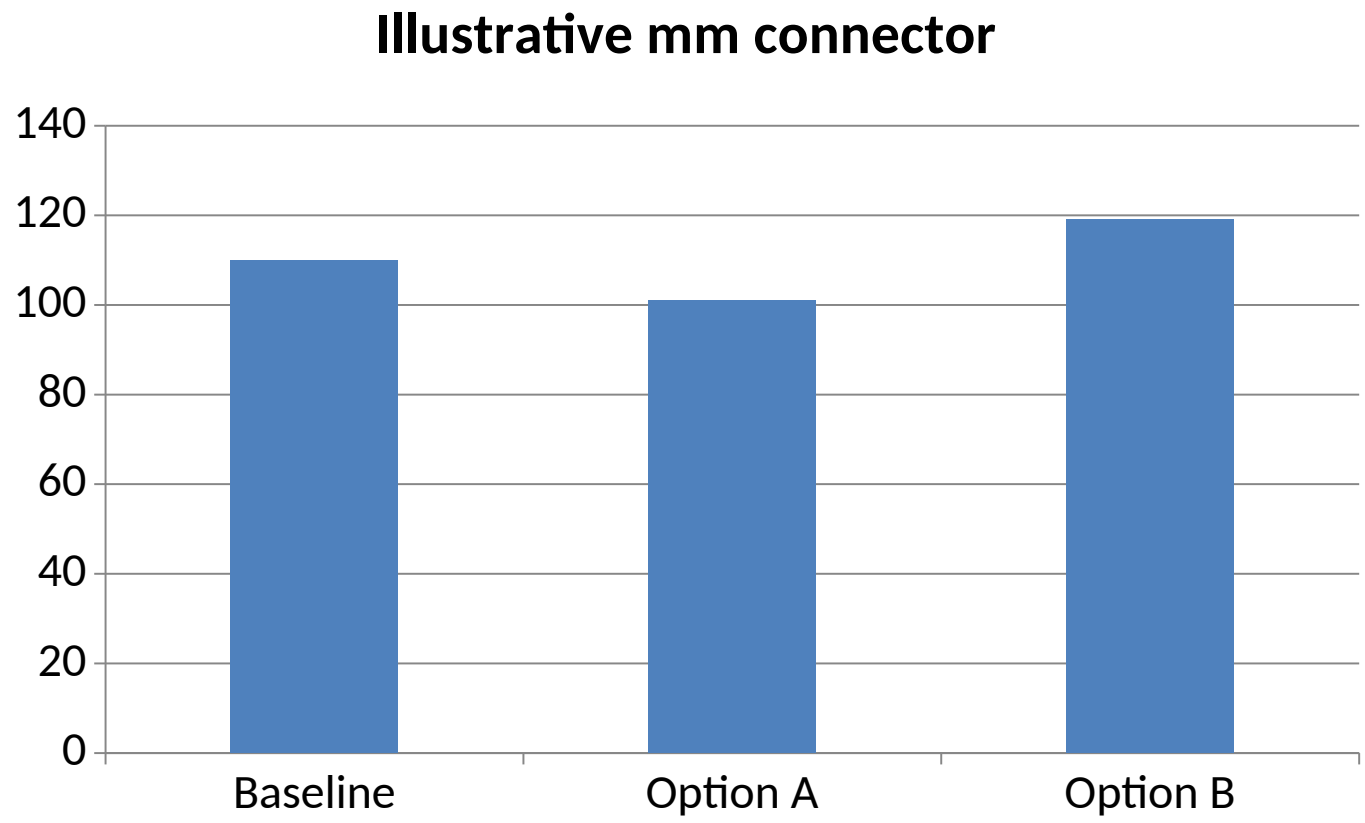
Accept Family category and part type only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Family category and part type — Quantitative review



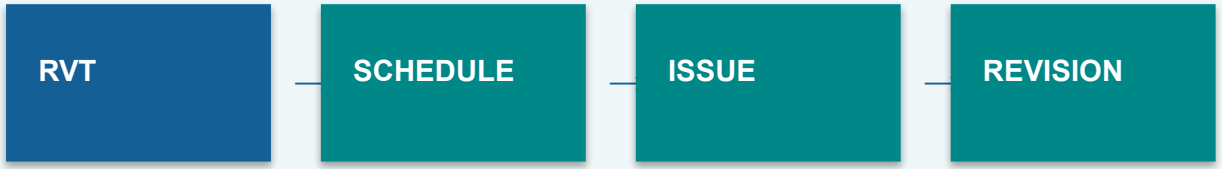
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Family category and part type — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Family category and part type.



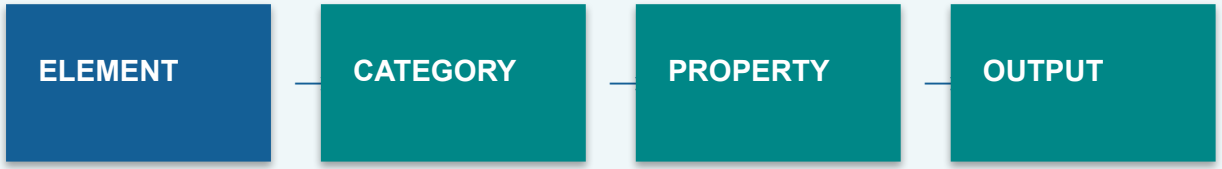
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 07.

Hosted versus non-hosted equipment — Model anatomy

MECHANISM / ARTIFACT

Input: Hosted versus non-hosted equipment.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



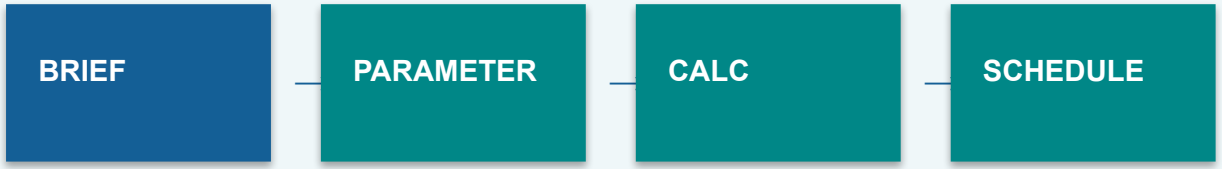
VERIFICATION

Check hosted / versus / non-hosted / equipment in its relevant Revit view, browser or schedule.

Hosted versus non-hosted equipment — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 120 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Hosted versus non-hosted equipment — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Hosted versus non-hosted equipment.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Hosted versus non-hosted equipment — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø120 mm: area = π×(0.12 m)²/4 = 0.011 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Hosted versus non-hosted equipment — Failure trace

MECHANISM / ARTIFACT

If Hosted versus non-hosted equipment is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Hosted versus non-hosted equipment — Decision rule

MECHANISM / ARTIFACT

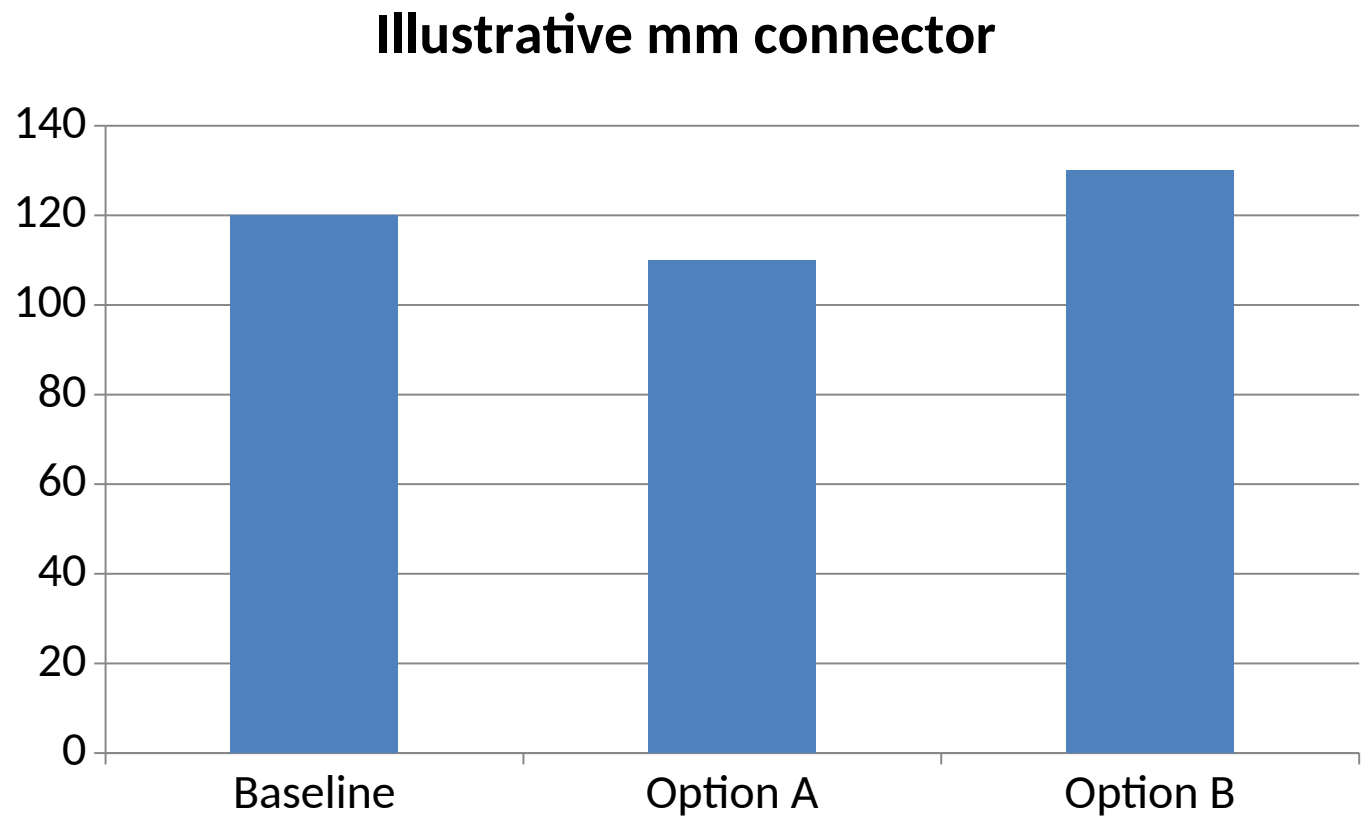
Accept Hosted versus non-hosted equipment only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Hosted versus non-hosted equipment — Quantitative review



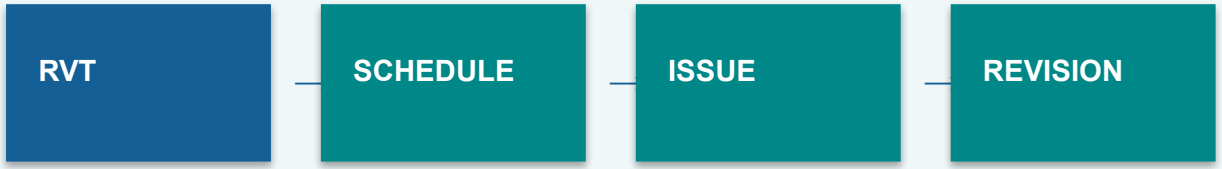
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Hosted versus non-hosted equipment — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Hosted versus non-hosted equipment.



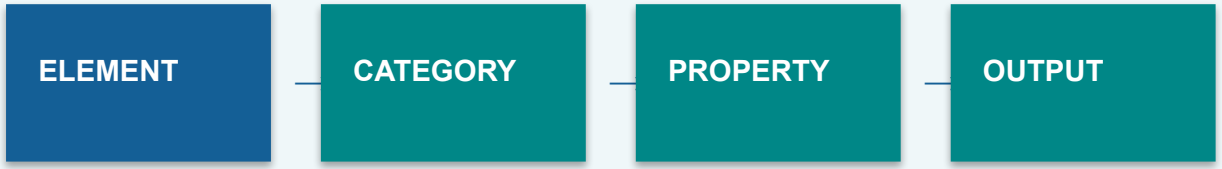
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 07.

Duct connectors and flow direction — Model anatomy

MECHANISM / ARTIFACT

Input: Duct connectors and flow direction.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



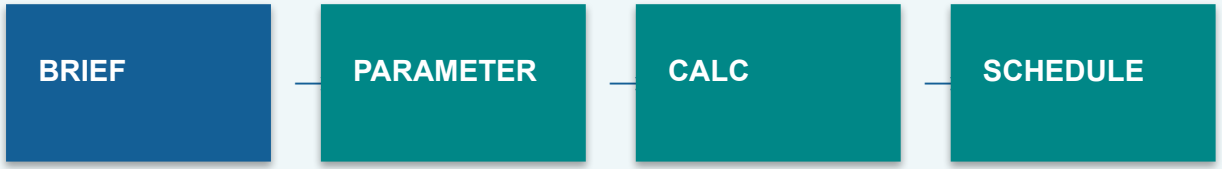
VERIFICATION

Check duct / connectors / and / flow in its relevant Revit view, browser or schedule.

Duct connectors and flow direction — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 130 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Duct connectors and flow direction — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Duct connectors and flow direction. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Duct connectors and flow direction — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø130 mm: area = π×(0.13 m)²/4 = 0.013 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Duct connectors and flow direction — Failure trace

MECHANISM / ARTIFACT

If Duct connectors and flow direction is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Duct connectors and flow direction — Decision rule

MECHANISM / ARTIFACT

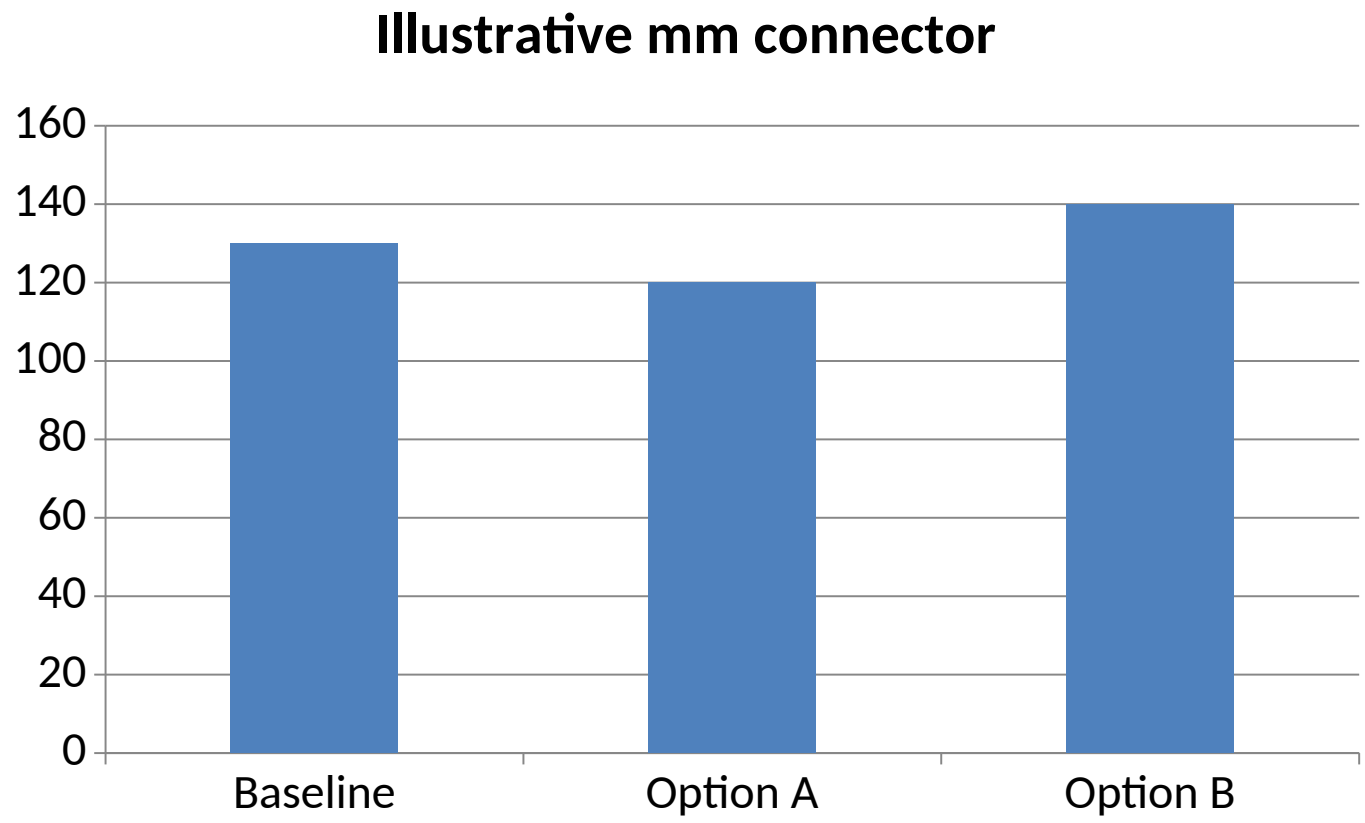
Accept Duct connectors and flow direction only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Duct connectors and flow direction — Quantitative review



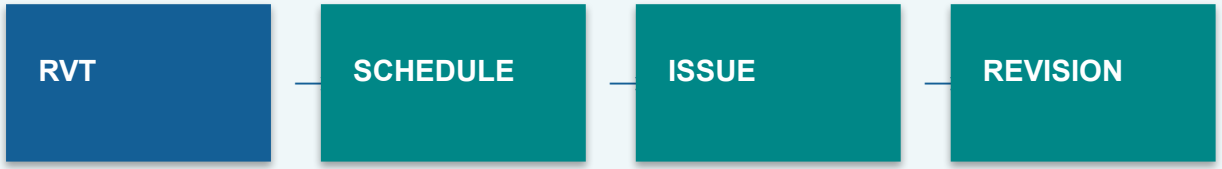
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Duct connectors and flow direction — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Duct connectors and flow direction.



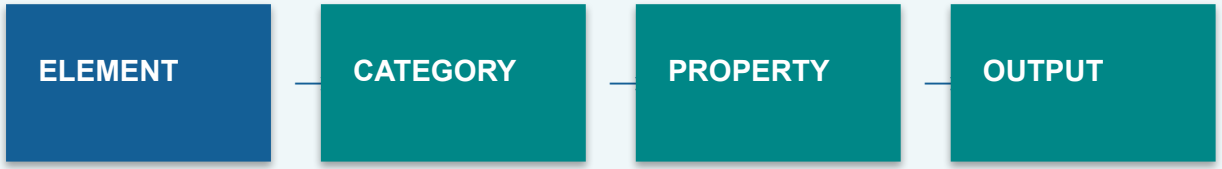
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 07.

Pipe connectors and system classification — Model anatomy

MECHANISM / ARTIFACT

Input: Pipe connectors and system classification.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



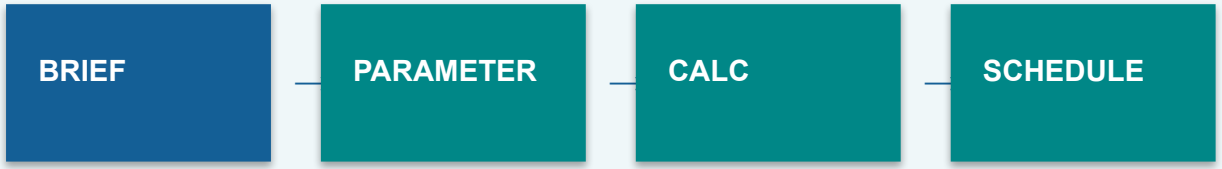
VERIFICATION

Check pipe / connectors / and / system in its relevant Revit view, browser or schedule.

Pipe connectors and system classification — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 140 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Pipe connectors and system classification — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Pipe connectors and system classification.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Pipe connectors and system classification — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø140 mm: area = π×(0.14 m)²/4 = 0.015 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Pipe connectors and system classification — Failure trace

MECHANISM / ARTIFACT

If Pipe connectors and system classification is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Pipe connectors and system classification — Decision rule

MECHANISM / ARTIFACT

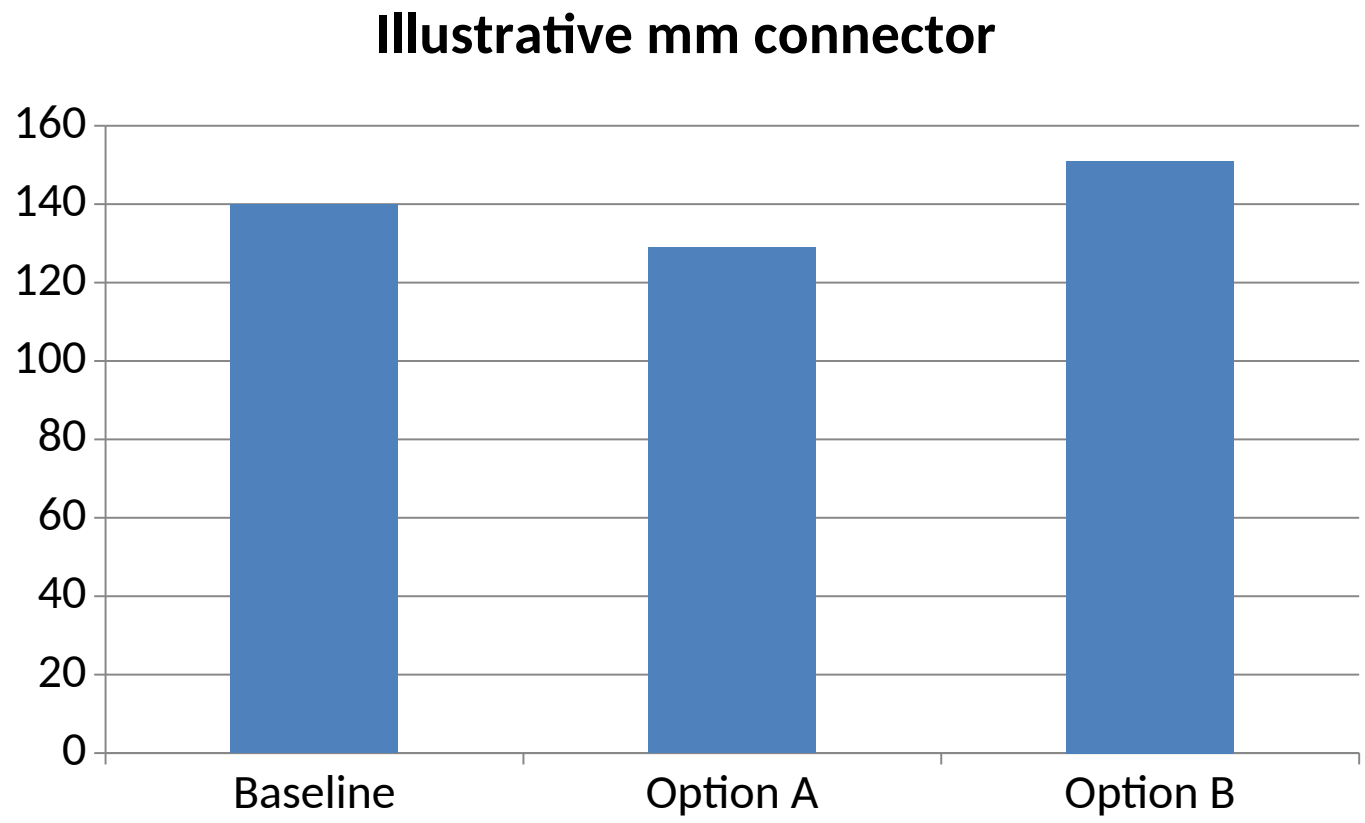
Accept Pipe connectors and system classification only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Pipe connectors and system classification — Quantitative review



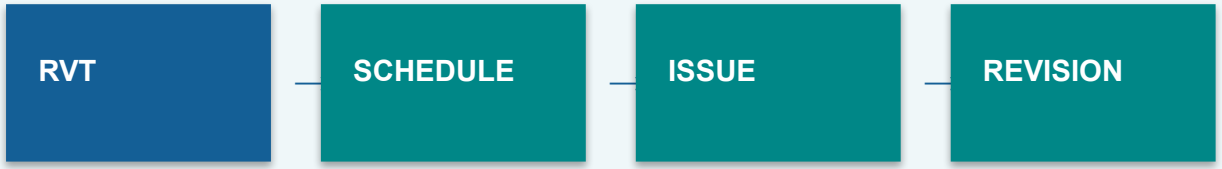
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Pipe connectors and system classification — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Pipe connectors and system classification.



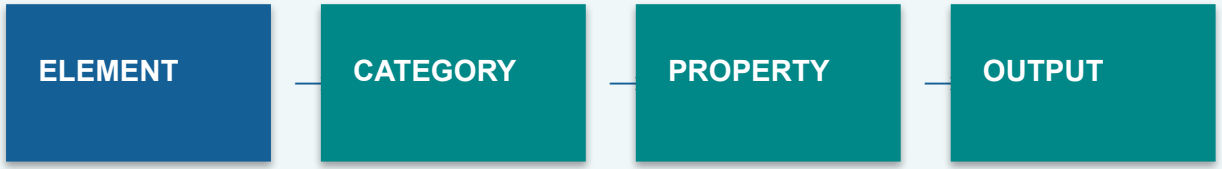
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 08.

Connector dimensions and associations — Model anatomy

MECHANISM / ARTIFACT

Input: Connector dimensions and associations.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



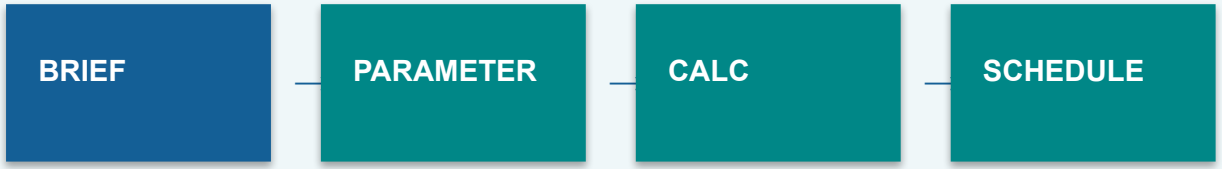
VERIFICATION

Check connector / dimensions / and / associations in its relevant Revit view, browser or schedule.

Connector dimensions and associations — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 150 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Connector dimensions and associations — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Connector dimensions and associations.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Connector dimensions and associations — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø150 mm: area = π×(0.15 m)²/4 = 0.018 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Connector dimensions and associations — Failure trace

MECHANISM / ARTIFACT

If Connector dimensions and associations is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Connector dimensions and associations — Decision rule

MECHANISM / ARTIFACT

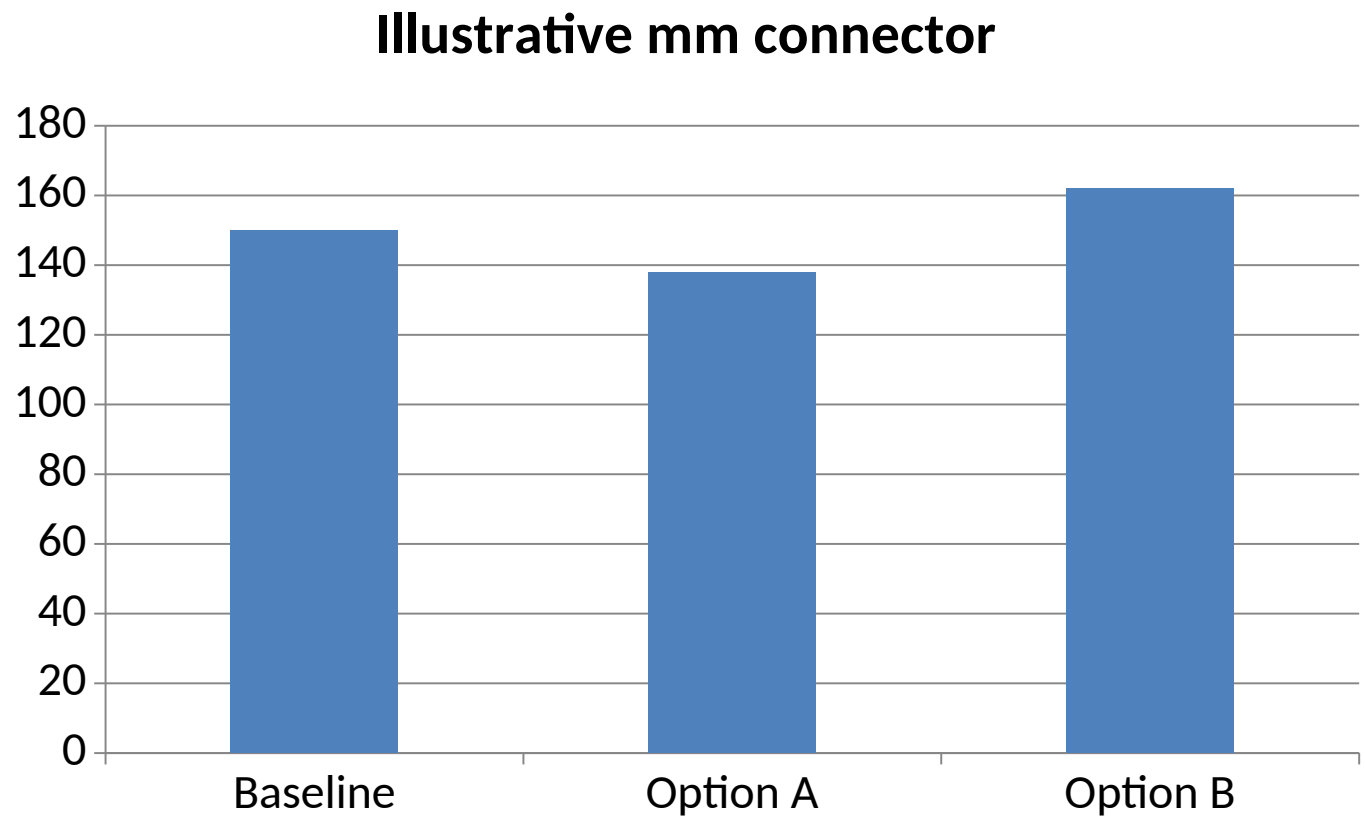
Accept Connector dimensions and associations only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Connector dimensions and associations — Quantitative review



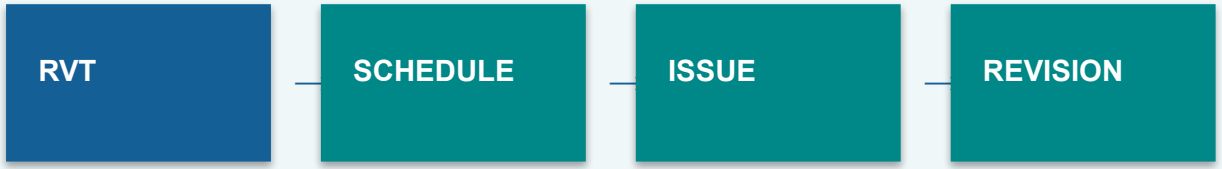
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Connector dimensions and associations — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Connector dimensions and associations.



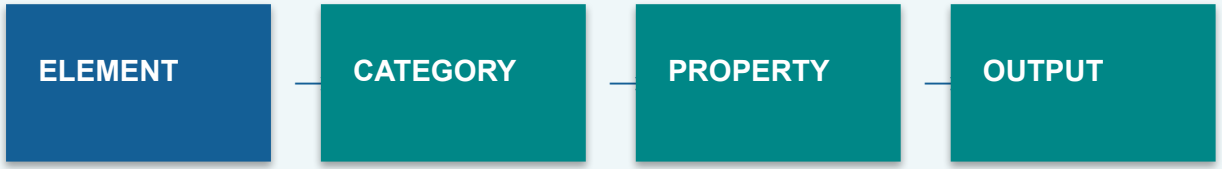
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 08.

Family type and instance parameters — Model anatomy

MECHANISM / ARTIFACT

Input: Family type and instance parameters.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



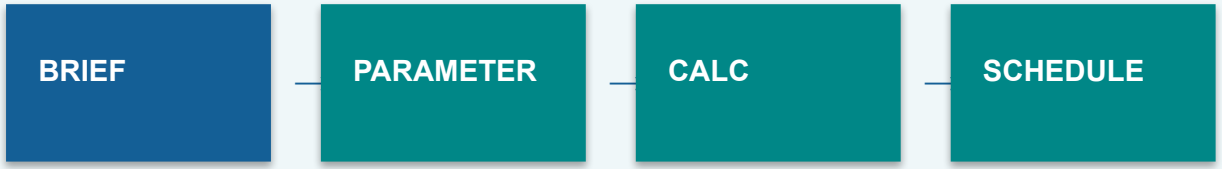
VERIFICATION

Check family / type / and / instance in its relevant Revit view, browser or schedule.

Family type and instance parameters — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 160 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Family type and instance parameters — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Family type and instance parameters.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Family type and instance parameters — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø160 mm: area = π×(0.16 m)²/4 = 0.020 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Family type and instance parameters — Failure trace

MECHANISM / ARTIFACT

If Family type and instance parameters is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Family type and instance parameters — Decision rule

MECHANISM / ARTIFACT

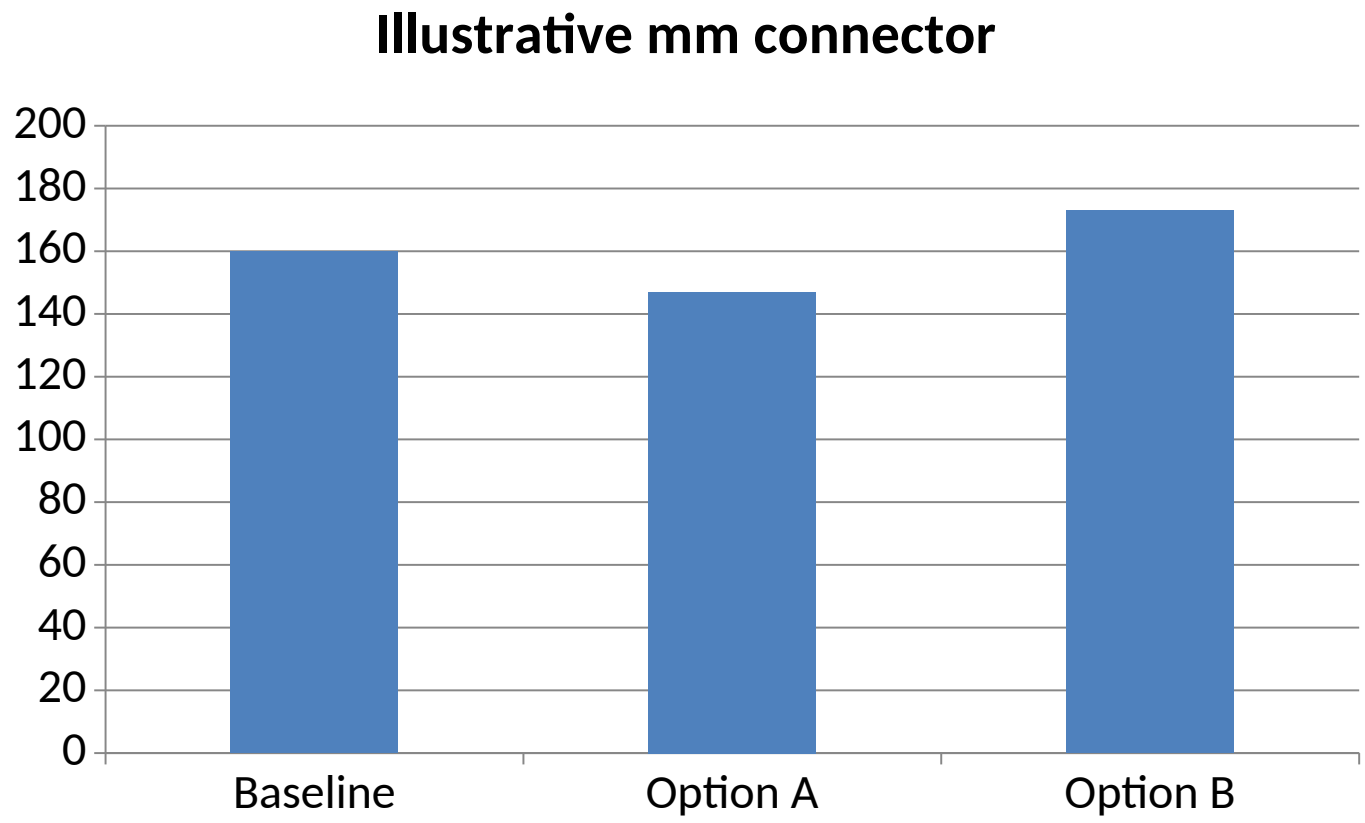
Accept Family type and instance parameters only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Family type and instance parameters — Quantitative review



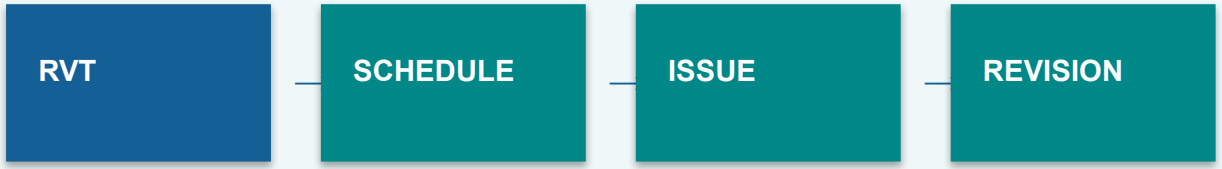
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Family type and instance parameters — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Family type and instance parameters.



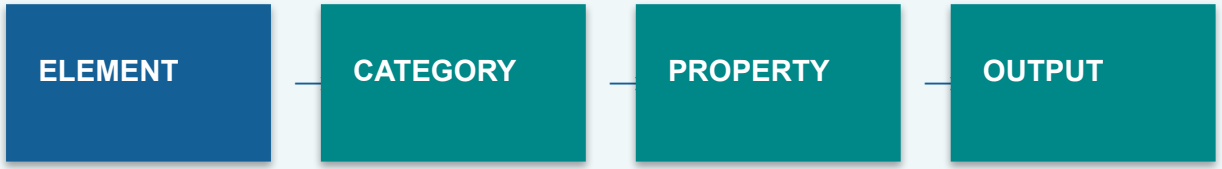
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 08.

Shared parameters and schedule identity — Model anatomy

MECHANISM / ARTIFACT

Input: Shared parameters and schedule identity.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



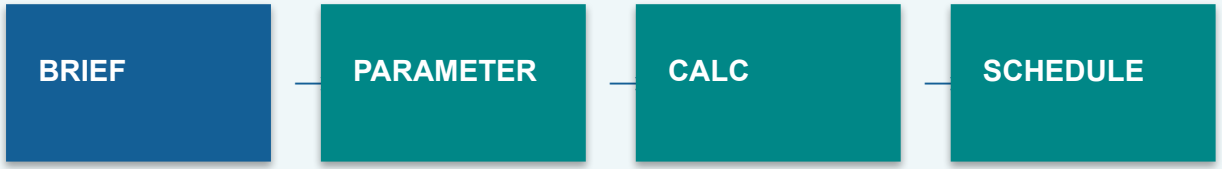
VERIFICATION

Check shared / parameters / and / schedule in its relevant Revit view, browser or schedule.

Shared parameters and schedule identity — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 170 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Shared parameters and schedule identity — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Shared parameters and schedule identity.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Shared parameters and schedule identity — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø170 mm: area = π×(0.17 m)²/4 = 0.023 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Shared parameters and schedule identity — Failure trace

MECHANISM / ARTIFACT

If Shared parameters and schedule identity is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Shared parameters and schedule identity — Decision rule

MECHANISM / ARTIFACT

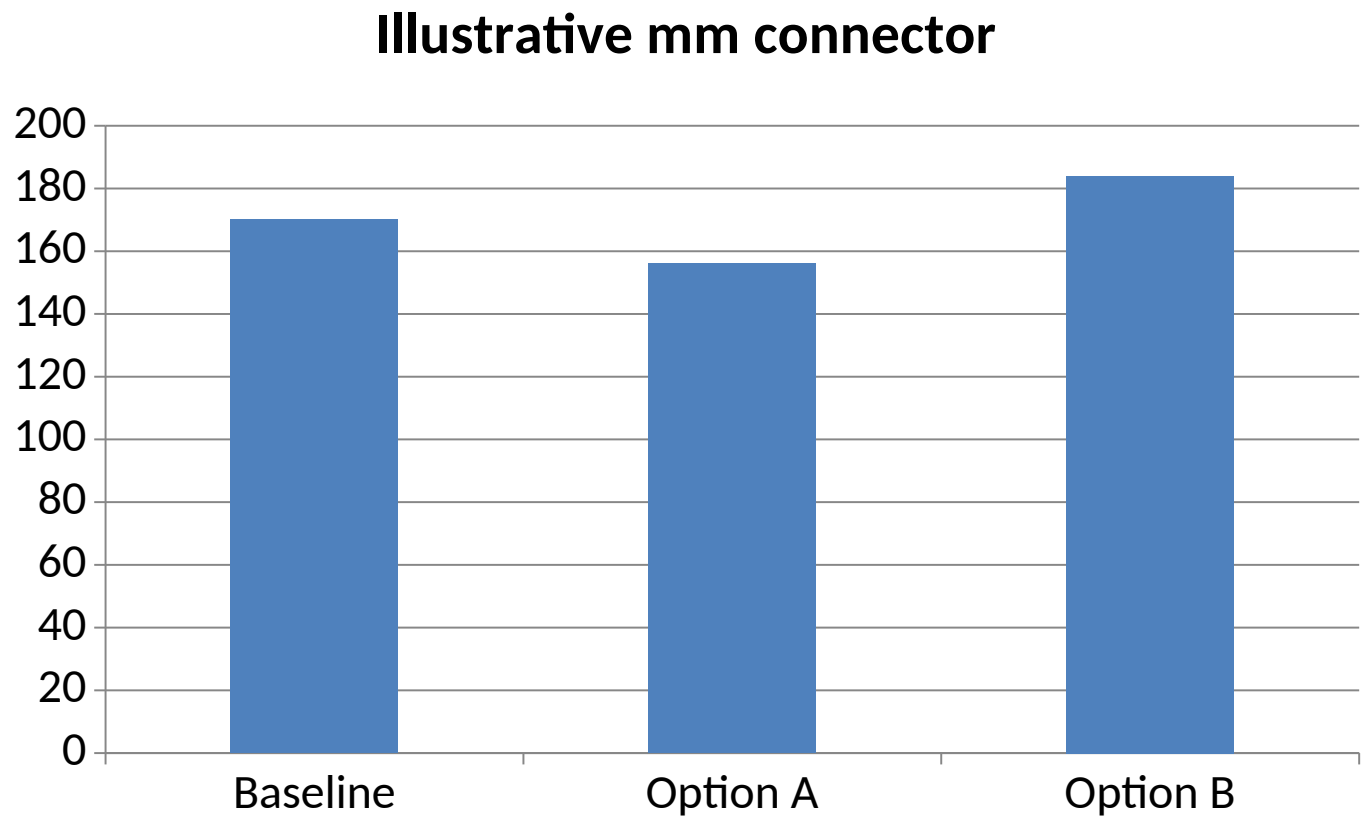
Accept Shared parameters and schedule identity only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Shared parameters and schedule identity — Quantitative review



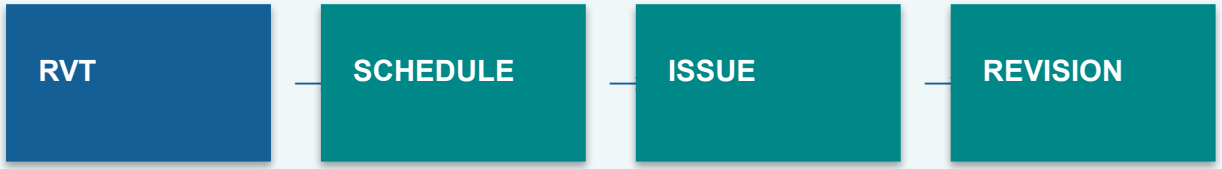
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Shared parameters and schedule identity — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Shared parameters and schedule identity.



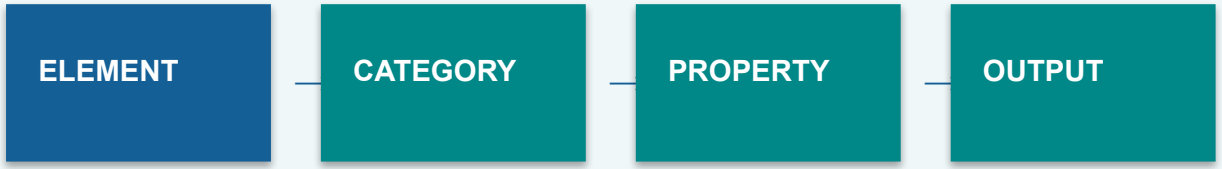
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 09.

Visibility controls and detail levels — Model anatomy

MECHANISM / ARTIFACT

Input: Visibility controls and detail levels.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



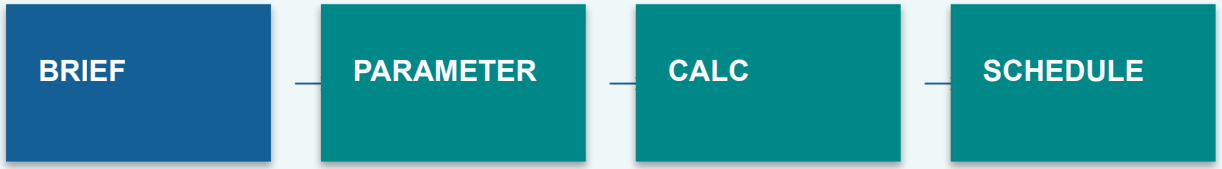
VERIFICATION

Check visibility / controls / and / detail in its relevant Revit view, browser or schedule.

Visibility controls and detail levels — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 180 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Visibility controls and detail levels — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Visibility controls and detail levels. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Visibility controls and detail levels — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø180 mm: area = π×(0.18 m)²/4 = 0.025 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Visibility controls and detail levels — Failure trace

MECHANISM / ARTIFACT

If Visibility controls and detail levels is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Visibility controls and detail levels — Decision rule

MECHANISM / ARTIFACT

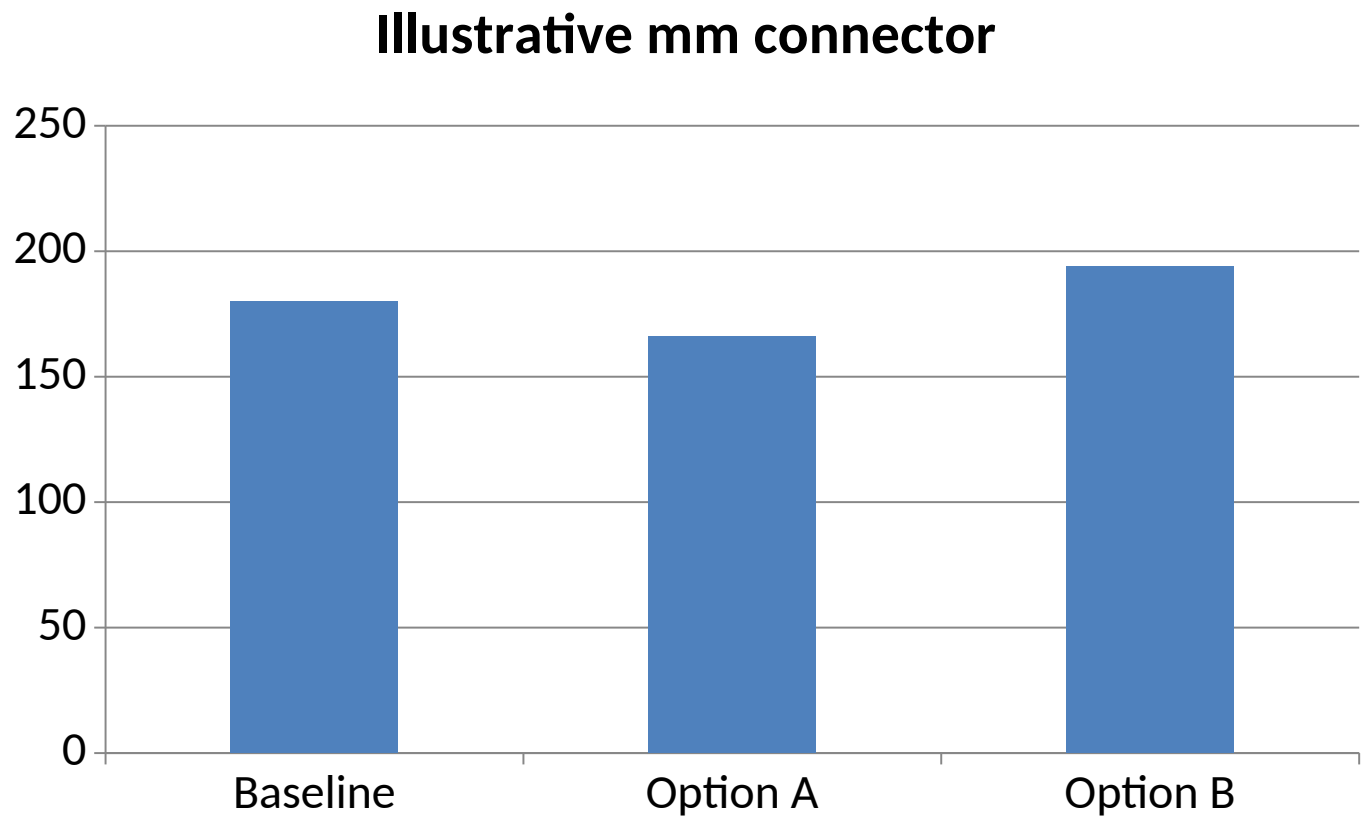
Accept Visibility controls and detail levels only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Visibility controls and detail levels — Quantitative review



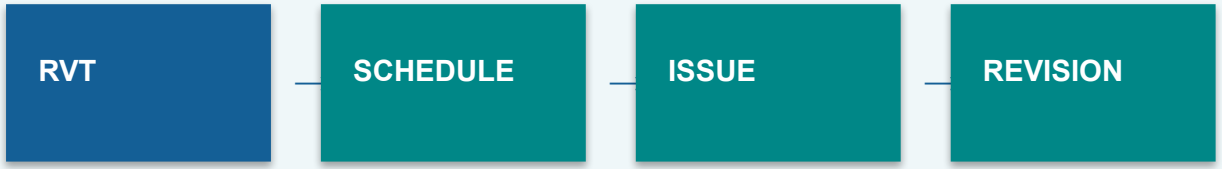
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Visibility controls and detail levels — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Visibility controls and detail levels.



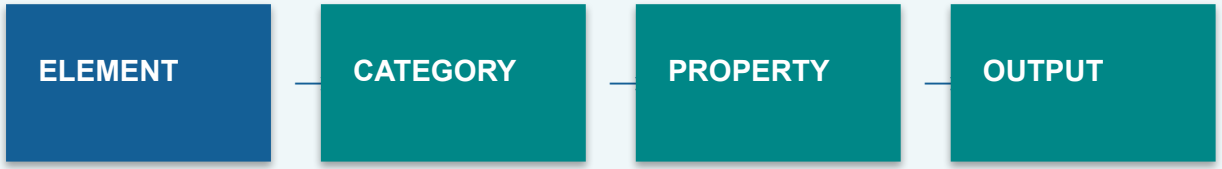
VERIFICATION

Criteria: K5 K6 A5 A6; Lab 09.

Family formulas and flex validation — Model anatomy

MECHANISM / ARTIFACT

Input: Family formulas and flex validation.
Control: family category, host, type parameter and connector.
Output: a named loadable family type with a traceable element ID.



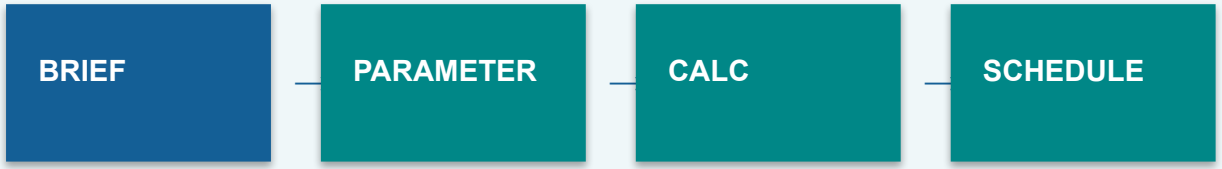
VERIFICATION

Check family / formulas / and / flex in its relevant Revit view, browser or schedule.

Family formulas and flex validation — Parameter chain

MECHANISM / ARTIFACT

Project brief → family category, host, type parameter and connector → model state → drawing or report.
Illustrative value: 190 mm connector.



VERIFICATION

Check units, discipline and type versus instance scope.

Family formulas and flex validation — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Family formulas and flex validation. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Family formulas and flex validation — Worked design example

MECHANISM / ARTIFACT

Illustrative round connector Ø190 mm: area = π×(0.19 m)²/4 = 0.028 m².



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Family formulas and flex validation — Failure trace

MECHANISM / ARTIFACT

If Family formulas and flex validation is missing or inconsistent, isolate connector classification, direction, host and family type association.



VERIFICATION

Capture element ID and exact property before changing settings.

Family formulas and flex validation — Decision rule

MECHANISM / ARTIFACT

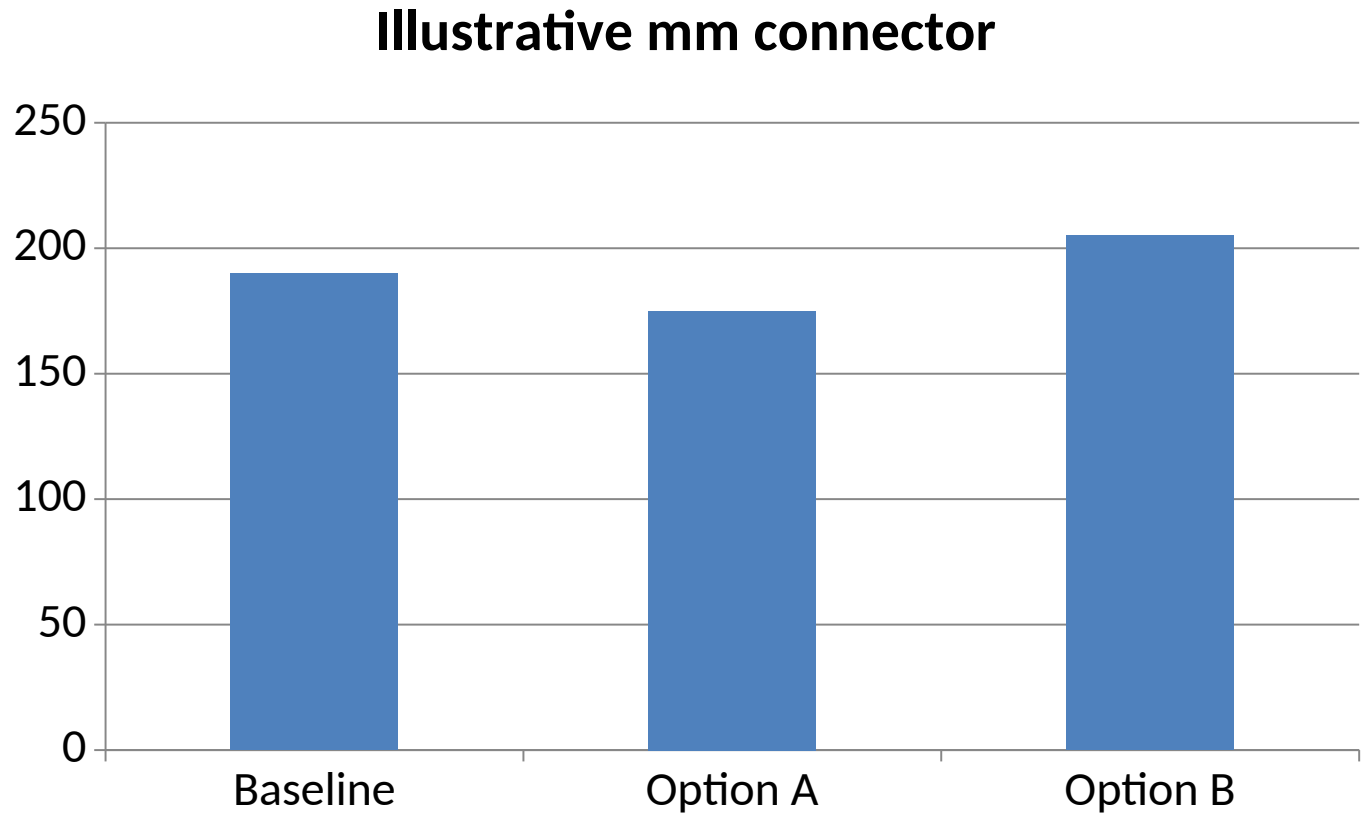
Accept Family formulas and flex validation only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Family formulas and flex validation — Quantitative review



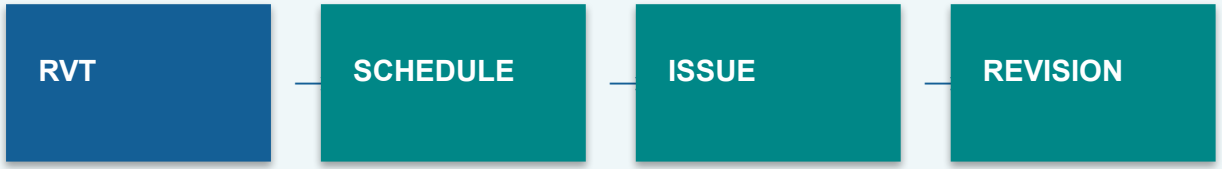
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Family formulas and flex validation — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Family formulas and flex validation.



VERIFICATION

Criteria: K5 K6 A5 A6; Lab 09.

Lab 07 — Build an equipment family

MODEL OUTPUT

loadable mechanical equipment family

EVIDENCE

family category, types and host; working RVT, schedule and PDF

Lab 07 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 08 — Configure HVAC and pipe connectors

MODEL OUTPUT

family with valid HVAC and piping connectors

EVIDENCE

connector classification, flow direction, dimensions; working RVT, schedule and PDF

Lab 08 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 09 — Flex test parameters and annotation

MODEL OUTPUT

flex-tested family and tag

EVIDENCE

shared parameter, formula, visibility; working RVT, schedule and PDF

Lab 09 — acceptance evidence

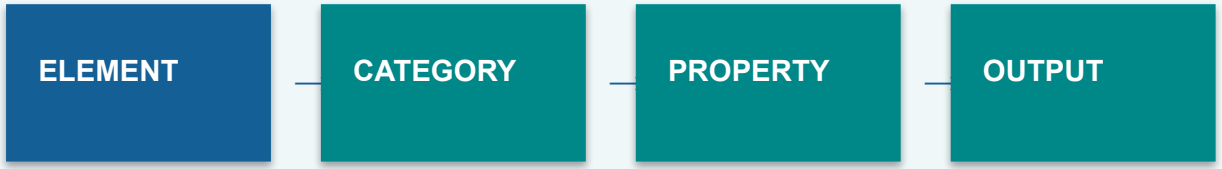
PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Energy settings and analytical model — Model anatomy

MECHANISM / ARTIFACT

Input: Energy settings and analytical model.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



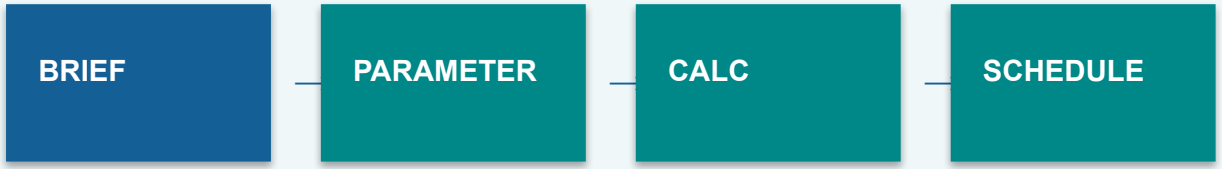
VERIFICATION

Check energy / settings / and / analytical in its relevant Revit view, browser or schedule.

Energy settings and analytical model — Parameter chain

MECHANISM / ARTIFACT

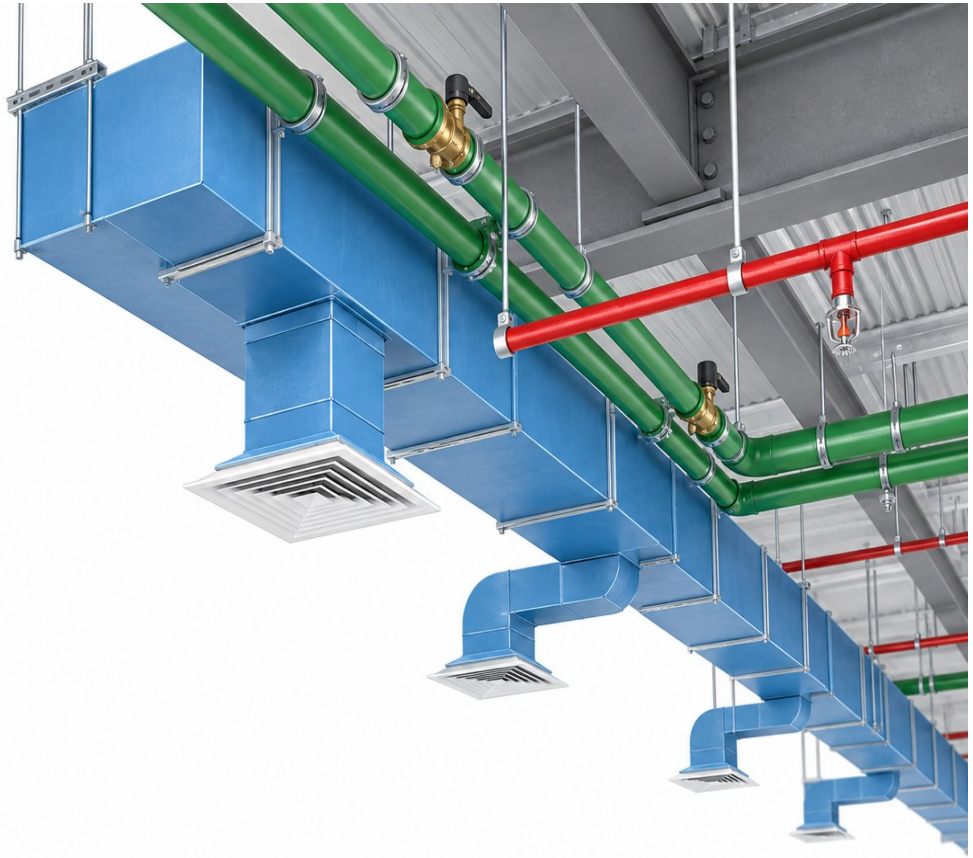
Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 13 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Energy settings and analytical model — Coordinated model state



COORDINATION CHECK

Architecture link provides levels and room bounds. Mechanical model owns Energy settings and analytical model.
A coordination view must expose both without duplicating source elements.
Check shared coordinates and linked-model visibility.

Energy settings and analytical model — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 13 kW at ΔT 10 K: airflow ≈ 1.08 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Energy settings and analytical model — Failure trace

MECHANISM / ARTIFACT

If Energy settings and analytical model is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Energy settings and analytical model — Decision rule

MECHANISM / ARTIFACT

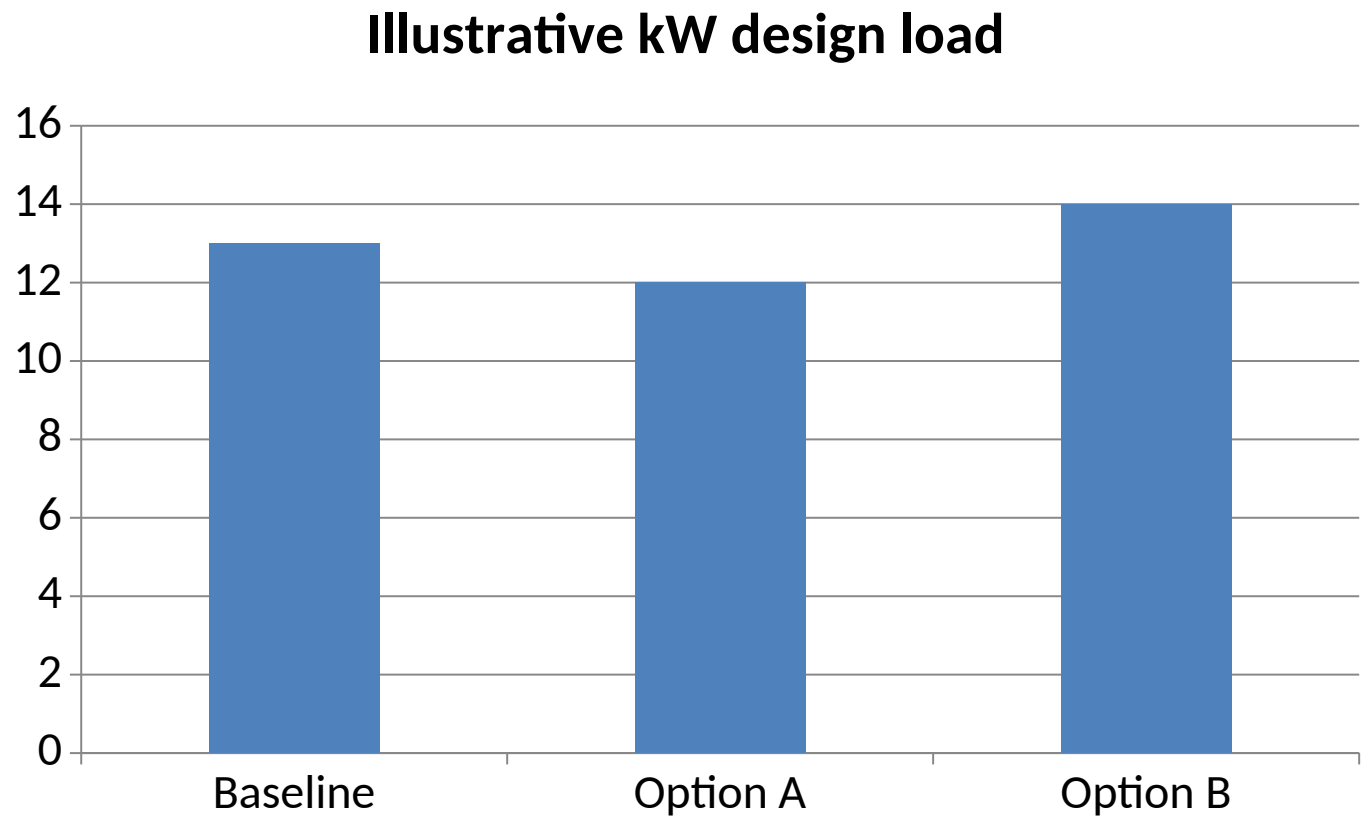
Accept Energy settings and analytical model only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Energy settings and analytical model — Quantitative review



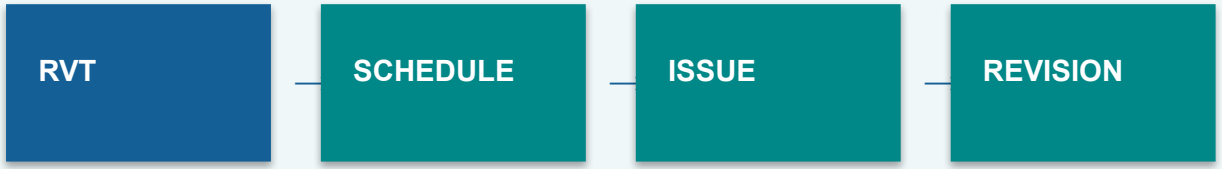
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Energy settings and analytical model — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Energy settings and analytical model.



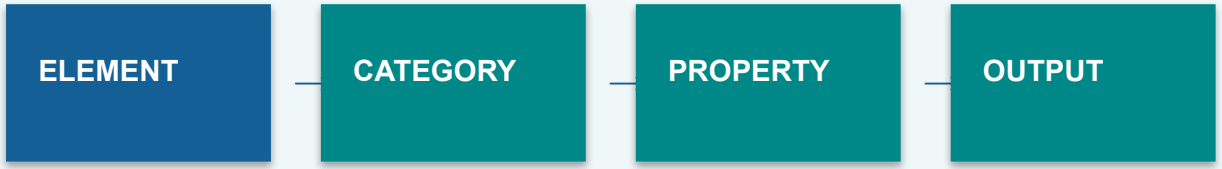
VERIFICATION

Criteria: K7 A7; Lab 10.

Rooms, spaces and analytical boundaries — Model anatomy

MECHANISM / ARTIFACT

Input: Rooms, spaces and analytical boundaries.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



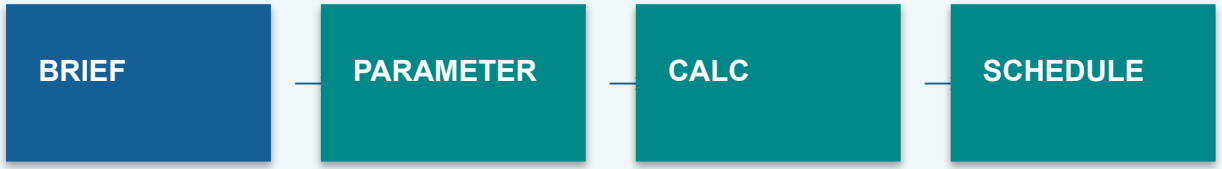
VERIFICATION

Check rooms, / spaces / and / analytical in its relevant Revit view, browser or schedule.

Rooms, spaces and analytical boundaries — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 14 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Rooms, spaces and analytical boundaries — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Rooms, spaces and analytical boundaries.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Rooms, spaces and analytical boundaries — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 14 kW at ΔT 10 K: airflow ≈ 1.16 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Rooms, spaces and analytical boundaries — Failure trace

MECHANISM / ARTIFACT

If Rooms, spaces and analytical boundaries is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Rooms, spaces and analytical boundaries — Decision rule

MECHANISM / ARTIFACT

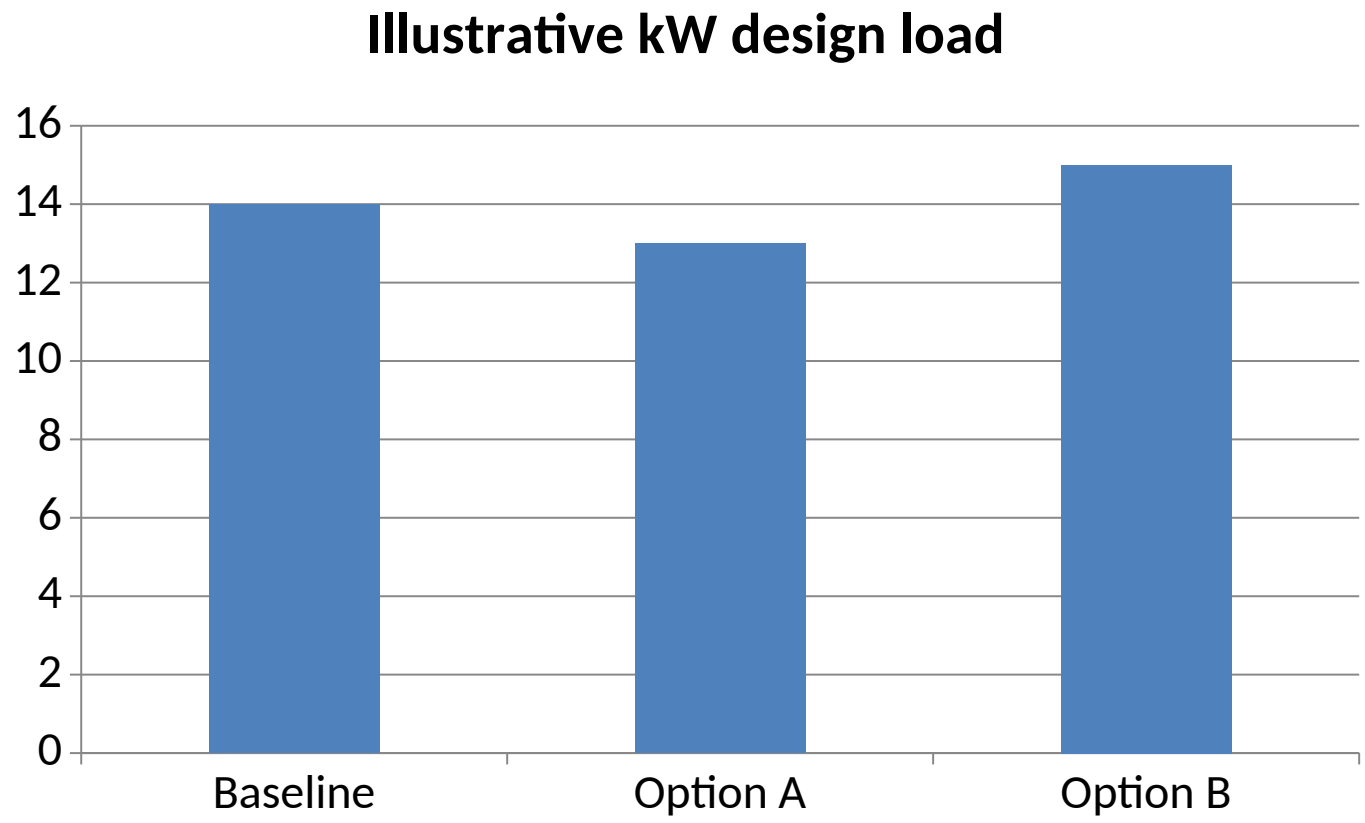
Accept Rooms, spaces and analytical boundaries only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Rooms, spaces and analytical boundaries — Quantitative review



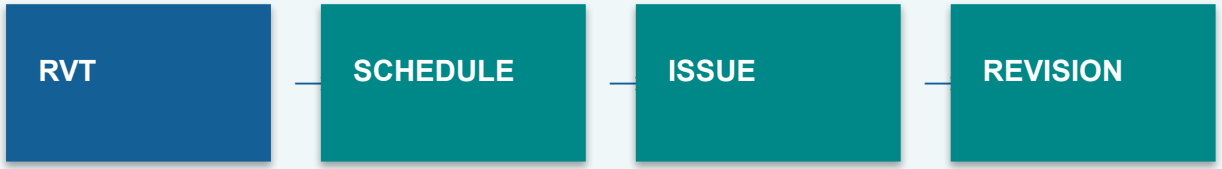
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Rooms, spaces and analytical boundaries — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Rooms, spaces and analytical boundaries.



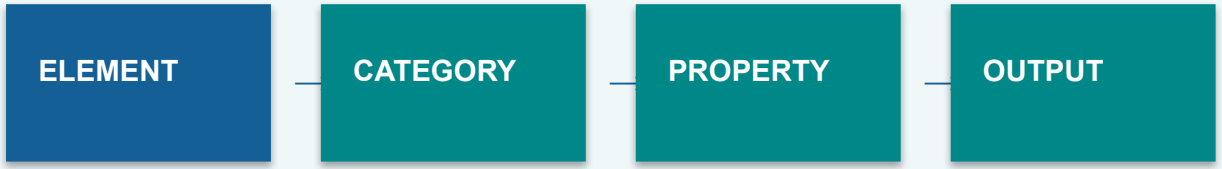
VERIFICATION

Criteria: K7 A7; Lab 10.

System zones and zone equipment — Model anatomy

MECHANISM / ARTIFACT

Input: System zones and zone equipment.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



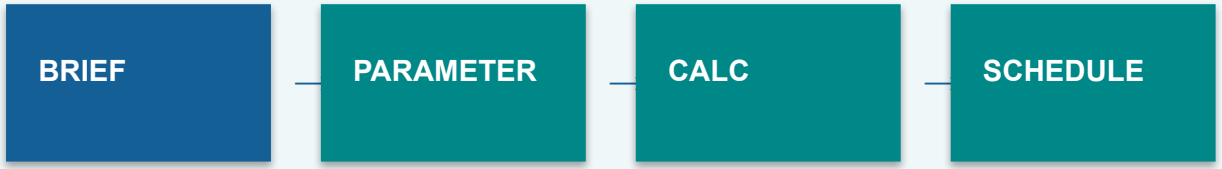
VERIFICATION

Check system / zones / and / zone in its relevant Revit view, browser or schedule.

System zones and zone equipment — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 15 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

System zones and zone equipment — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns System zones and zone equipment.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

System zones and zone equipment — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 15 kW at ΔT 10 K: airflow ≈ 1.24 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

System zones and zone equipment — Failure trace

MECHANISM / ARTIFACT

If System zones and zone equipment is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

System zones and zone equipment — Decision rule

MECHANISM / ARTIFACT

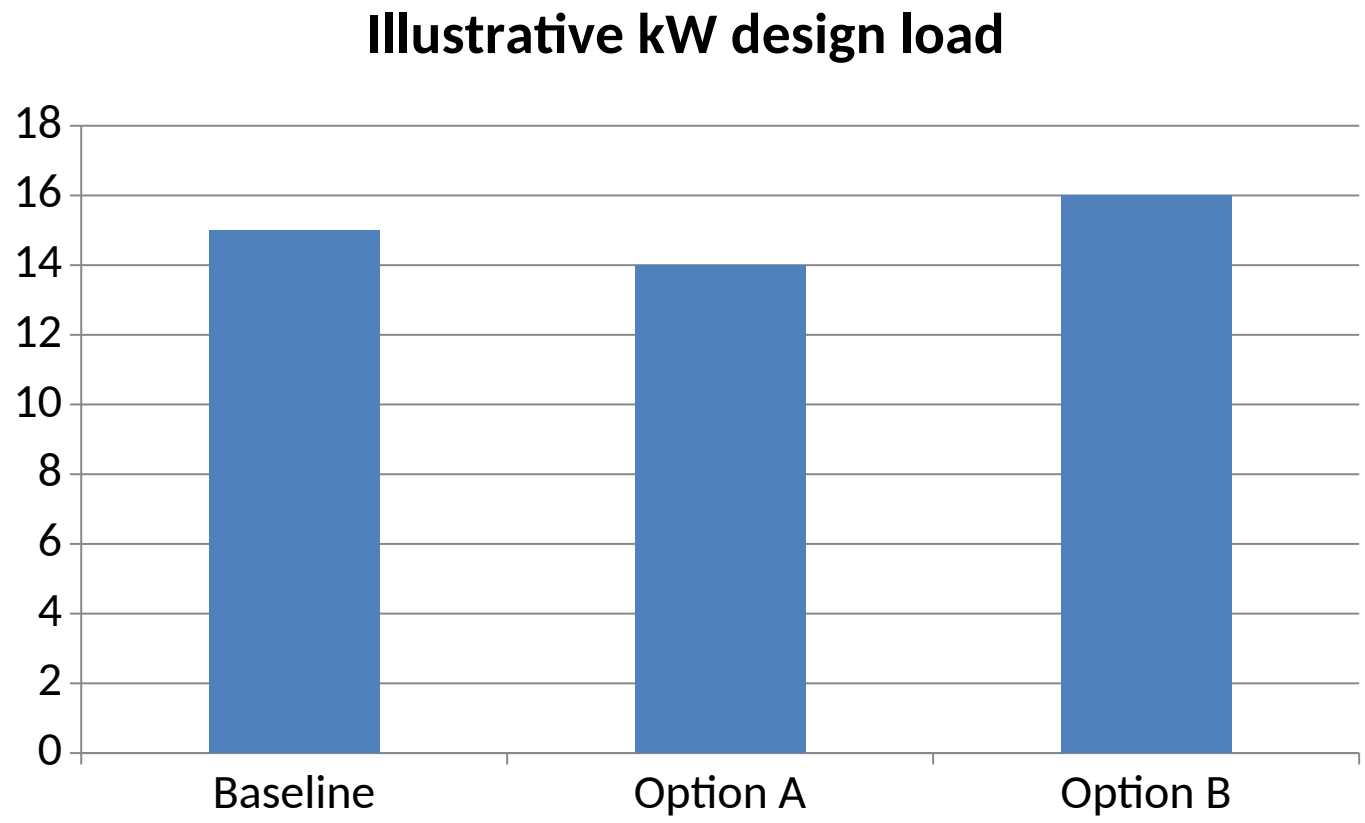
Accept System zones and zone equipment only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

System zones and zone equipment — Quantitative review



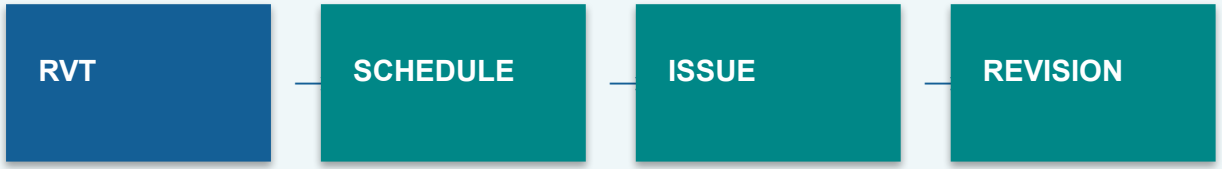
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

System zones and zone equipment — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for System zones and zone equipment.



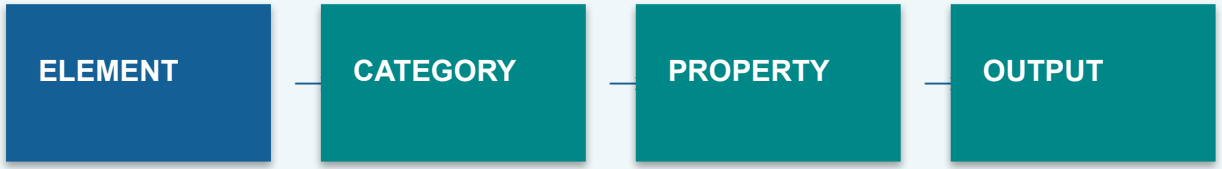
VERIFICATION

Criteria: K7 A7; Lab 10.

Supply air and ventilation loads — Model anatomy

MECHANISM / ARTIFACT

Input: Supply air and ventilation loads.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



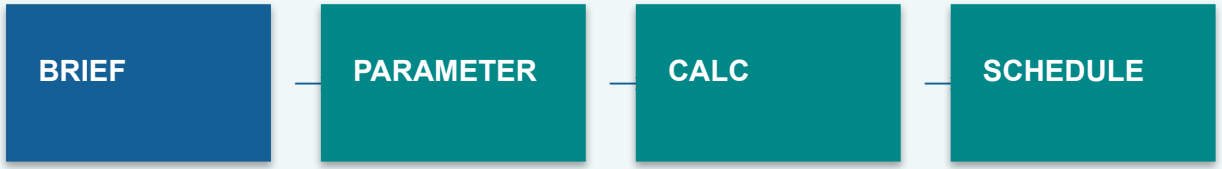
VERIFICATION

Check supply / air / and / ventilation in its relevant Revit view, browser or schedule.

Supply air and ventilation loads — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 16 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Supply air and ventilation loads — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Supply air and ventilation loads. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Supply air and ventilation loads — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 16 kW at ΔT 10 K: airflow ≈ 1.33 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Supply air and ventilation loads — Failure trace

MECHANISM / ARTIFACT

If Supply air and ventilation loads is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Supply air and ventilation loads — Decision rule

MECHANISM / ARTIFACT

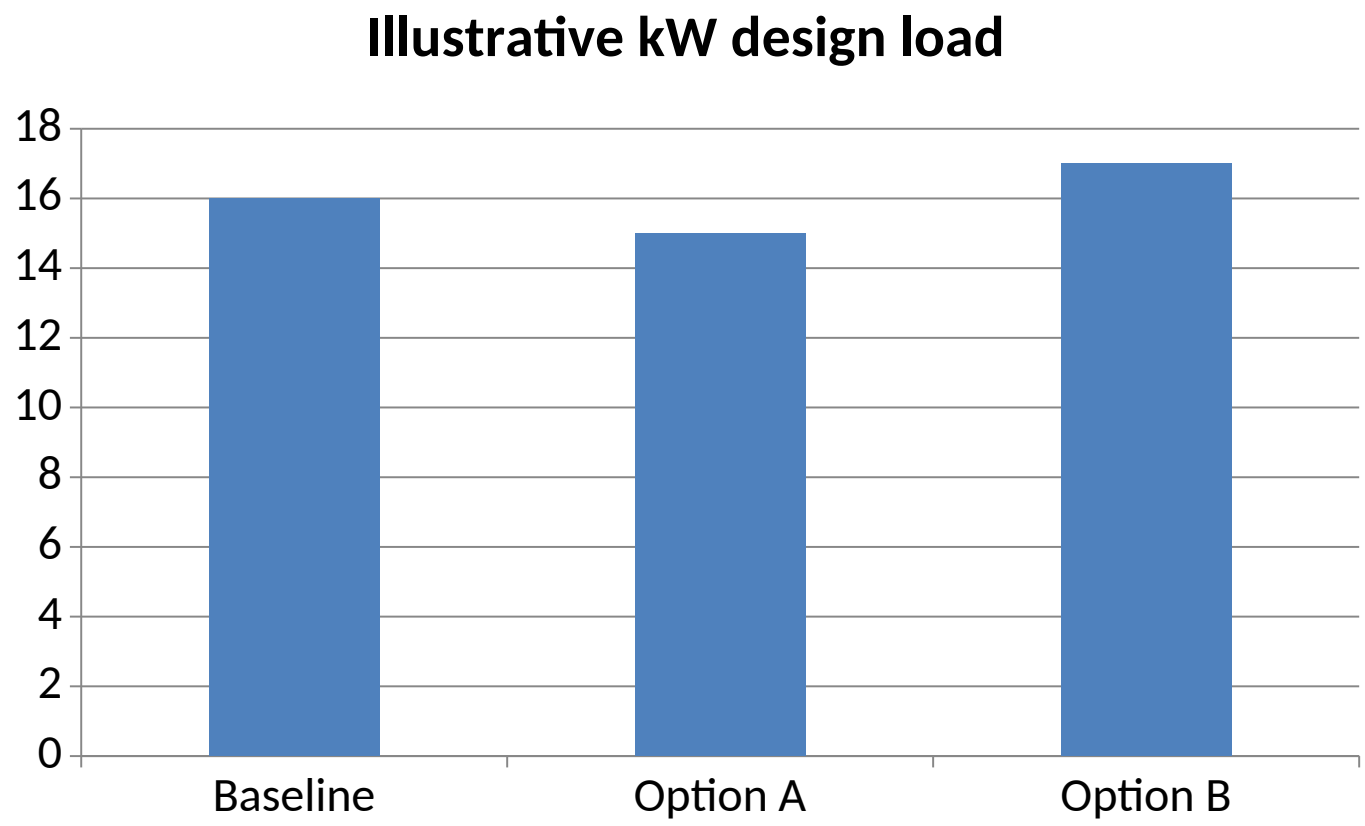
Accept Supply air and ventilation loads only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Supply air and ventilation loads — Quantitative review



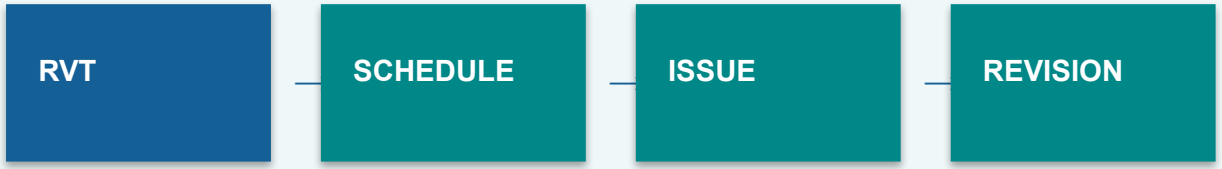
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Supply air and ventilation loads — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Supply air and ventilation loads.



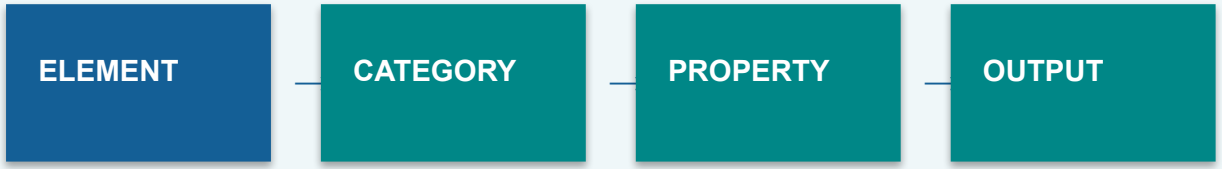
VERIFICATION

Criteria: K7 A7; Lab 11.

Heating and cooling load checks — Model anatomy

MECHANISM / ARTIFACT

Input: Heating and cooling load checks.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



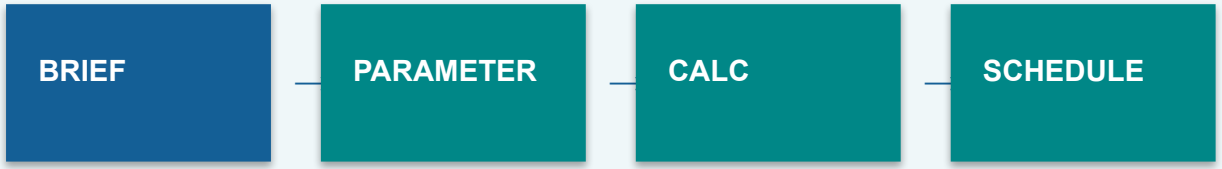
VERIFICATION

Check heating / and / cooling / load in its relevant Revit view, browser or schedule.

Heating and cooling load checks — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 17 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Heating and cooling load checks — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Heating and cooling load checks. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Heating and cooling load checks — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 17 kW at ΔT 10 K: airflow ≈ 1.41 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Heating and cooling load checks — Failure trace

MECHANISM / ARTIFACT

If Heating and cooling load checks is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Heating and cooling load checks — Decision rule

MECHANISM / ARTIFACT

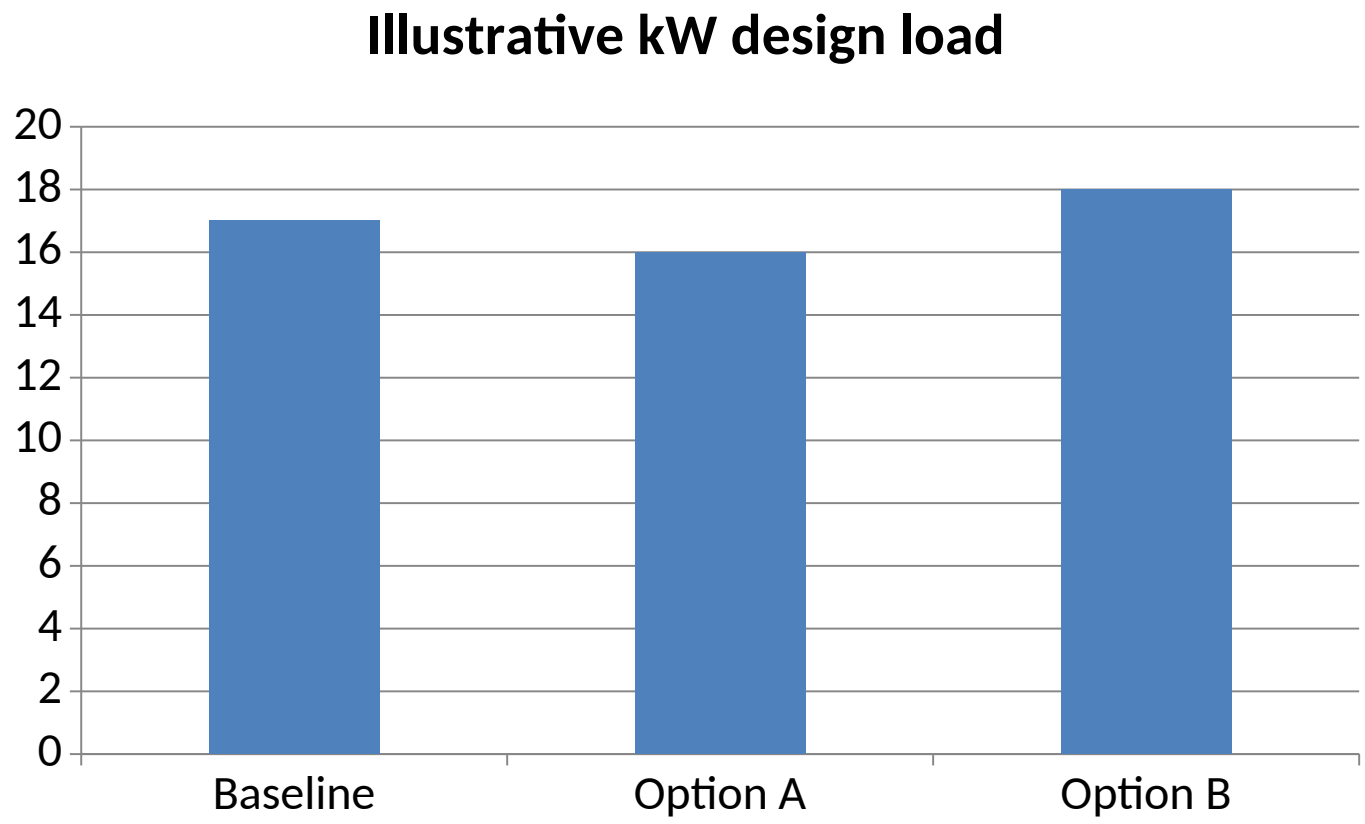
Accept Heating and cooling load checks only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Heating and cooling load checks — Quantitative review



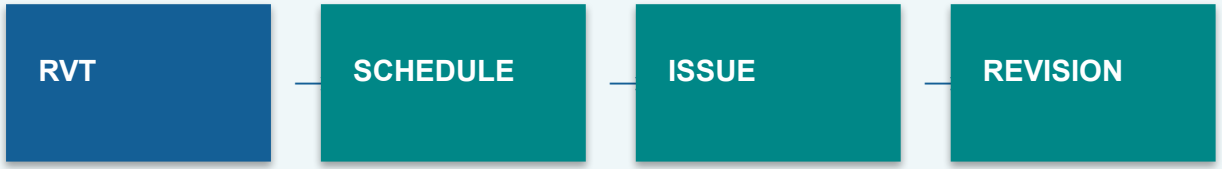
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Heating and cooling load checks — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Heating and cooling load checks.



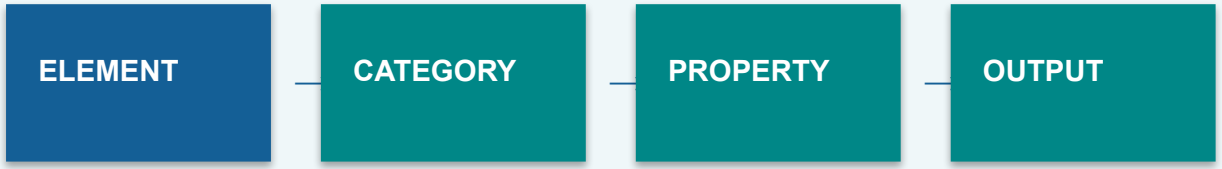
VERIFICATION

Criteria: K7 A7; Lab 11.

Duct velocity and pressure-drop checks — Model anatomy

MECHANISM / ARTIFACT

Input: Duct velocity and pressure-drop checks.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



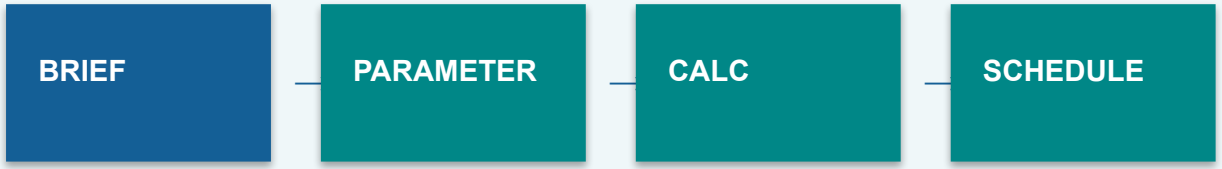
VERIFICATION

Check duct / velocity / and / pressure-drop in its relevant Revit view, browser or schedule.

Duct velocity and pressure-drop checks — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 18 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Duct velocity and pressure-drop checks — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Duct velocity and pressure-drop checks.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Duct velocity and pressure-drop checks — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 18 kW at ΔT 10 K: airflow ≈ 1.49 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Duct velocity and pressure-drop checks — Failure trace

MECHANISM / ARTIFACT

If Duct velocity and pressure-drop checks is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Duct velocity and pressure-drop checks — Decision rule

MECHANISM / ARTIFACT

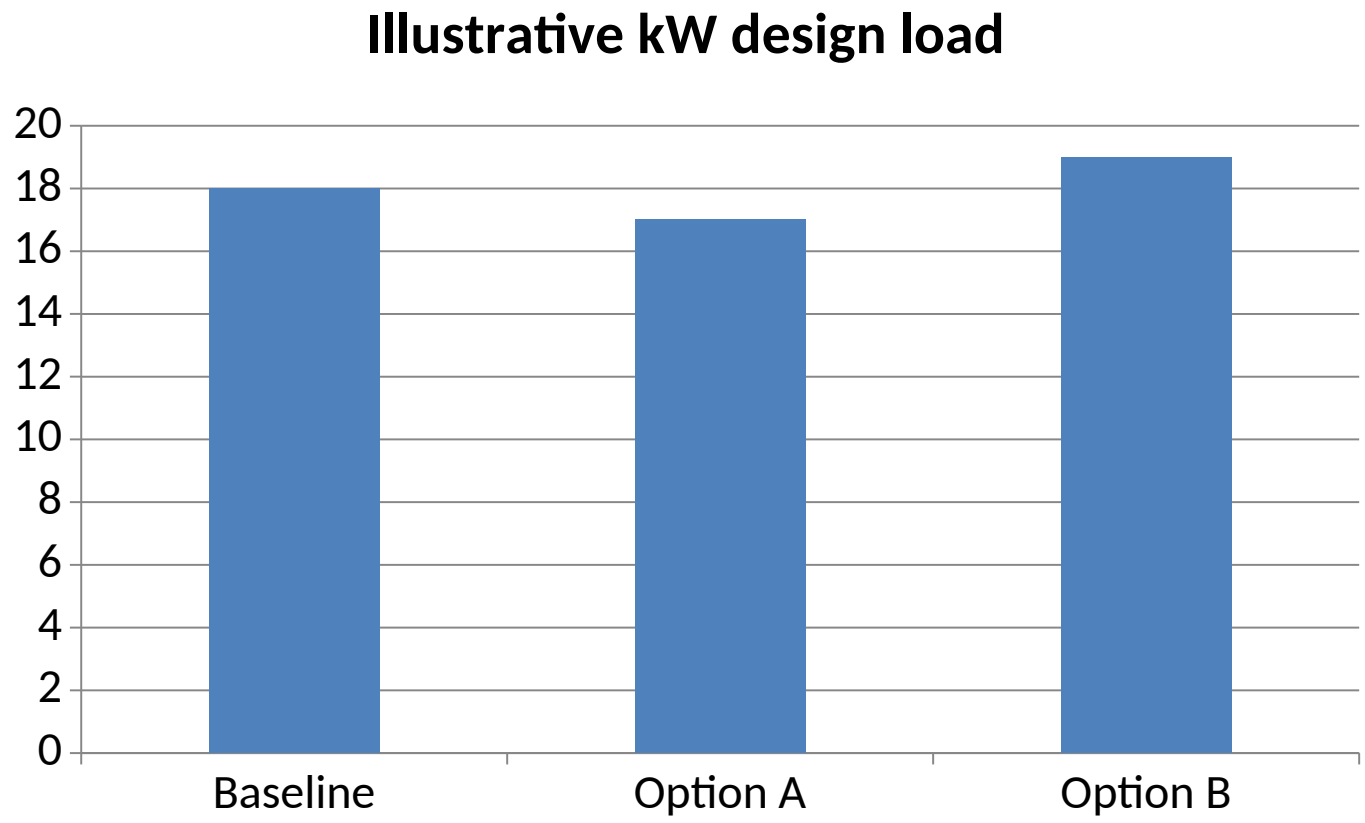
Accept Duct velocity and pressure-drop checks only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Duct velocity and pressure-drop checks — Quantitative review



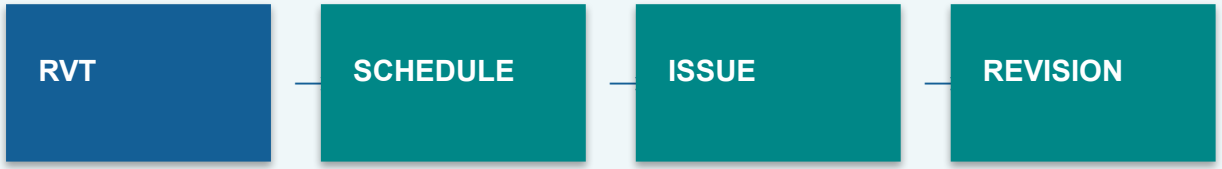
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Duct velocity and pressure-drop checks — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Duct velocity and pressure-drop checks.



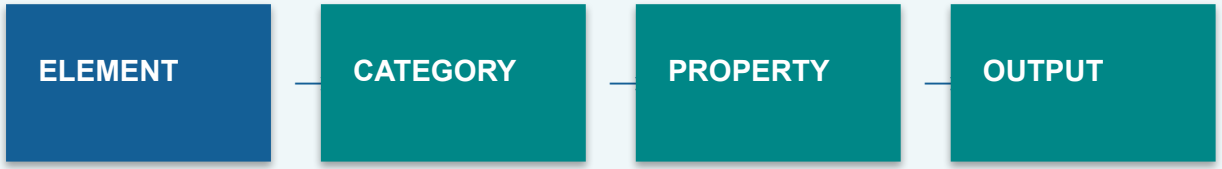
VERIFICATION

Criteria: K7 A7; Lab 11.

Hydronic flow and pipe loss checks — Model anatomy

MECHANISM / ARTIFACT

Input: Hydronic flow and pipe loss checks.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



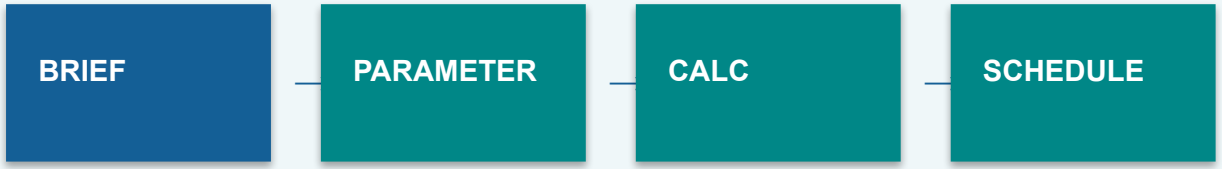
VERIFICATION

Check hydronic / flow / and / pipe in its relevant Revit view, browser or schedule.

Hydronic flow and pipe loss checks — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 19 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Hydronic flow and pipe loss checks — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Hydronic flow and pipe loss checks. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Hydronic flow and pipe loss checks — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 19 kW at ΔT 10 K: airflow ≈ 1.58 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Hydronic flow and pipe loss checks — Failure trace

MECHANISM / ARTIFACT

If Hydronic flow and pipe loss checks is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Hydronic flow and pipe loss checks — Decision rule

MECHANISM / ARTIFACT

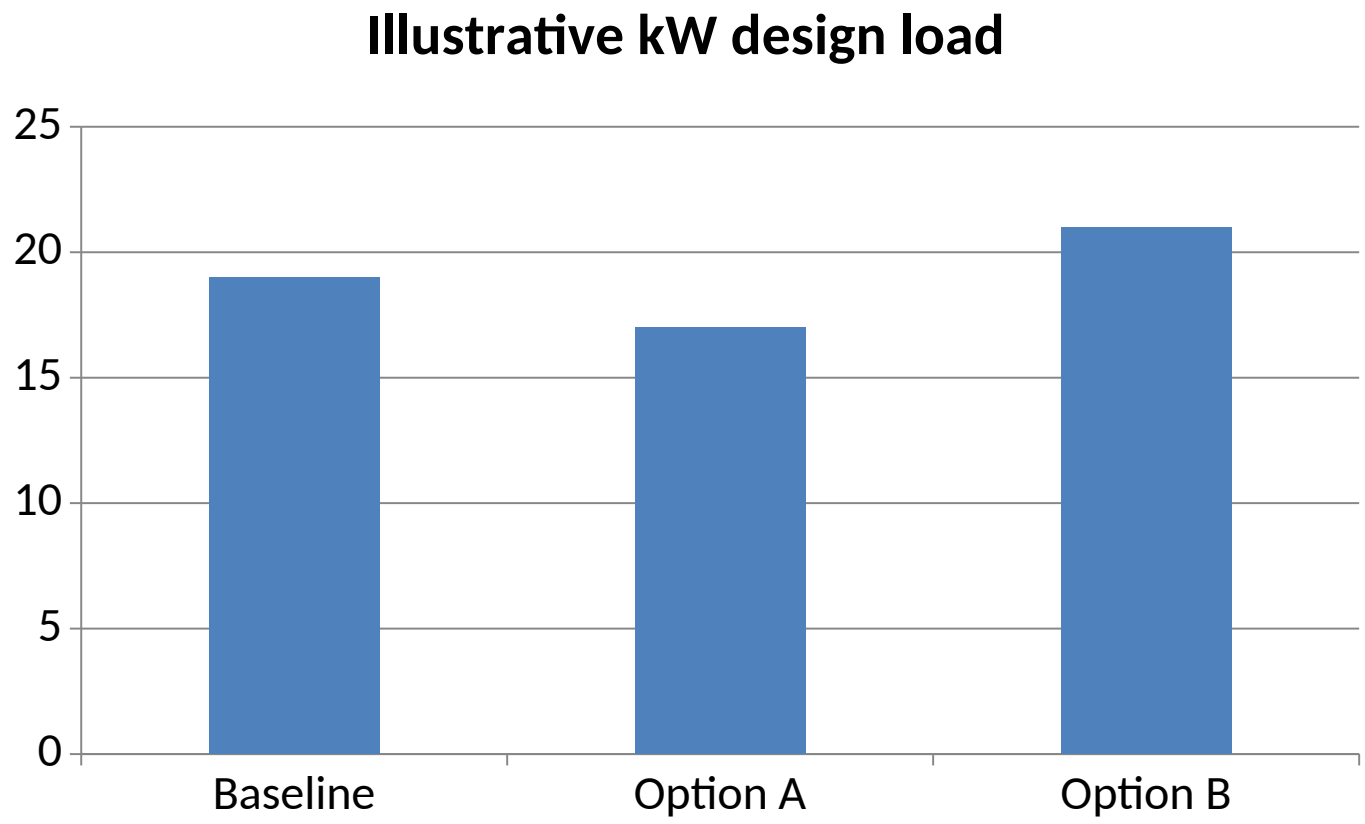
Accept Hydronic flow and pipe loss checks only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Hydronic flow and pipe loss checks — Quantitative review



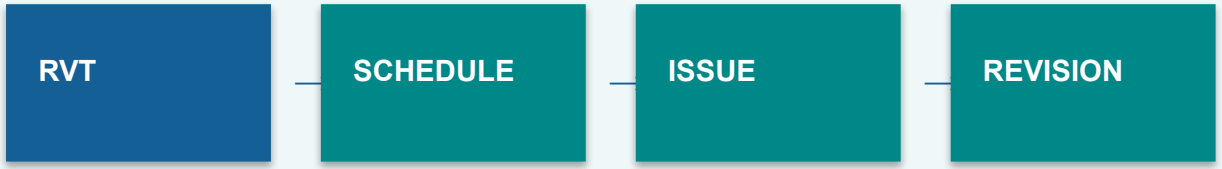
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Hydronic flow and pipe loss checks — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Hydronic flow and pipe loss checks.



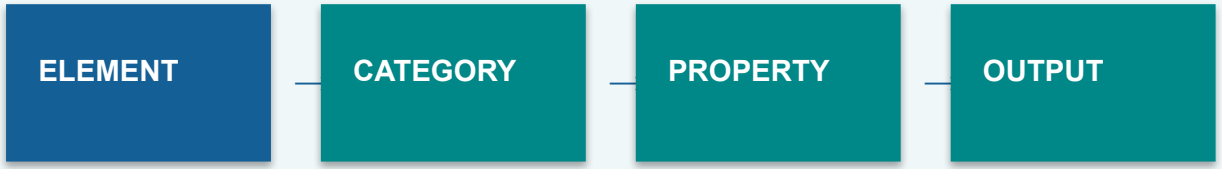
VERIFICATION

Criteria: K7 A7; Lab 12.

Thermal properties of constructions — Model anatomy

MECHANISM / ARTIFACT

Input: Thermal properties of constructions.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



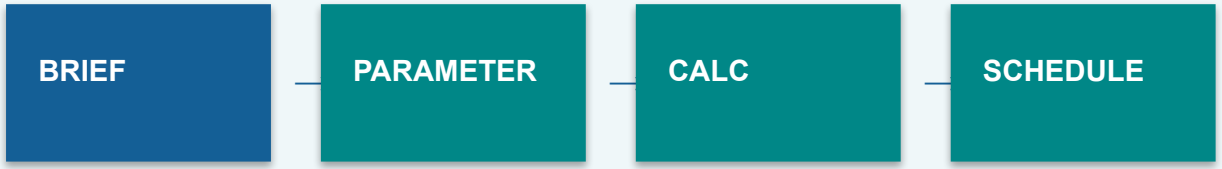
VERIFICATION

Check thermal / properties / of / constructions in its relevant Revit view, browser or schedule.

Thermal properties of constructions — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 20 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Thermal properties of constructions — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Thermal properties of constructions.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Thermal properties of constructions — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 20 kW at ΔT 10 K: airflow ≈ 1.66 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Thermal properties of constructions — Failure trace

MECHANISM / ARTIFACT

If Thermal properties of constructions is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Thermal properties of constructions — Decision rule

MECHANISM / ARTIFACT

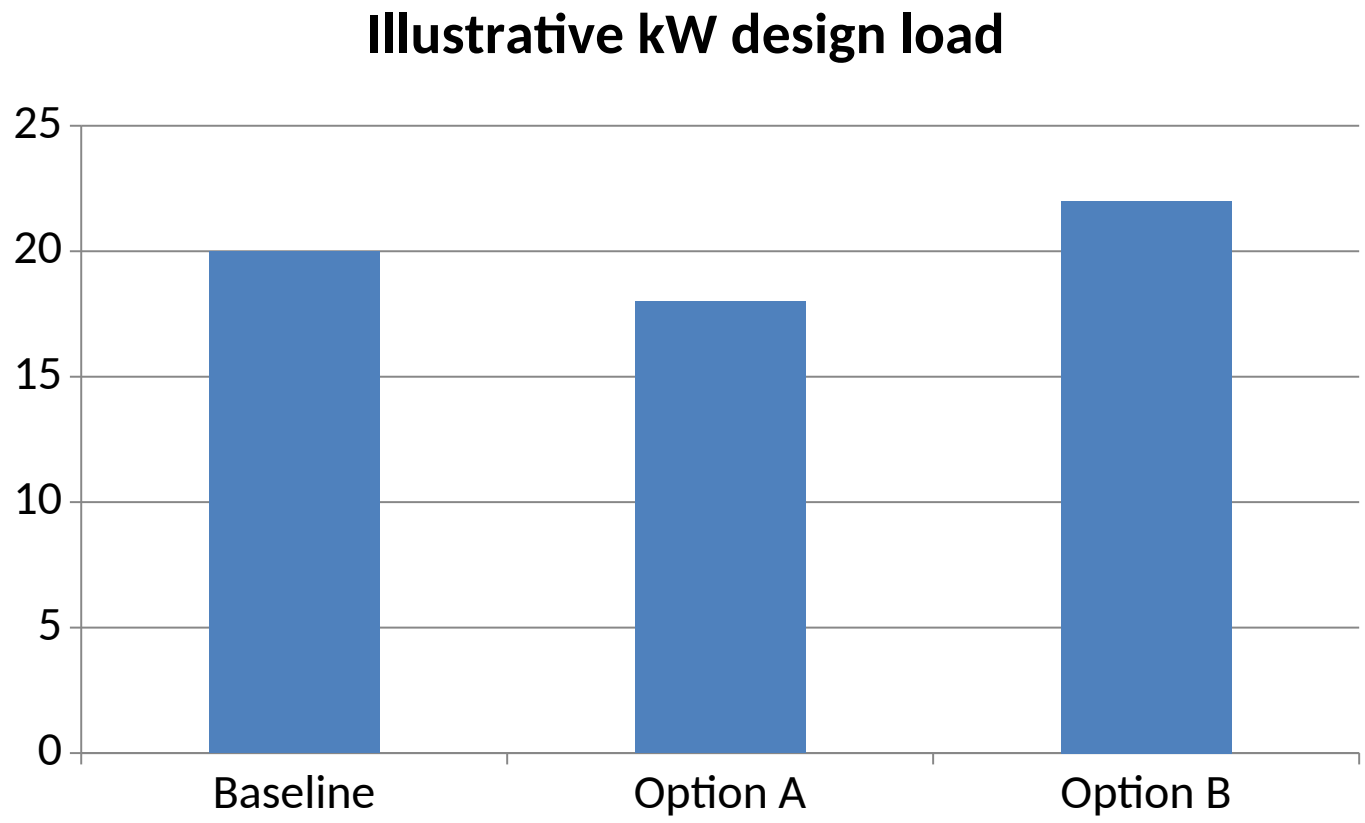
Accept Thermal properties of constructions only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Thermal properties of constructions — Quantitative review



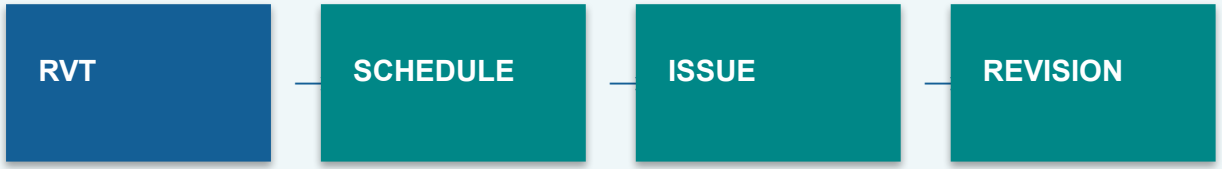
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Thermal properties of constructions — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Thermal properties of constructions.



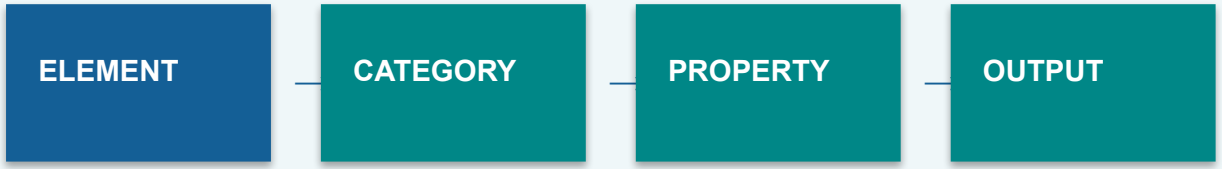
VERIFICATION

Criteria: K7 A7; Lab 12.

Calculated values and exception review — Model anatomy

MECHANISM / ARTIFACT

Input: Calculated values and exception review.
Control: space boundary, zone, analytical settings and thermal property.
Output: a named analytical space, zone or calculated result with a traceable element ID.



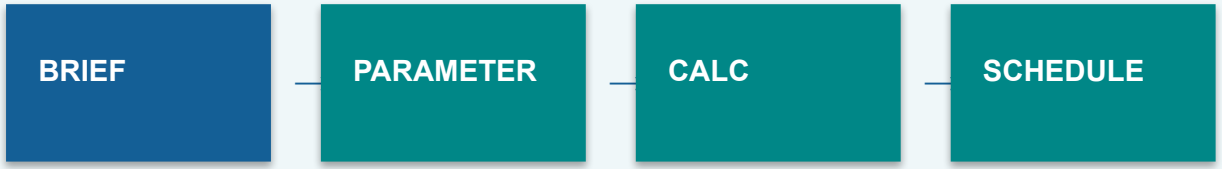
VERIFICATION

Check calculated / values / and / exception in its relevant Revit view, browser or schedule.

Calculated values and exception review — Parameter chain

MECHANISM / ARTIFACT

Project brief → space boundary, zone, analytical settings and thermal property → model state → drawing or report. Illustrative value: 21 kW design load.



VERIFICATION

Check units, discipline and type versus instance scope.

Calculated values and exception review — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Calculated values and exception review.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Calculated values and exception review — Worked design example

MECHANISM / ARTIFACT

Illustrative sensible load 21 kW at ΔT 10 K: airflow ≈ 1.74 m³/s using ρ=1.2 kg/m³ and cp=1.005 kJ/kg·K.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Calculated values and exception review — Failure trace

MECHANISM / ARTIFACT

If Calculated values and exception review is missing or inconsistent, isolate room boundary, zone assignment, energy setting and material thermal input.



VERIFICATION

Capture element ID and exact property before changing settings.

Calculated values and exception review — Decision rule

MECHANISM / ARTIFACT

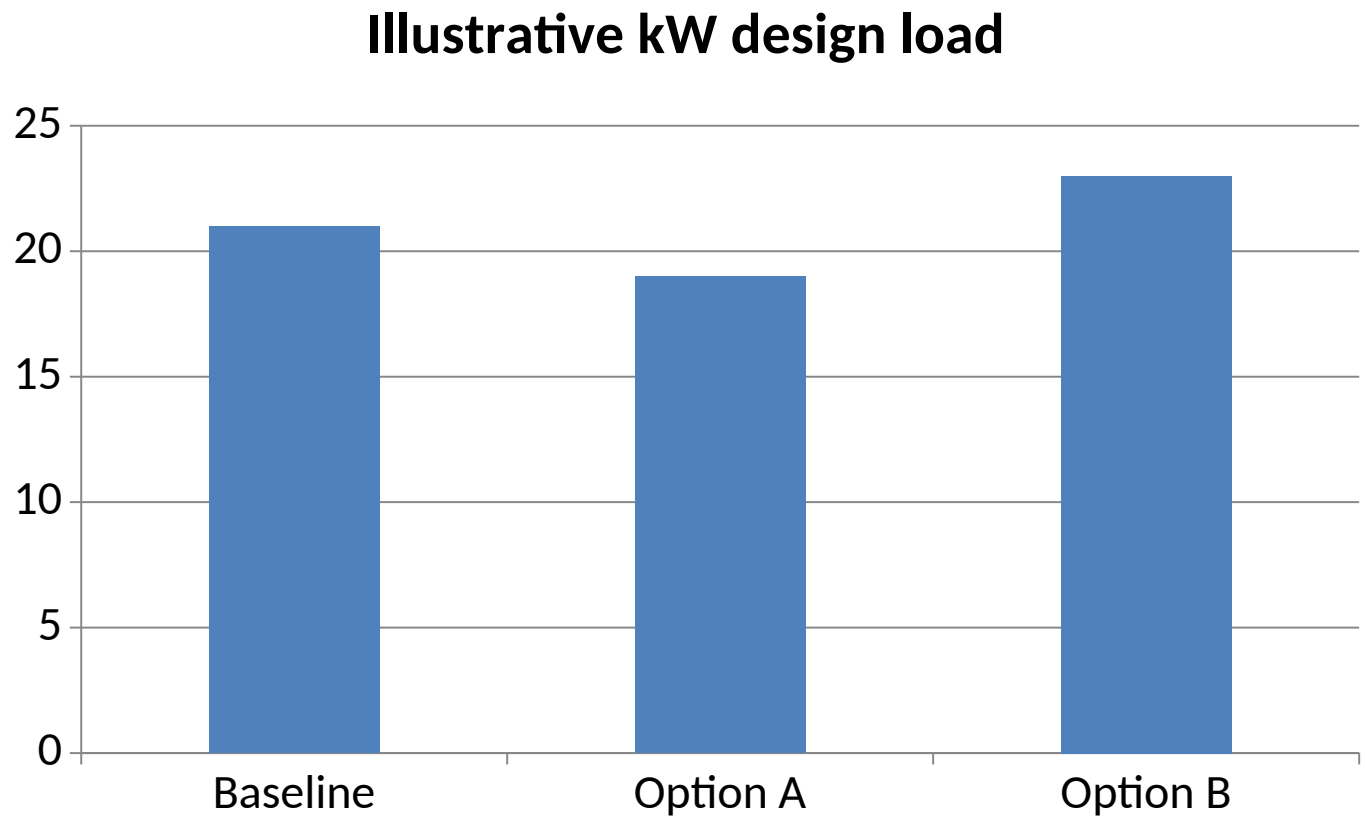
Accept Calculated values and exception review only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Calculated values and exception review — Quantitative review



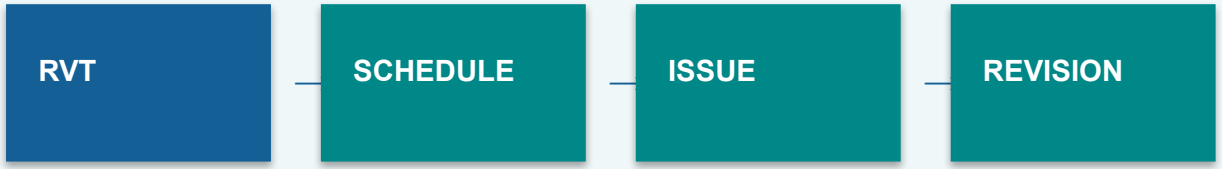
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Calculated values and exception review — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Calculated values and exception review.



VERIFICATION

Criteria: K7 A7; Lab 12.

Lab 10 — Create spaces zones and energy model

MODEL OUTPUT

energy model and zone mapping

EVIDENCE

rooms, spaces, analytical surfaces and zones; working RVT, schedule and PDF

Lab 10 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 11 — Review airflow loads and duct sizing

MODEL OUTPUT

load and airflow reconciliation sheet

EVIDENCE

terminal flow, zone loads, sizing constraints; working RVT, schedule and PDF

Lab 11 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 12 — Build hydronic piping and check pressure

MODEL OUTPUT

piping network and pressure review

EVIDENCE

pump, hydronic circuit, pipe sizing; working RVT, schedule and PDF

Lab 12 — acceptance evidence

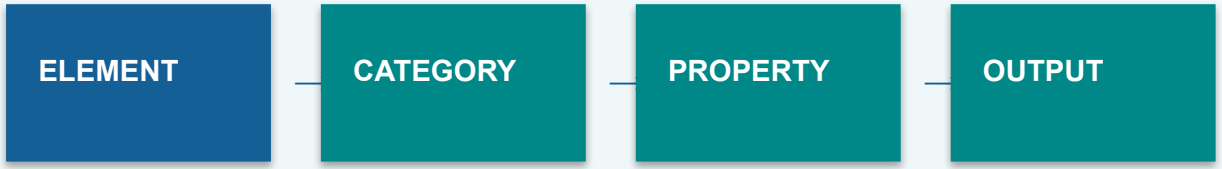
PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Link architecture and shared coordinates — Model anatomy

MECHANISM / ARTIFACT

Input: Link architecture and shared coordinates.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



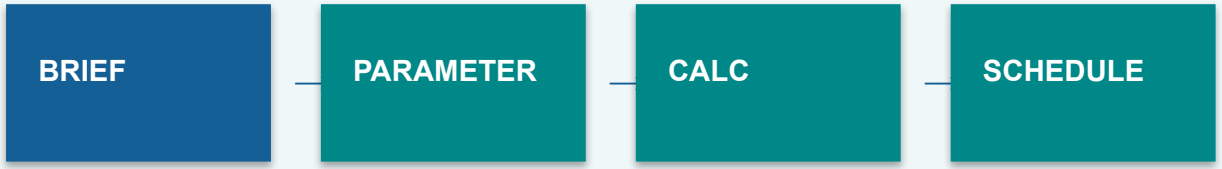
VERIFICATION

Check link / architecture / and / shared in its relevant Revit view, browser or schedule.

Link architecture and shared coordinates — Parameter chain

MECHANISM / ARTIFACT

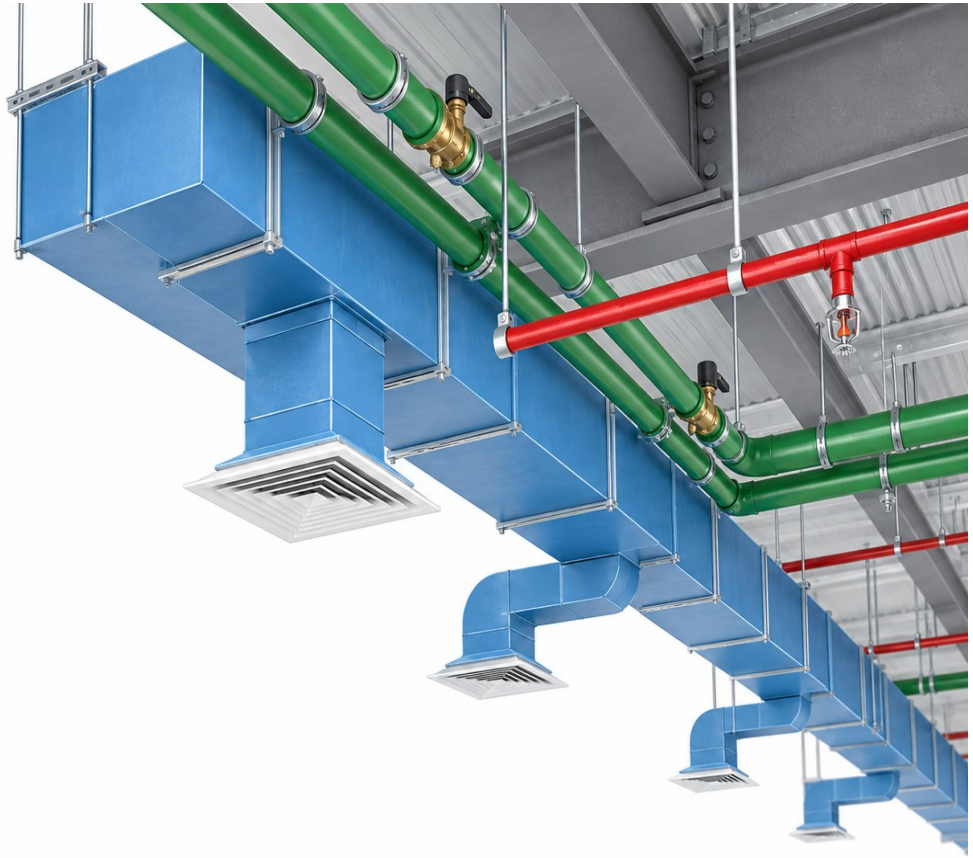
Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 4 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Link architecture and shared coordinates — Coordinated model state



COORDINATION CHECK

Architecture link provides levels and room bounds. Mechanical model owns Link architecture and shared coordinates.
A coordination view must expose both without duplicating source elements.
Check shared coordinates and linked-model visibility.

Link architecture and shared coordinates — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 4 baseline issues; Option A resolves 2 leaving 2; Option B resolves 3 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Link architecture and shared coordinates — Failure trace

MECHANISM / ARTIFACT

If Link architecture and shared coordinates is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Link architecture and shared coordinates — Decision rule

MECHANISM / ARTIFACT

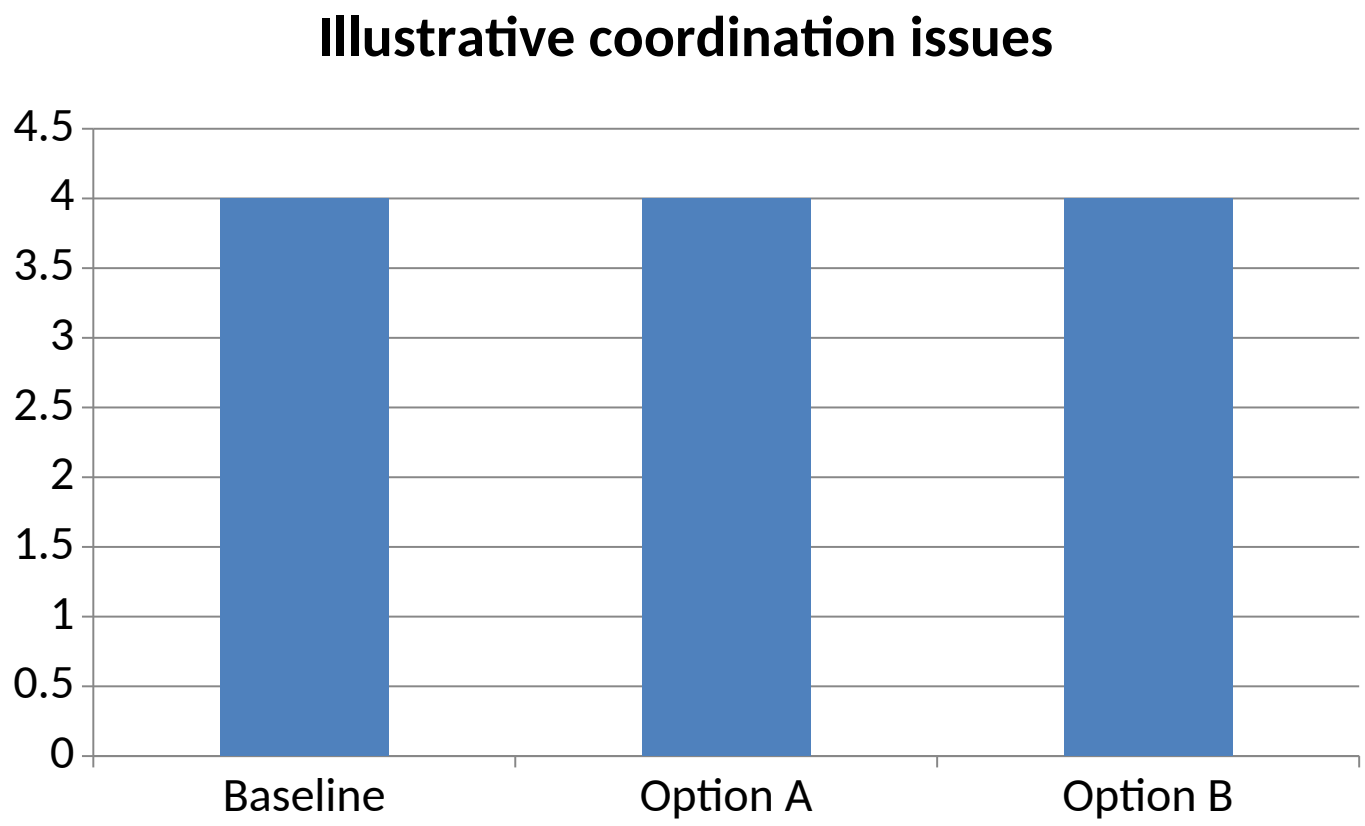
Accept Link architecture and shared coordinates only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Link architecture and shared coordinates — Quantitative review



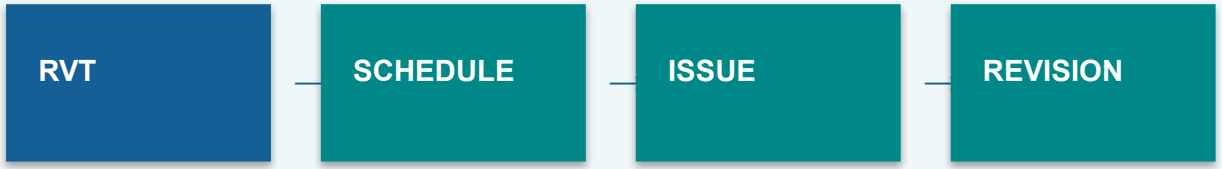
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Link architecture and shared coordinates — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Link architecture and shared coordinates.



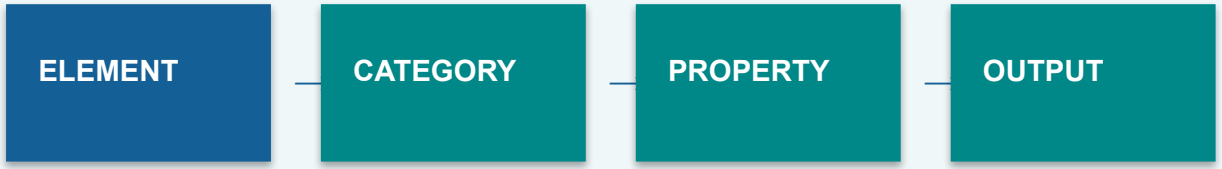
VERIFICATION

Criteria: K8 A8 A9; Lab 13.

Worksets and central model ownership — Model anatomy

MECHANISM / ARTIFACT

Input: Worksets and central model ownership.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



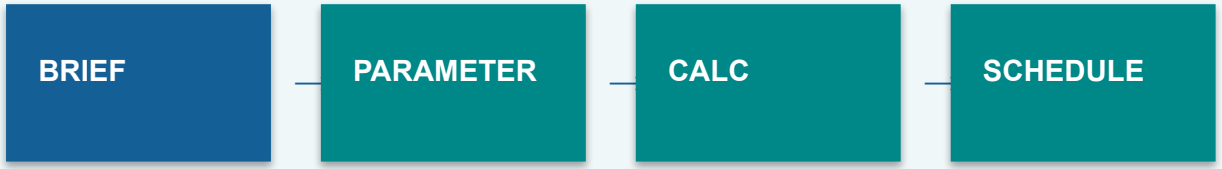
VERIFICATION

Check worksets / and / central / model in its relevant Revit view, browser or schedule.

Worksets and central model ownership — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 5 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Worksets and central model ownership — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Worksets and central model ownership.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Worksets and central model ownership — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 5 baseline issues; Option A resolves 3 leaving 2; Option B resolves 4 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Worksets and central model ownership — Failure trace

MECHANISM / ARTIFACT

If Worksets and central model ownership is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Worksets and central model ownership — Decision rule

MECHANISM / ARTIFACT

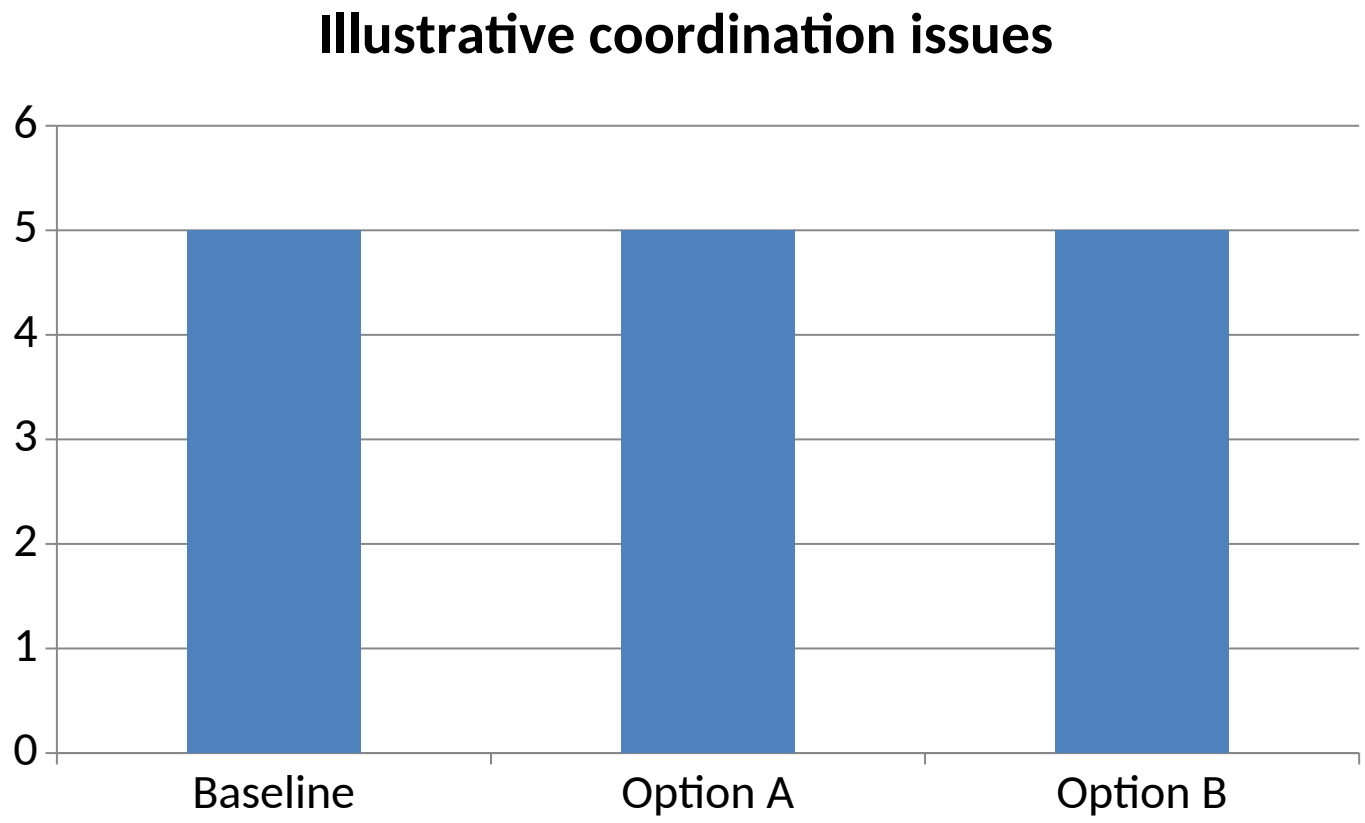
Accept Worksets and central model ownership only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Worksets and central model ownership — Quantitative review



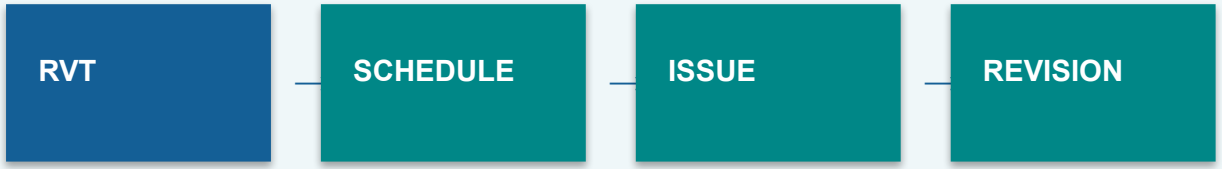
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Worksets and central model ownership — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Worksets and central model ownership.



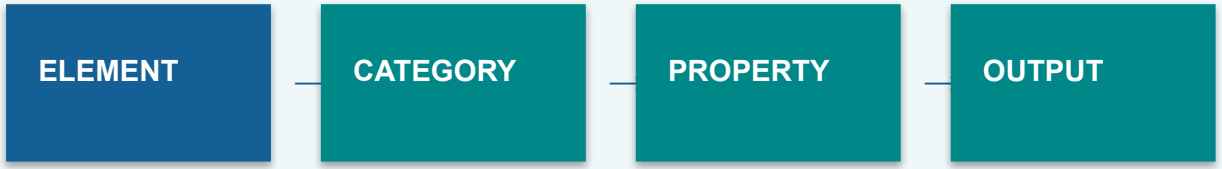
VERIFICATION

Criteria: K8 A8 A9; Lab 13.

Copy/Monitor of levels and grids — Model anatomy

MECHANISM / ARTIFACT

Input: Copy/Monitor of levels and grids.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



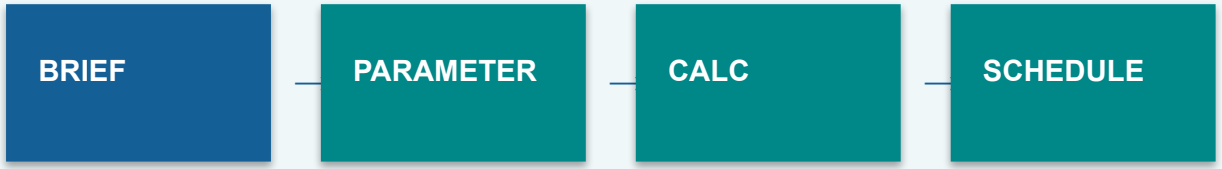
VERIFICATION

Check copy/monitor / of / levels / and in its relevant Revit view, browser or schedule.

Copy/Monitor of levels and grids — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 6 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Copy/Monitor of levels and grids — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds. Mechanical model owns Copy/Monitor of levels and grids. A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Copy/Monitor of levels and grids — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 6 baseline issues; Option A resolves 4 leaving 2; Option B resolves 5 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Copy/Monitor of levels and grids — Failure trace

MECHANISM / ARTIFACT

If Copy/Monitor of levels and grids is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Copy/Monitor of levels and grids — Decision rule

MECHANISM / ARTIFACT

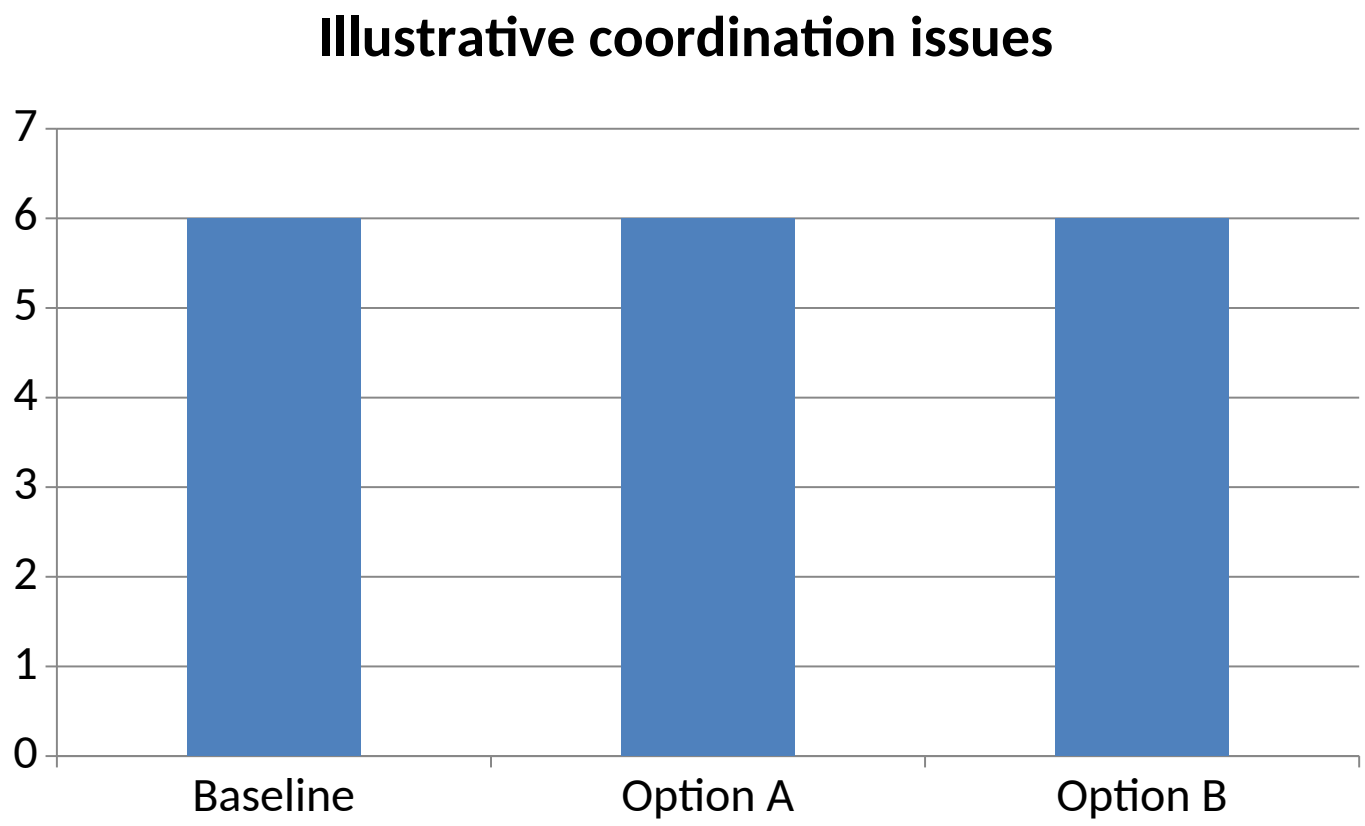
Accept Copy/Monitor of levels and grids only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Copy/Monitor of levels and grids — Quantitative review



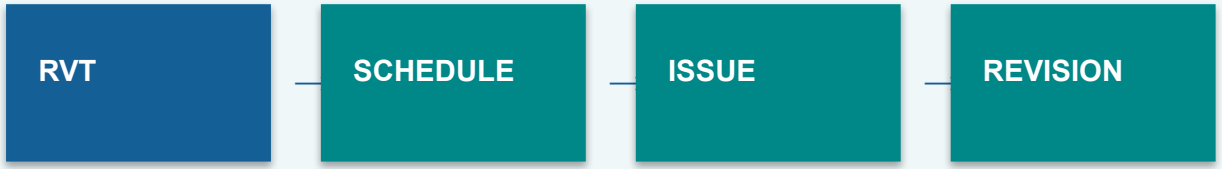
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Copy/Monitor of levels and grids — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Copy/Monitor of levels and grids.



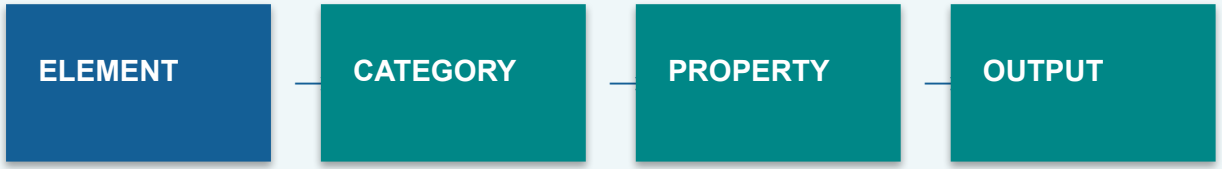
VERIFICATION

Criteria: K8 A8 A9; Lab 13.

Interference checks and clash records — Model anatomy

MECHANISM / ARTIFACT

Input: Interference checks and clash records.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



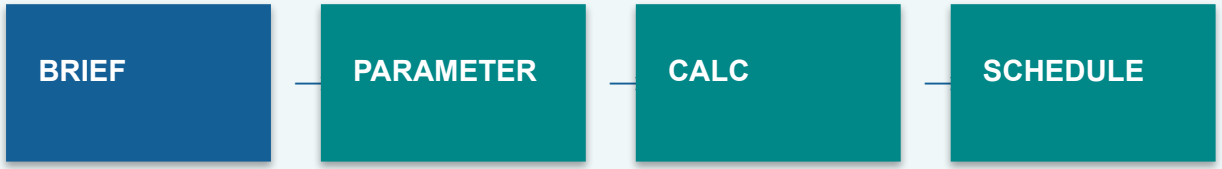
VERIFICATION

Check interference / checks / and / clash in its relevant Revit view, browser or schedule.

Interference checks and clash records — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 7 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Interference checks and clash records — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Interference checks and clash records.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Interference checks and clash records — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 7 baseline issues; Option A resolves 5 leaving 2; Option B resolves 6 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Interference checks and clash records — Failure trace

MECHANISM / ARTIFACT

If Interference checks and clash records is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Interference checks and clash records — Decision rule

MECHANISM / ARTIFACT

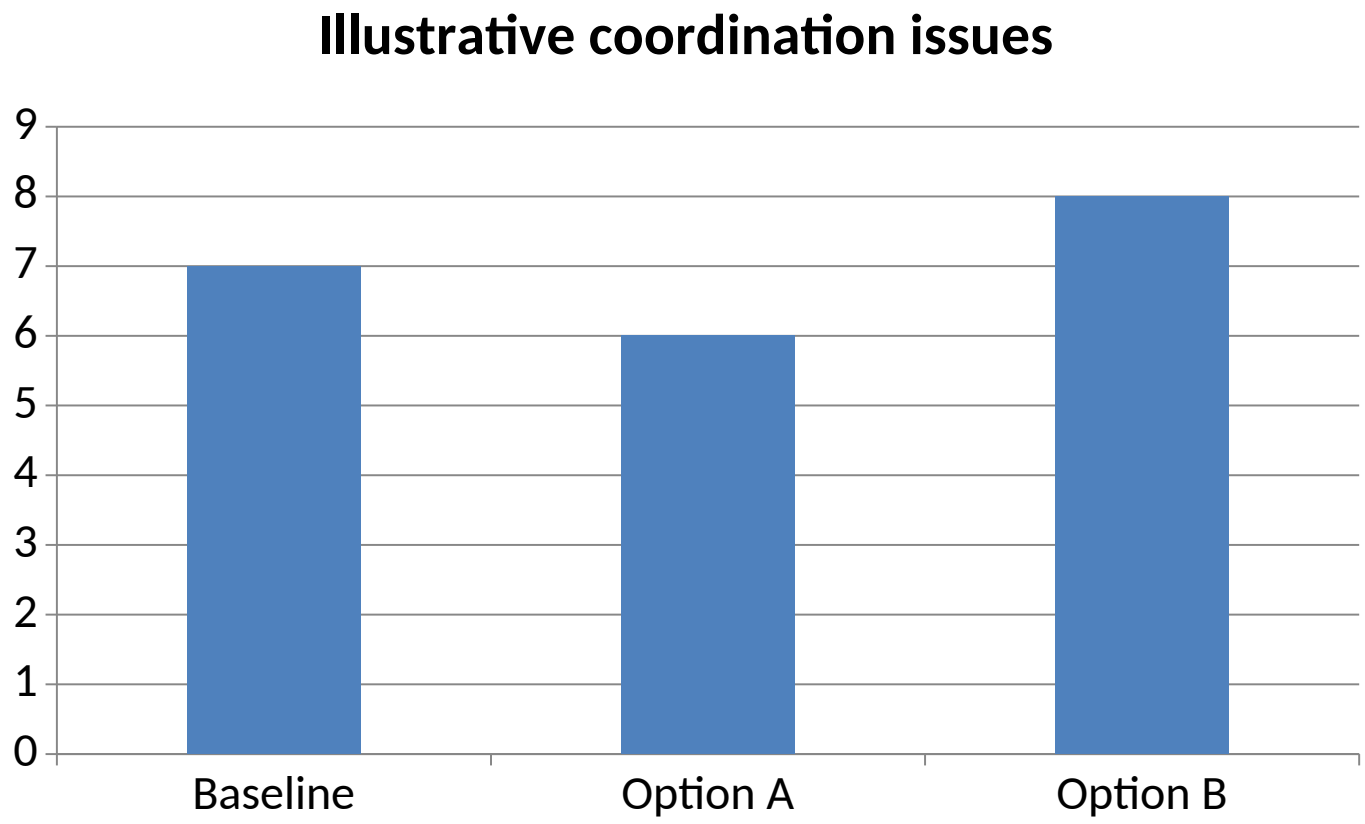
Accept Interference checks and clash records only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Interference checks and clash records — Quantitative review



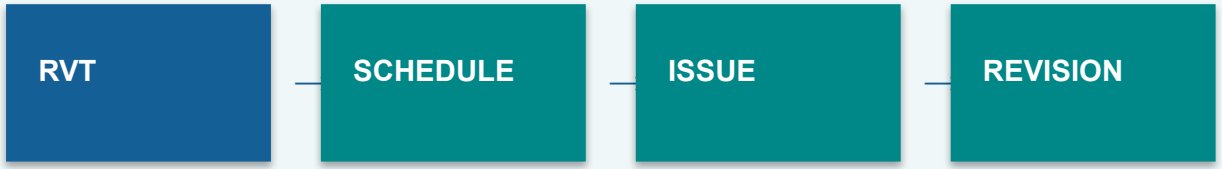
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Interference checks and clash records — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Interference checks and clash records.



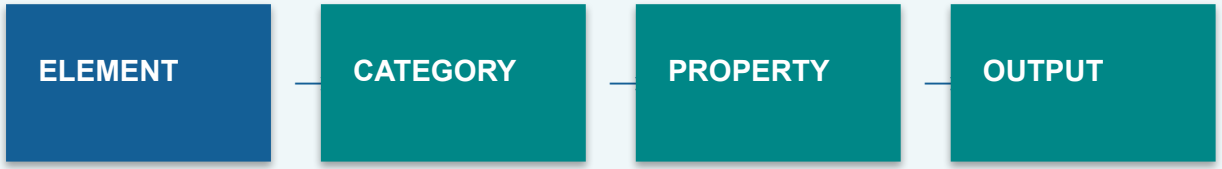
VERIFICATION

Criteria: K8 A8 A9; Lab 14.

Model review and issue assignment — Model anatomy

MECHANISM / ARTIFACT

Input: Model review and issue assignment.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



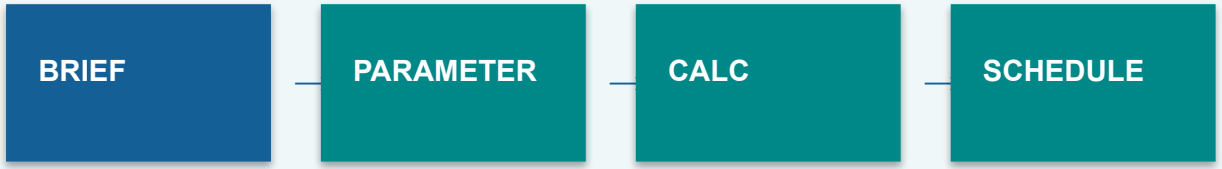
VERIFICATION

Check model / review / and / issue in its relevant Revit view, browser or schedule.

Model review and issue assignment — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 8 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Model review and issue assignment — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Model review and issue assignment.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Model review and issue assignment — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 8 baseline issues; Option A resolves 6 leaving 2; Option B resolves 7 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Model review and issue assignment — Failure trace

MECHANISM / ARTIFACT

If Model review and issue assignment is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Model review and issue assignment — Decision rule

MECHANISM / ARTIFACT

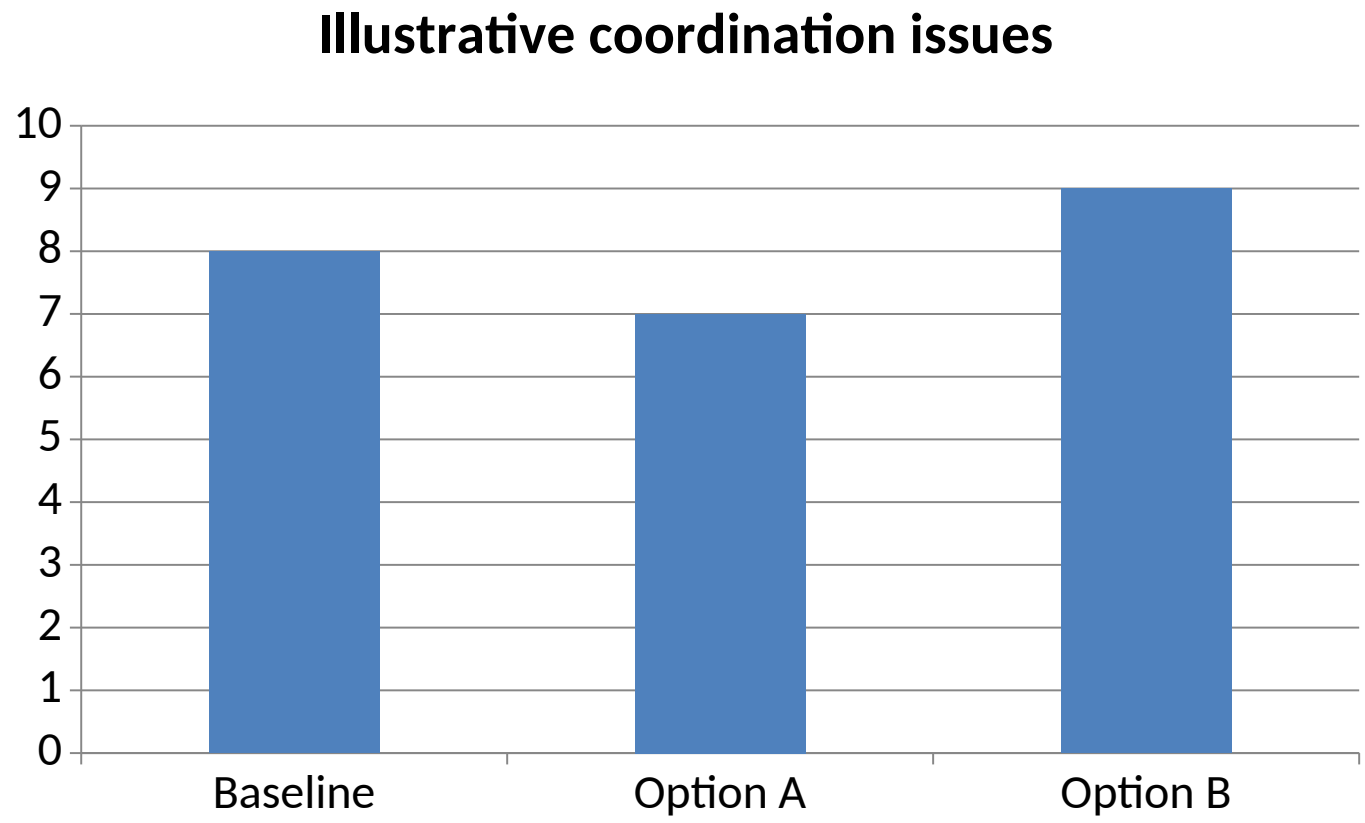
Accept Model review and issue assignment only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Model review and issue assignment — Quantitative review



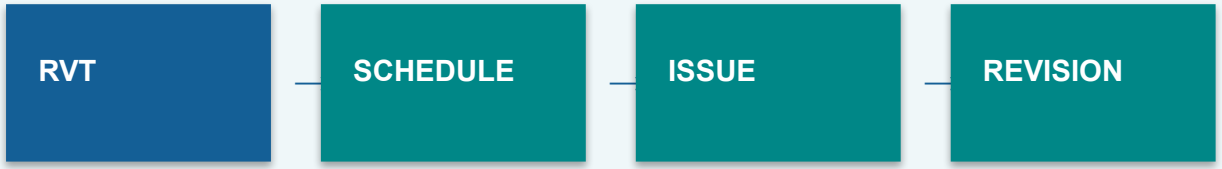
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Model review and issue assignment — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Model review and issue assignment.



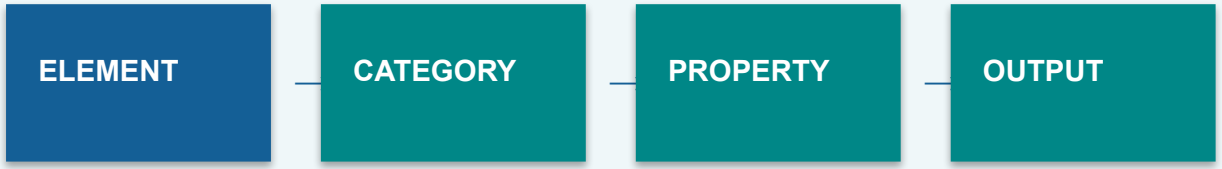
VERIFICATION

Criteria: K8 A8 A9; Lab 14.

Design options for route alternatives — Model anatomy

MECHANISM / ARTIFACT

Input: Design options for route alternatives.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



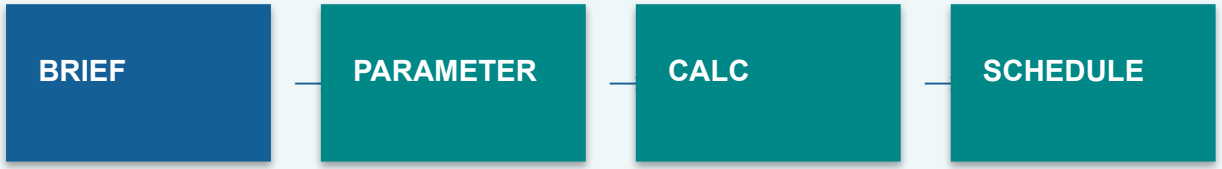
VERIFICATION

Check design / options / for / route in its relevant Revit view, browser or schedule.

Design options for route alternatives — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 9 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Design options for route alternatives — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Design options for route alternatives.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Design options for route alternatives — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 9 baseline issues; Option A resolves 7 leaving 2; Option B resolves 8 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Design options for route alternatives — Failure trace

MECHANISM / ARTIFACT

If Design options for route alternatives is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Design options for route alternatives — Decision rule

MECHANISM / ARTIFACT

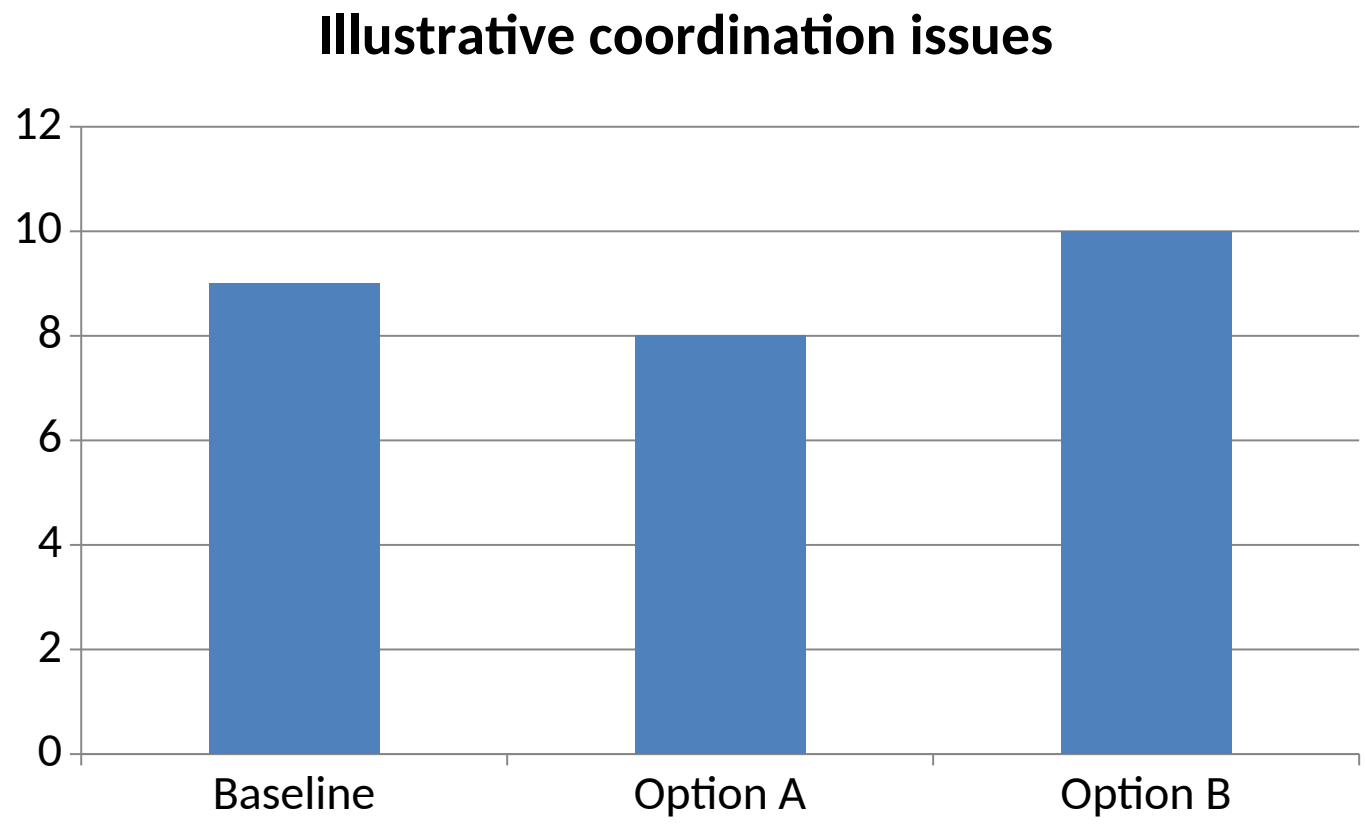
Accept Design options for route alternatives only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Design options for route alternatives — Quantitative review



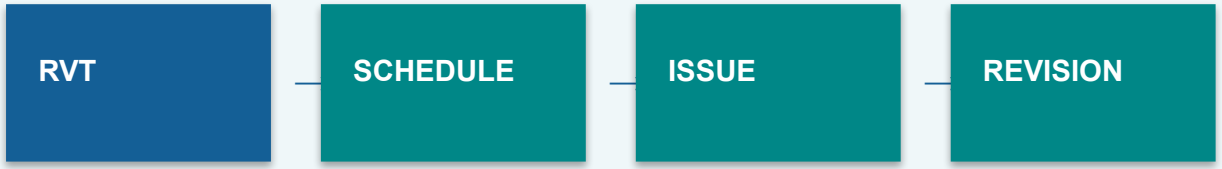
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Design options for route alternatives — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Design options for route alternatives.



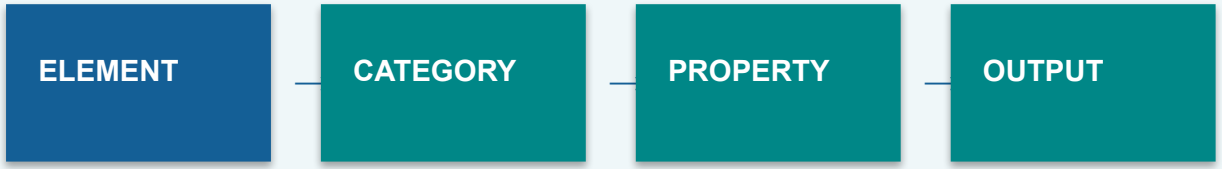
VERIFICATION

Criteria: K8 A8 A9; Lab 14.

Project standards transfer — Model anatomy

MECHANISM / ARTIFACT

Input: Project standards transfer.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



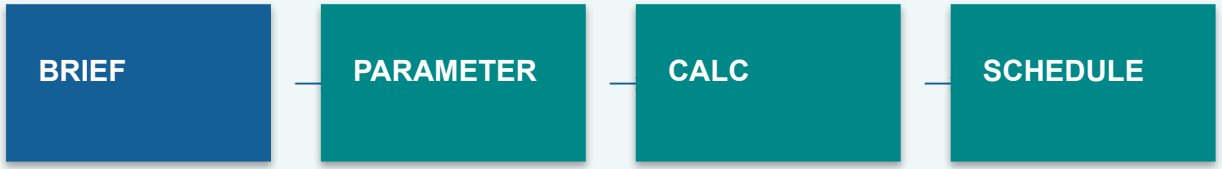
VERIFICATION

Check project / standards / transfer in its relevant Revit view, browser or schedule.

Project standards transfer — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 10 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Project standards transfer — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Project standards transfer.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Project standards transfer — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 10 baseline issues; Option A resolves 8 leaving 2; Option B resolves 9 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Project standards transfer — Failure trace

MECHANISM / ARTIFACT

If Project standards transfer is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Project standards transfer — Decision rule

MECHANISM / ARTIFACT

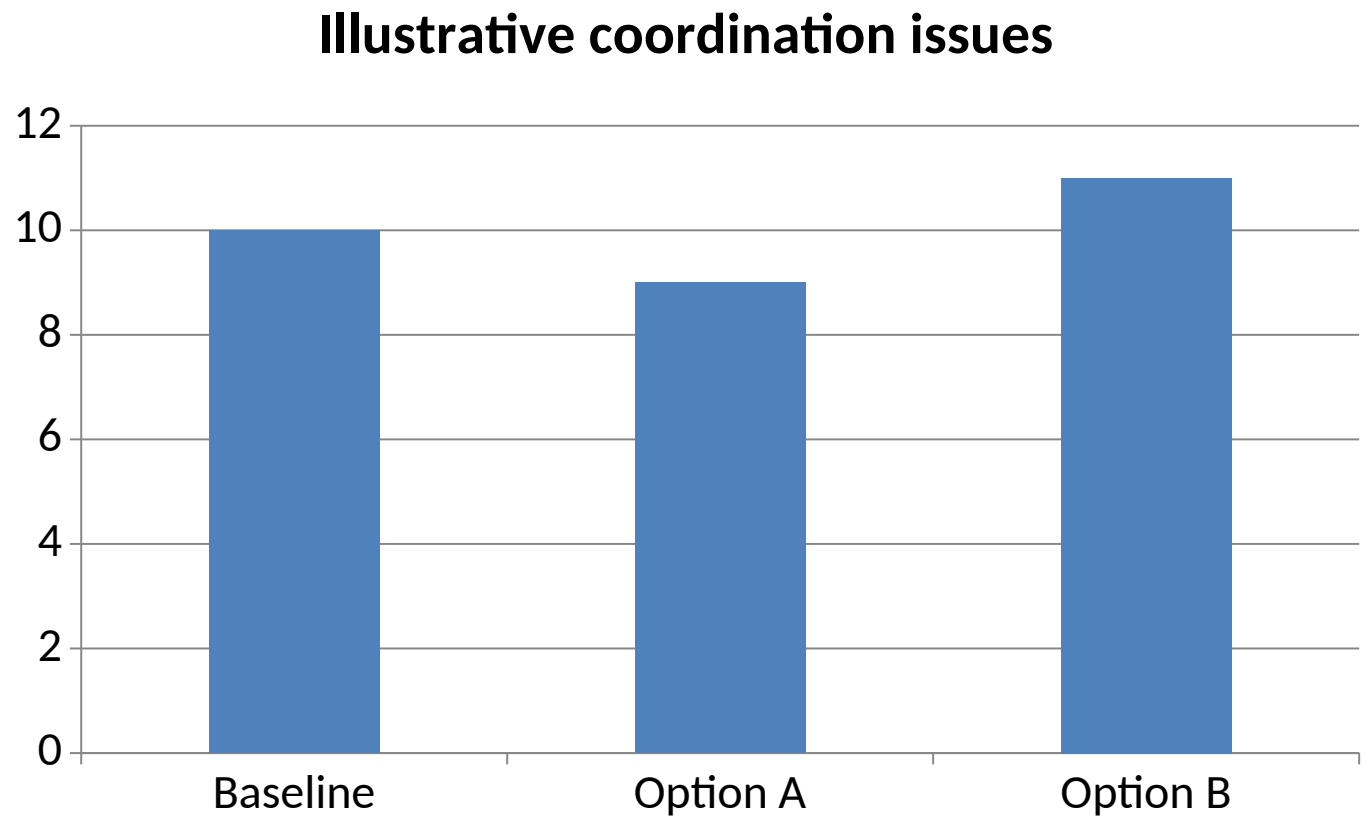
Accept Project standards transfer only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Project standards transfer — Quantitative review



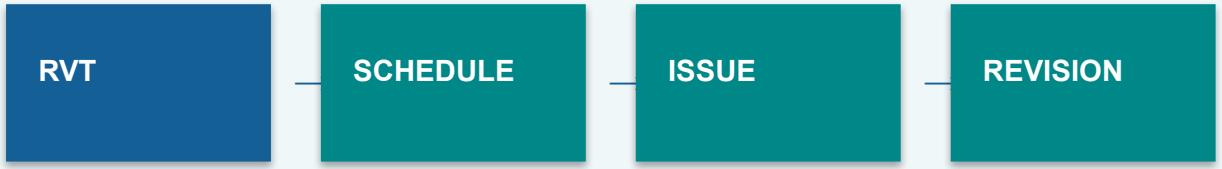
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Project standards transfer — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Project standards transfer.



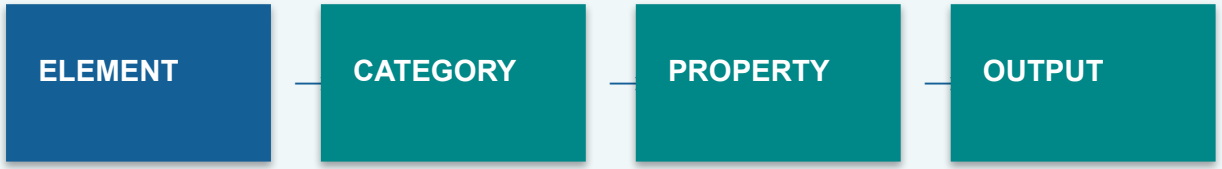
VERIFICATION

Criteria: K8 A8 A9; Lab 15.

IFC, PDF and schedule export — Model anatomy

MECHANISM / ARTIFACT

Input: IFC, PDF and schedule export.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



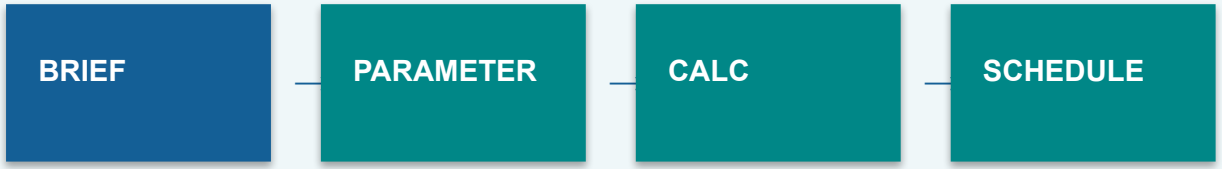
VERIFICATION

Check ifc, / pdf / and / schedule in its relevant Revit view, browser or schedule.

IFC, PDF and schedule export — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 11 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

IFC, PDF and schedule export — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns IFC, PDF and schedule export.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

IFC, PDF and schedule export — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 11 baseline issues; Option A resolves 9 leaving 2; Option B resolves 10 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

IFC, PDF and schedule export — Failure trace

MECHANISM / ARTIFACT

If IFC, PDF and schedule export is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

IFC, PDF and schedule export — Decision rule

MECHANISM / ARTIFACT

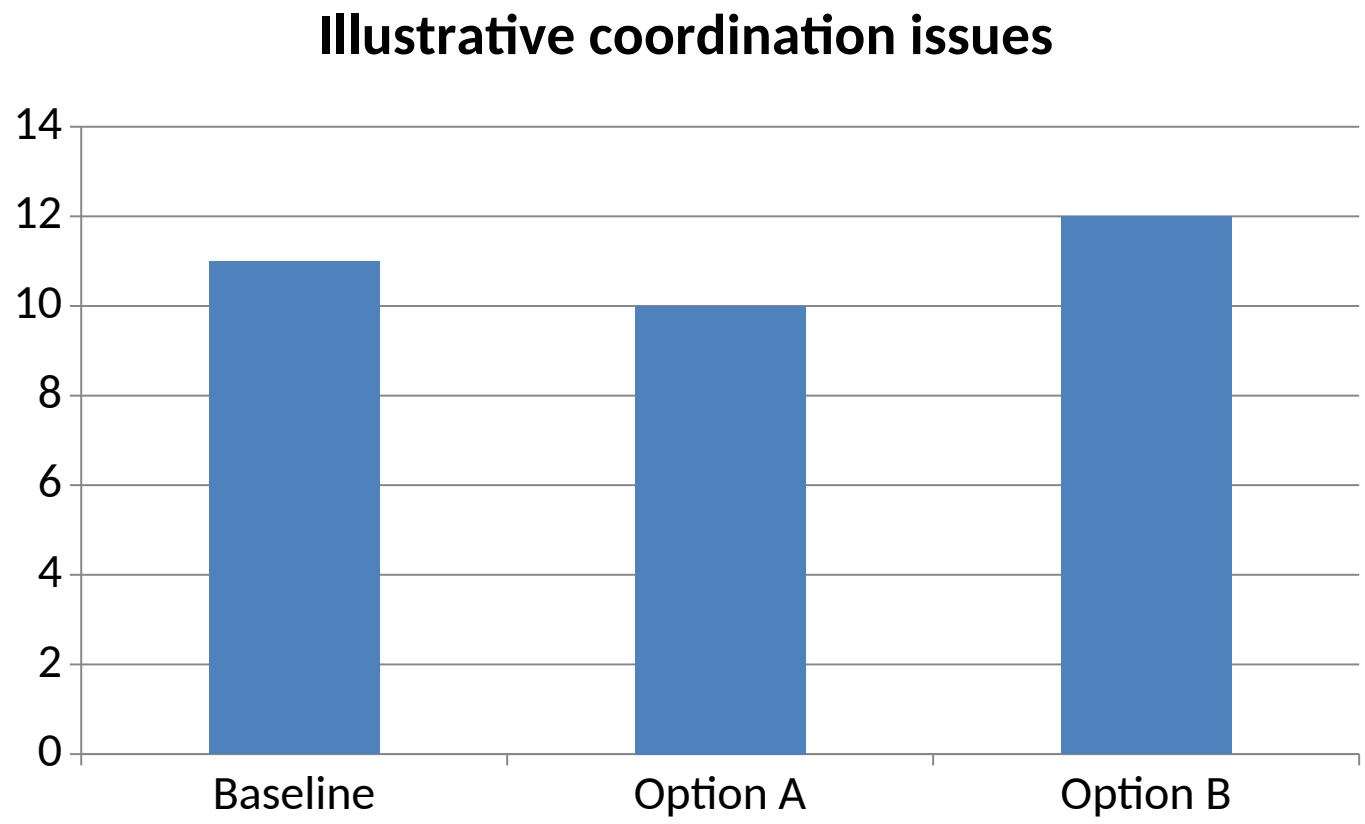
Accept IFC, PDF and schedule export only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

IFC, PDF and schedule export — Quantitative review



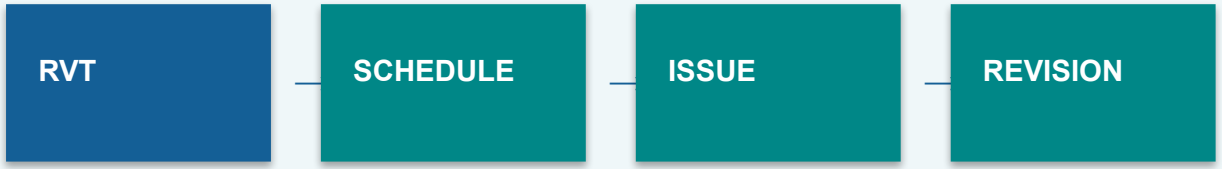
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

IFC, PDF and schedule export — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for IFC, PDF and schedule export.



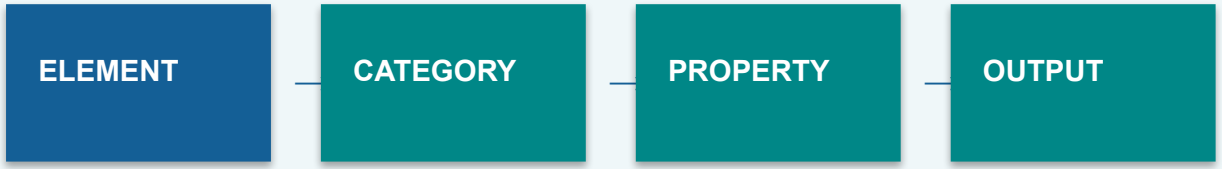
VERIFICATION

Criteria: K8 A8 A9; Lab 15.

Revision handover and evidence register — Model anatomy

MECHANISM / ARTIFACT

Input: Revision handover and evidence register.
Control: link coordinates, workset, issue ID and revision.
Output: a named linked model, coordination issue or export with a traceable element ID.



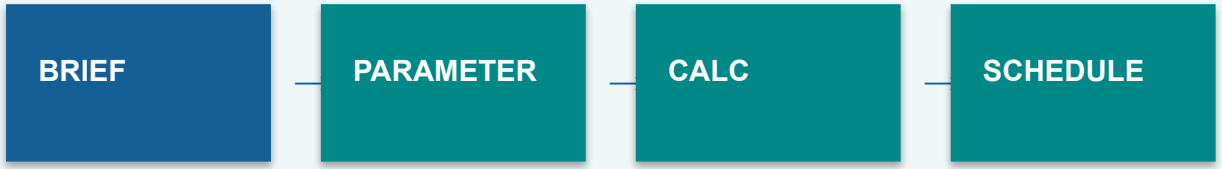
VERIFICATION

Check revision / handover / and / evidence in its relevant Revit view, browser or schedule.

Revision handover and evidence register — Parameter chain

MECHANISM / ARTIFACT

Project brief → link coordinates, workset, issue ID and revision → model state → drawing or report.
Illustrative value: 12 coordination issues.



VERIFICATION

Check units, discipline and type versus instance scope.

Revision handover and evidence register — Coordinated model state

MECHANISM / ARTIFACT

Architecture link provides levels and room bounds.
Mechanical model owns Revision handover and evidence register.
A coordination view must expose both without duplicating source elements.



VERIFICATION

Check shared coordinates and linked-model visibility.

Revision handover and evidence register — Worked design example

MECHANISM / ARTIFACT

Illustrative coordination review: 12 baseline issues; Option A resolves 10 leaving 2; Option B resolves 11 leaving 1. Compare cost and effort.



VERIFICATION

Validate the illustrative calculation against project criteria and model evidence.

Revision handover and evidence register — Failure trace

MECHANISM / ARTIFACT

If Revision handover and evidence register is missing or inconsistent, isolate shared coordinates, link reload, workset ownership and issue status.



VERIFICATION

Capture element ID and exact property before changing settings.

Revision handover and evidence register — Decision rule

MECHANISM / ARTIFACT

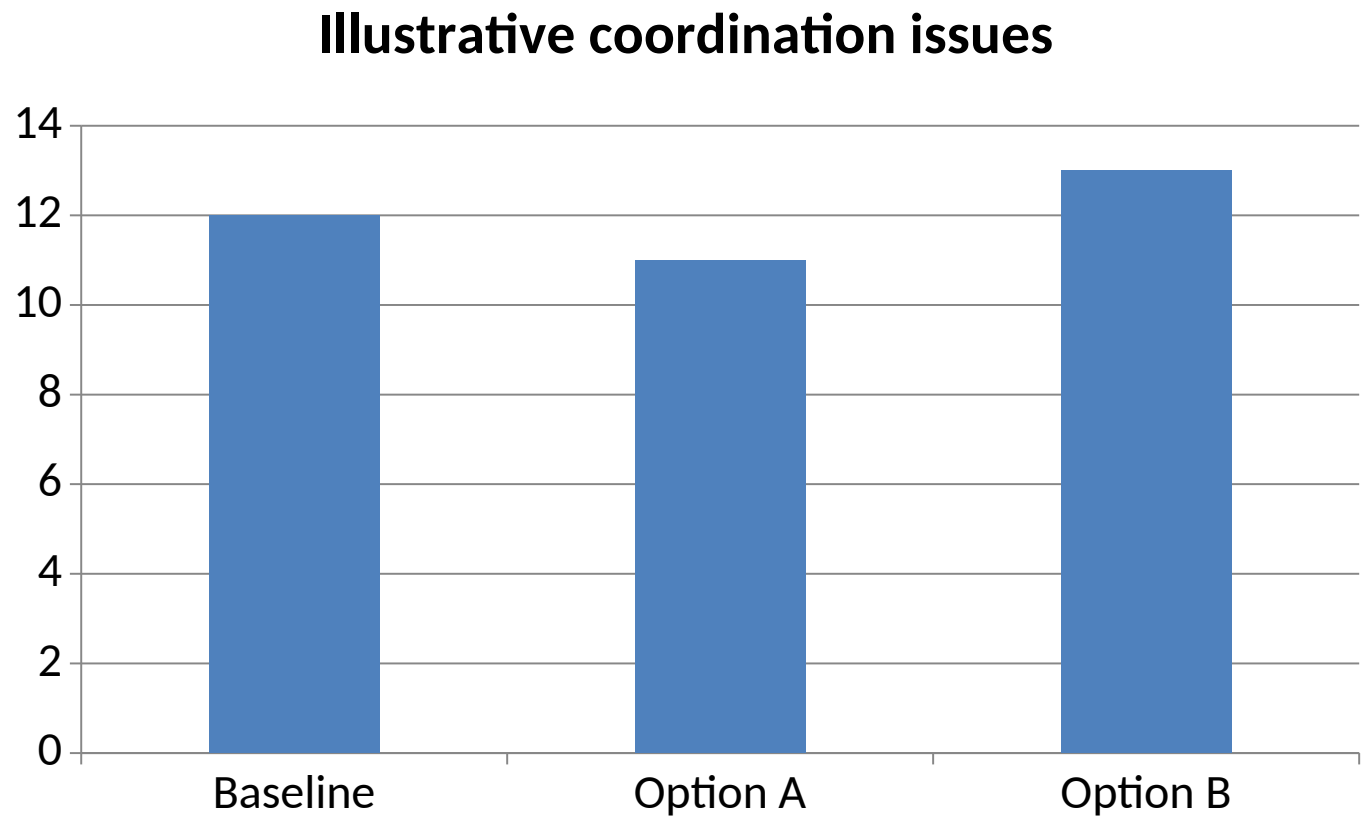
Accept Revision handover and evidence register only when the model object, schedule total, drawing annotation and Basis of Design agree.



VERIFICATION

Escalate mismatches with revision and owner.

Revision handover and evidence register — Quantitative review



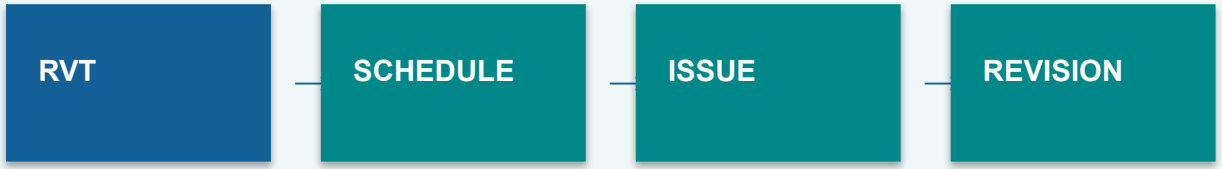
DESIGN GATE

Choose only after domain criteria and project requirements are checked.

Revision handover and evidence register — Evidence handoff

MECHANISM / ARTIFACT

Package: relevant RVT view + schedule extract + issue record + revision on sheet for Revision handover and evidence register.



VERIFICATION

Criteria: K8 A8 A9; Lab 15.

Lab 13 — Link and coordinate discipline models

MODEL OUTPUT

linked coordination model and change log

EVIDENCE

architecture link, workset, copy-monitor; working RVT, schedule and PDF

Lab 13 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 14 — Resolve clashes and evaluate options

MODEL OUTPUT

resolved clash register with option decision

EVIDENCE

interference report, design options, issue owner; working RVT, schedule and PDF

Lab 14 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Lab 15 — Export and defend the mechanical package

MODEL OUTPUT

handover bundle and signed evidence register

EVIDENCE

project standards, IFC, PDF, final QA; working RVT, schedule and PDF

Lab 15 — acceptance evidence

PASS CONDITIONS

System Browser assignment · model schedule reconciliation · linked architecture · revisioned sheet

Assessment

WA + PP

WA: eight open-ended questions, 60 min. PP: three integrated tasks, 120 min. Submit through LMS-TMS.

Assessment flow and sign-off

EVIDENCE

Question paper → assessor review → Competent / Not Yet Competent → signed record

Digital attendance

TRAQOM

Record assessment attendance in the approved digital attendance system.

Thank you

NEXT

Retain your RVT, exported sheet and evidence register for review.